

Income Approach to Valuation

Student Reference Manual



IAAO

INTERNATIONAL ASSOCIATION *of* **ASSESSING OFFICERS**

WE are a professional membership organization of government assessment officials and others interested in the administration of the property tax. We were founded in 1934, and we have more than 8,000 members worldwide from governmental, business, and academic communities.

WE are the internationally recognized leader and preeminent source for innovation, education, and research in property appraisal, assessment administration and property tax policy.

WE are IAAO, and **WE** value the world!



www.iaao.org



Contents

Curriculum Provisos.....	i
About the Course and IAAO	i
International Association of Assessing Officers Code of Ethics and Standards of Professional Conduct	ii
<i>Uniform Standards of Professional Appraisal Practice of the Appraisal Foundation</i>	vi
IAAO Education Policies.....	vii
Course Description.....	ix
Course Alignment with the IAAO <i>Apendium</i> Knowledge Areas (KAs)	xi
Chapter 1 Assessment and Appraisal Theory	1
Chapter 2 Development of the Net Operating Income Estimate	23
Chapter 3 Development of Capitalization Rates.....	115
Chapter 4 Contemporary Capitalization Methods.....	179
Chapter 5 Historical Capitalization Methods (Straight-Line Capitalization)	221
Addendum Fee Simple and Related Information	261
Appendix	267
A Note on Rounding.....	283
Glossary	284
We are IAAO	293
IAAO Designations.....	294
Courses offered by IAAO	296

COPYRIGHT

© 2022 by the International Association of Assessing Officers (IAAO), 314 W. 10th Street, Kansas City, MO 64105.

All rights reserved. No part of this document may be reprinted or reproduced or utilized in any form or by any electronic, mechanical, or other means, now known or hereafter invented including photocopying and recording, or in any information storage or retrieval system, without permission in writing from IAAO.

Designated trademarks and brands are the property of their respective owners. Other trademarks and trade names may be used in this document to refer to either the entities claiming the marks and names or their products. IAAO disclaims any proprietary interests in trademarks and trade names other than its own.

Information in this document is subject to change. No liability is assumed for technical or editorial omissions contained herein.

PAGE LEFT BLANK INTENTIONALLY

Curriculum Provisos

MATERIALS PROVISIO

The International Association of Assessing Officers has provided the accompanying course curriculum materials solely as educational information on the subject matter covered.

IAAO does not endorse any specific application, process, or form used in the accompanying curriculum materials, nor does it endorse any specific appraisal application, technique, methodology, or standard used in determining valuations for any assessment jurisdiction. Moreover, if any legal advice or other expert assistance is required, the services of a competent professional should be sought.

INSTRUCTOR PROVISIO

IAAO maintains a list of approved course instructors. IAAO is not responsible for instructors who are not approved by IAAO.

About the Course and IAAO

ABOUT THE COURSE

This course is presented by the Professional Development Department of IAAO. It was developed under the guidance of the Education Committee as part of a curriculum of almost 50 programs on appraisal procedures and assessment administration offered throughout the world.

This course consists of classroom instruction and a final examination. A certified IAAO instructor will guide students through learning aids, which may include lectures, discussions, case problems, review quizzes, and demonstrations. If the final examination is completed successfully, this course may fulfill part of the requirements for an IAAO professional designation. Talk to the course instructor or contact IAAO for additional information about the Professional Designation Program.

IAAO will send notification of the results of the final examination to students at the address indicated on the Education Record Form filled out the first day of class. Students will be notified of the total score. Seventy percent is the minimum passing score for all courses except the 2-day National USPAP workshop. Note that the examination pass/fail results will also be forwarded to the local sponsor and to the instructor. If a student fails the final examination, he or she is permitted a reexamination for a fee. Information about this reexamination opportunity will accompany the examination results.

IAAO attendance policy, adopted by the Education Sub-Committee, went into effect April 1, 2010. The policy states that students must attend 90% of the class hours to achieve the full number of educational hours offered for the course. Anything less than 90% attendance will result in zero hours of credit.

ABOUT IAAO

IAAO is a nonprofit, educational, and research association. The mission of IAAO is to promote innovation and excellence in property appraisal, assessment administration, and property tax policy through professional development, education, research, and technical assistance throughout the world. IAAO offers courses, workshops, and forums and presents the Annual International Conference on Assessment Administration. It also conducts research, produces publications, and provides technical assistance.

Members of IAAO subscribe to the IAAO Code of Ethics and Standards of Professional Conduct and to the *Uniform Standards of Professional Appraisal Practice*.

The six internationally recognized designations offered by IAAO represent the highest level of achievement in the field: Certified Assessment Evaluator (CAE), Assessment Administration Specialist (AAS), Cadastral Mapping Specialist (CMS), Personal Property Specialist (PPS), Residential Evaluation Specialist (RES) and Mass Appraisal Specialist (MAS).

Further information about publications, membership services, educational programs, professional designations, and requirements for membership can be obtained by writing to IAAO, 314 W. 10th Street, Kansas City, MO 64105-1616 or by calling 816/701-8100 or through the Internet home page, <http://www.iaao.org>.

International Association of Assessing Officers Code of Ethics and Standards of Professional Conduct

PREAMBLE

As a matter of fundamental principle, IAAO members should adhere to the highest ethical standards. Public trust in our performance is the foundation of our credibility. Assessment professionals support IAAO because they trust us to be good stewards of their resources, to uphold rigorous standards of conduct and to serve as a catalyst for excellence in the assessment profession.

Nonprofit organizations must earn this trust every day. It is up to all members of the IAAO – Executive Board members, committee members, volunteers, staff, and the general membership – to demonstrate their ongoing commitment to the core values of integrity, honesty, fairness, openness, respect, and responsibility.

The purpose of this Code of Ethics and Standards of Professional Conduct is to establish guidelines for assessing officials and all members of the International Association of Assessing Officers (IAAO) and set forth standards by which to judge an IAAO member whose conduct is in question. Members shall conduct themselves in a professional manner that reflects favorably upon themselves, the organization, the appraisal profession, and the property tax system, and avoid any action that could discredit themselves or these entities.

Adherence to the IAAO Constitution, Bylaws, Procedural Rules and Code of Ethics is the minimum standard of expected behavior. We must do more, however, than simply obey the rules. We must embrace the spirit of the governing documents, and go beyond stated requirements, making sure that what we do is matched by what the membership perceives and expects. Transparency, openness, and responsiveness to member's concerns must be integral to our behavior.

STATEMENT OF VALUES

The Code of Ethics of the International Association of Assessing Officers is built on a foundation of widely shared values. These values include our:

- Commitment to the improvement of the property tax system worldwide;
- Accountability to the public good;
- Commitment to excellence in assessment administration beyond property tax law;
- Respect for the worth and dignity of all individuals;
- Promotion of inclusiveness, fairness, and diversity;
- Obligation to organizational transparency, integrity, and honesty in all professional activities;
- Practice of responsible stewardship of resources; and
- Dedication to excellence and maintenance of the public trust.

The values are reflected in the following Code of Ethics of the International Association of Assessing Officers.

DEFINITIONS

For definitions of terms relating to appraisal practice, refer to the definitions section of the Uniform Standards of Professional Appraisal Practice (USPAP).

EXCEPTIONS

If compliance with or adherence to any Canon or Ethical Rule set forth in the IAAO Code of Ethics and Standards of Professional Conduct would constitute a violation of the law of any jurisdiction, such Canon or Ethical Rule shall be void and of no force or effect in such jurisdiction.

In stating each individual Canon or Ethical Rule, no attempt has been made to enumerate all of the various circumstances and conditions that will excuse an IAAO member from strict observance; however, the IAAO recognizes that illness, acts of God, and various other events beyond the control of an IAAO member may make it inequitable to insist upon a strict observance in a particular case. When an IAAO member, in the exercise of reasonable care, commits a violation due to illness, acts of God, or other circumstances beyond his or her control, it is expected that the Ethics Committee will act in a manner that will avoid an inequitable result.

Inasmuch as there are other remedies under applicable federal, state/provincial, and local laws, nothing in this Code shall apply to the conduct of a member toward his or her employees and other workers in the member's workplace including, but not limited to, employment discrimination and harassment.

CANON 1: (PROFESSIONAL DUTIES)

Members shall conduct their professional duties and any activities as a member of IAAO in a manner that reflects credit upon themselves, their profession, and the organization.

Ethical Rules

ER 1-1

It is unethical for members to conduct their professional duties in a manner that could reasonably be expected to create the appearance of impropriety.

ER 1-2

It is unethical for members to accept an appraisal or assessment -related assignment which they are not qualified to perform.

ER 1-3

It is unethical for members to knowingly violate laws and regulations or in the uniform application of such laws and regulations in the performance of their duties or to apply such laws and regulations in an inequitable manner.

ER 1-4

It is unethical for members to refuse (by intent or omission) to make available all public records in their custody for public review, unless access to such records is specifically limited or prohibited by law, or the information has been obtained on a confidential basis and the law permits such information to be treated confidentially. Assessing officers must make every reasonable effort to inform the public about their rights and responsibilities under the law and the property tax system.

ER 1-5

It is unethical for members to refuse to cooperate with public officials to improve the efficiency and effectiveness of the property tax in particular and public administration in general.

ER 1-6

It is unethical to engage in misconduct of any kind that leads to a conviction for a crime involving fraud, dishonesty, false statements, or moral turpitude.

ER 1-7

It is unethical to perform any appraisal, assessment, or consulting service that is not in compliance with the IAAO governing documents or the Uniform Standards of Professional Appraisal Practice.

CANON 2: (TRUTHFULNESS)

Members shall not make public statements (written or oral) that are untrue or tend to mislead or deceive the public in the course of performing their professional duties.

Ethical Rules

ER 2-1

It is unethical to provide inaccurate, untruthful, or misleading information to solicit assessment-related assignments or to use misleading claims or promises of relief that could lead to loss of confidence in appraisal or assessment professionals by the public.

ER 2-2

It is unethical to claim an IAAO professional designation unless authorized, whether the claim is verbal or written, or to claim qualifications that are not factual or may be misleading.

ER 2-3

It is unethical to fail to recognize the source (s) of any materials quoted or cited in writings or speeches.

CANON 3: (CONFLICT OF INTEREST)

Members shall not engage in any activities in which they have or may reasonably be considered by the public as having, a conflict of interest.

Ethical Rules

ER 3-1

It is unethical for members to accept an appraisal or assessment-related assignment that can reasonably be construed as being in conflict with their responsibility to their jurisdiction, employer, or client, or in which they have an unrevealed personal interest or bias.

ER 3-2

It is unethical to accept an assignment or responsibility in which there is a personal interest without full disclosure of that interest.

ER 3-3

It is unethical to accept an assignment or participate in an activity where a conflict of interest exists and could be perceived as a bias or impair objectivity.

CANON 4: (SUPPORT OF IAAO)

Members shall abide by and support the provisions of the IAAO Constitution, Bylaws, and Procedural Rules.

Ethical Rules

ER 4-1

It is unethical for an IAAO member to:

(a) Knowingly to make false statements or submit misleading information when completing a membership application, or to refrain from promptly submitting any significant information in the possession of such member when requested to do so as part of an IAAO membership application. (b) Knowingly to submit misleading information to the duly authorized Ethics Committee or subcommittee; to refrain from promptly submitting any significant information in the possession of the member as requested by the committee or subcommittee; to refuse to appear for a personal interview or participate in an interview conducted by telephone as scheduled by the committee or subcommittee; or to refuse to answer promptly all relevant questions concerning an appraisal or assessment-related assignment or related testimony being investigated by the committee or subcommittee. Any member who has submitted misleading information to the Ethics

Committee may be subject to ethical charges filed by the committee. (c) Fail or refuse to submit promptly to an authorized IAAO committee a written appraisal report or file memorandum containing data, reasoning, and conclusions, or to fail or refuse to permit an authorized committee to review an appraisal report, assessment-related assignment, or file memorandum when requested to do so by a person or persons authorized to review such material. (d) Fail or refuse to submit promptly any significant information in the possession of a member concerning the status of litigation related to an ethics matter when requested to do so by the chair of the Ethics Committee; or knowingly to submit misleading information to the chair of the Ethics Committee concerning the status of litigation.

ER 4-2

It is unethical to fail to comply with the terms of a summons issued by the Ethics Committee.

ER 4-3

It is unethical to refuse to cooperate fully with the IAAO Executive Board, Ethics Committee and the staff of IAAO in all matters related to the enforcement of this Code, as set forth in the Ethics Committee's Rules and Procedures, as amended from time to time.

ER 4-4

It is unethical to violate the IAAO Constitution, Bylaws, or Procedural Rules.

ER 4-5

Any member who has submitted misleading information to the Ethics Committee or does not comply with the terms of these Canons may be subject to ethical charges by the Committee.

CANON 5: (PROFESSIONAL DUTIES)

Members shall comply with the requirements of the Uniform Standards of Professional Appraisal Practice.

Ethical Rules**ER 5-1**

It is unethical to fail to observe the requirements of the Uniform Standards of Professional Appraisal Practice. Members residing outside the United States must follow appraisal standards that govern appraisers within their jurisdiction.

Uniform Standards of Professional Appraisal Practice of the Appraisal Foundation

The International Association of Assessing Officers has adopted standards I through I 0 of the Uniform Standards of Professional Appraisal Practice. (Definitions, Preamble, Ethics Rule, Record Keeping Rule, Competency Rule, Scope of Work Rule, Jurisdictional Exception Rule, Standards, Standards Rules, Statements, and Comments, although not included below, are incorporated by reference.)

Standard 1. Real Property Appraisal, Development

In developing a real property appraisal, an appraiser must identify the problem to be solved, determine the scope of work necessary to solve the problem, and correctly complete research and analyses necessary to produce a credible appraisal.

Standard 2. Real Property Appraisal, Reporting

In reporting the results of a real property appraisal, an appraiser must communicate each analysis, opinion, and conclusion in a manner that is not misleading.

Standard 3. Appraisal Review, Development

In developing an appraisal review, an appraiser must identify the problem to be solved, determine the scope of work necessary to solve the problem, and correctly complete research and analyses necessary to produce a credible appraisal review.

Standard 4. Appraisal Review, Reporting

In reporting the results of an appraisal review, an appraiser must communicate each analysis, opinion, and conclusion in a manner that is not misleading.

Standard 5. Mass Appraisal, Development

In developing a mass appraisal, an appraiser must identify the problem to be solved, determine the scope of work necessary to solve the problem, and correctly complete research and analyses necessary to produce a credible mass appraisal.

Standard 6. Mass Appraisal, Reporting

In reporting the results of a mass appraisal, an appraiser must communicate each analysis, opinion, and conclusion in a manner that is not misleading.

Standard 7. Personal Property Appraisal, Development

In developing a personal property appraisal, an appraiser must identify the problem to be solved, determine the scope of work necessary to solve the problem, and correctly complete research and analyses necessary to produce a credible appraisal.

Standard 8. Personal Property Appraisal, Reporting

In reporting the results of a personal property appraisal, an appraiser must communicate each analysis, opinion, and conclusion in a manner that is not misleading.

Standard 9. Business Appraisal, Development

In developing an appraisal of an interest in a business enterprise or intangible asset, an appraiser must identify the problem to be solved, determine the scope of work necessary to solve the problem, and correctly complete the research and analyses necessary to produce a credible appraisal.

Standard 10. Business Appraisal, Reporting

In reporting the results of an appraisal of an interest in a business enterprise or intangible asset, an appraiser must communicate each analysis, opinion, and conclusion in a manner that is not misleading.

IAAO Education Policies

Use of Educational Materials

This educational material was developed, published and copyrighted by the International Association of Assessing Officers (IAAO). The material is to be used exclusively for educational purposes.

Cost of Program

The cost of the program has been determined by the program provider for this educational program. They have purchased these IAAO copyrighted materials from the IAAO.

Selection of Instructor

This program is not sponsored by the IAAO, and the IAAO has not taken part in selecting the instructor. If your instructor qualifies as an in-state instructor, he or she has attended an IAAO Instructor Evaluation Workshop and has attended and successfully passed the course he or she is teaching. An instructor that has achieved the status of a senior, senior specialty, or regular instructor has successfully completed a 3-day Instructor Evaluation Workshop and training assignment, independently taught IAAO courses and holds an IAAO designation or candidacy in the IAAO designation program.

Only senior, senior specialty or regular instructors are on the IAAO Approved Instructor List. IAAO takes no responsibility for the quality of instruction, delivery or interpretation of material in this course for instructors not on the IAAO Approved Instructor List. Therefore, any questions related to the quality of instruction, or delivery and interpretation of material should be directed to the local coordinator.

Questions related to the printed materials provided by the IAAO, however, should be addressed to the IAAO Professional Development Department.

Continuing Education Credit

If you are applying for continuing education credit for the IAAO Professional Designation Program, you must attend the course for the minimum number of hours required by the IAAO and pass the IAAO course examination.

Continuing education credit is not automatic and application for it is the responsibility of the student and/or the program provider.

Exam Results

Students will be notified by email at the address provided on their Scantron form 2-4 weeks after the program. Please refrain from phone calls requesting your results; it is against IAAO Policy to release results over the phone. If you have not received your results after 4 weeks, please email results@iaao.org for assistance.

IAAO Attendance Policy

The Education Committee has responded to the higher standards of The Appraisal Foundation by adopting a new attendance policy for all educational classes offered through IAAO. Effective April 1, 2010, the policy states that students must attend 90% of classroom hours to achieve the full number of educational hours offered for the course. Anything less than 90% attendance will result in zero hours of credit.

Attendance will be verified by IAAO using the Attendance Sheet returned by the Instructor. Students are responsible for initialing the sheet for every $\frac{1}{2}$ day they attend in class. Missing initials will be counted as an absence for that period of time.

Ten percent (10%) of a 30-hour class would be equal to 3 hours, however, if a student misses a $\frac{1}{2}$ day of a 30-hour class, this would actually equal 3.75 hours. Because of the difficulty in measuring less than a $\frac{1}{2}$ day of class, it has been determined that a student will be allowed to miss up to $\frac{1}{2}$ day and still obtain the maximum offered 30-hour credit. Anything over $\frac{1}{2}$ day absence will result in the loss of the full 30 hours credit. An absence from any of the IAAO Workshops will result in the student earning zero continuing education hours.

Course Description

This course is designed to provide students with an understanding and working knowledge of the procedures and techniques required to estimate the market value of vacant or improved properties by the income approach. This course covers real estate finance and investment, capitalization methods and techniques, analysis of income and expenses to estimate operating income, selection of capitalization rates, and application of the approach.

NOTE: The timetable and day covered info applies to the classroom in-person courses only, not our online offerings.

Topic	Timetable	Day Covered
Chapter 1		
Overview and Introduction to Income Capitalization	15 minutes	Monday AM
Real Estate Investment	15 minutes	Monday AM
Real Estate Financing	60 minutes	Monday AM
Primary Rates	30 minutes	Monday AM
Generic Capitalization Formulas	40 minutes	Monday AM
Review Questions	20 minutes	Monday AM
Chapter 2		
Sources of Income Data	15 minutes	Monday PM
Basis of Income	90 minutes	Monday PM
Considerations When Comparing Rental Properties	20 minutes	Monday PM
Commercial Property Classifications	40 minutes	Monday PM
Rental Units of Comparison	30 minutes	Monday PM
Income and Expense Statements	150 minutes	Monday PM/Tuesday AM
Review Questions	25 minutes	Tuesday AM

Chapter 3		
Introduction to Relationship of Capitalization Rates and Value Estimate	30 minutes	Tuesday AM
Overall Capitalization Rate (R_0) or (OAR)	15 minutes	Tuesday AM
Five Methods of Developing an Overall Capitalization Rate (R_0)	210 minutes	Tuesday AM/PM
Capitalization Rate Individual Components	95 minutes	Tuesday PM/Wednesday AM
Review Questions	35 minutes	Wednesday AM
Quiz #1	45 minutes	Wednesday AM
Chapter 4		
Capitalization Methods	30 minutes	Wednesday AM
Application of Direct Capitalization	200 minutes	Wednesday PM
Introduction to Compound Interest Tables	30 minutes	Thursday AM
Review Questions	20 minutes	Thursday AM
Chapter 5		
Classic Straight-Line Capitalization Method	30 minutes	Thursday AM
Straight-Line Capitalization Assumptions	15 minutes	Thursday AM
Components of the Land and Building Capitalization Rates	55 minutes	Thursday AM
Methods of Developing the Land Capitalization Rate and the Building Capitalization Rate	60 minutes	Thursday AM/PM
Land Residual Technique	105 minutes	Thursday PM
Building Residual Technique	105 minutes	Thursday PM
Review Questions	20 minutes	Thursday PM
Quiz #2	60 minutes	Thursday PM

Course Alignment with the IAAO *Apendium* Knowledge Areas (KAs)

The IAAO *Apendium* is a compilation of key knowledge and skills for the mass appraisal profession. The IAAO Body of Knowledge image below identifies the eight broad Knowledge Areas (KAs).



As you work through this Student Reference Manual, you will see one or more of the Knowledge Area icons at the start of each chapter. These indicate the knowledge areas of the *Apendium* that align with the knowledge presented in that chapter.

For example, if a chapter covers information from *Apendium* Knowledge Area 4 (Appraising Property), the icon you will see is this:



PAGE LEFT BLANK INTENTIONALLY

Chapter 1 | Assessment and Appraisal Theory

APPENDIX KNOWLEDGE AREAS



SUGGESTED READING

Source	Chapter	Pages
Property Assessment Valuation	Chapter 11	303-315
	Chapter 13	341-345

CHAPTER TOPICS AND TIMETABLE

Topic	Time	Percentage
Overview and Introduction to Income Capitalization	15 minutes	8
Real Estate Investment	15 minutes	8
Real Estate Financing	1 hour	34
Primary Rates	30 minutes	17
Generic Capitalization Formulas	40 minutes	22
Review Questions	20 minutes	11

OBJECTIVES

Upon completion of Chapter 1, the student should be able to:

- Recognize the definition of income-producing property.
- Recognize the difference between market value and investment value and know why these differences are significant in deriving value estimates by the income approach.
- Explain how the appraiser's understanding of anticipation, change, and the other basic appraisal principles aids in the analysis of market data necessary in the application of the income approach.
- Recognize the different investor requirements for income-producing properties.
- Recognize the factors that influence the behavior of investors in real estate.
- Identify and explain the various types of financing available to investors in real estate.
- List the sources of financing for purchasing real estate.
- Identify and be able to explain the following components of capitalization rates: overall yield rate (discount rate); recapture rate; and effective tax rate.
- Understand the basic concepts of the income approach model and how supply and demand factors interact in specific markets.
- Understand the income capitalization formulas, IRV, and VIF in appraising income-producing properties.

Chapter 1 | Assessment and Appraisal Theory

I. Introduction to Income Capitalization Approach

A. The income capitalization approach is one of the three approaches to value. In this approach, the value of an income-producing property is estimated by converting anticipated benefits (income or rent) arising from the ownership of the income producing property. This is also, referred to as the capitalization process. **Components of the capitalization process are value, net operating income and capitalization rate.**

B. Income capitalization is based upon the economic principles of the following:

1. Anticipation – Value is created by the expectation of benefits to be derived in the future. This is the underlying principle which provides the basis of the income capitalization approach.
2. Change – Investor's expectations of changes in income levels, the expenses required to ensure income, and probable increases or decreases in property that must be addressed and forecast.
3. Competition – Competition means that an excess of one type of facility will decrease the value of all such facilities. Excess competition destroys balance. Competition is created by the potential for profits, which attracts new buyers and sellers to a market. Competition among buyers may lead to shortages, which increases prices and therefore profits to sellers. Conversely, competition among sellers may lead to oversupply.
4. Substitution – The prices, rents, and rates-of-return of property tend to be set by the current prices, rents, and rates-of-return for equally desirable substitute properties. The principle of substitution is market-oriented and provides the basis for estimating rents and expenses and selecting the proper discount rate or capitalization rate for the subject property.
5. Balance – A suitable balance among types and locations of income-producing properties affects value; an imbalance in use may result in declining value.
6. Contribution – The value of a component of real estate can be measured by the amount it contributes to net operating income because net operating income can be capitalized into value.

7. Supply – The amount of product that producers are willing to sell under various conditions during a given period.
 8. Demand – The amount of product a buyer is willing and able to buy during some period, given the choices available to them.
- C. **Market value** (from IAAO Glossary) – Market value (also referred to as value in exchange) means the most probable price a property should bring in a competitive and open market under all conditions requisite to a fair sale, the buyer and seller each acting prudently and knowledgeably, and assuming the price is not affected by undue stimulus. Implicit in this definition are the consummation of a sale as of a specified date and the passing of title from seller to buyer under conditions whereby:
1. Buyer and seller are typically motivated.
 2. Both parties are well informed or well advised and acting in what they consider their own best interests.
 3. A reasonable time is allowed for exposure in the open market.
 4. Payment is made in terms of cash in U.S. dollars or in terms of financial arrangements comparable thereto.
 5. The price represents the normal consideration for the property sold unaffected by special or creative financing or sales concessions granted by anyone associated with the sale.
- D. **Investment value** – The worth of an investment property to an investor. Investment value may or may not coincide with market value depending on the requirements of the specific investor.
- E. Components of real property rights.
1. Ownership entity – individual, corporate shareholders, partnership interests
 2. Financial interest – equity, debt (mortgage)
 3. Legal estate – fee simple *, leased fee, leasehold

***SEE ADDENDUM FOR COMPLETE AND ACCURATE DESCRIPTION OF FEE SIMPLE**

II. Real Estate Investments

- A. Real estate competes with other investments for the investor's dollars. Investments are generally divided into two categories:
 - 1. Fixed-dollar investments – Not much risk and very little gain. Example would be a savings account.
 - 2. Growth investments – More risk but greater opportunity for growth. Examples would be stocks and real estate purchase.
- B. An investor analyzes the various opportunities available and asks the following questions:
 - 1. How much will it cost?
 - 2. How much will I get back?
 - 3. When will I get it back?
 - 4. What are the risks?
 - 5. What is the return of a real estate investment compared to other investments with similar risks?
- C. The objectives of the investor vary.
 - 1. All investors want a return **of** their investment (recapture rate or getting back the amount invested).
 - 2. All investors want a return **on** their investment (overall yield or discount rate; profit realized in addition to getting back the amount invested).
- D. Investors are different.
 - 1. Some investors require a substantial annual return on investment.
 - 2. Some investors require potential growth for their investment (capital gain).
 - 3. Some investors are interested in both an annual return on investment and potential for growth.
- E. Factors influencing the decisions of investors are diverse. To better understand these factors, Illustration 1-1 compares an investment in a savings account and an investment in real estate.

Illustration 1-1: Factors Influencing Investor Decisions

Factor	Savings Account	Real Estate
Safety	Safe	Can be risky
Liquidity	Good as cash	Time to convert
Size	Can be small	Relatively large
Collateral	Full amount	Limitations
Time	Short or long	Short or long
Management	Very little	Requires management
Appreciation	May lag inflation	May increase/decrease in value
Income Tax	Taxed directly	Tax advantages
Leverage	No chance	Possible

1. **Safety** of the investment – Investors are averse to risk. A savings account insured by the government is relatively free of risk. Real estate may be a risky investment due to a variety of factors affecting its economic environment.
2. **Liquidity** of the investment – A savings account is accessible on demand for conversion to cash. Real estate investment requires time to convert to cash because it needs time to secure a contract and sell. This may take weeks or even months before the title transfers and the seller is paid.
3. **Size** of the investment – A savings account can be opened with a very small amount of money. Real estate generally requires a substantial commitment of capital.
4. **Collateral** – The entire investment in a savings account can be used as collateral for a loan. However, lending institutions generally will only allow a percentage of the real estate's market value to be used as collateral.
5. **Time** of the investment (holding period) – A savings account can be opened and closed in a very short period. Real estate generally requires a longer-term commitment of funds but can be shorter for value-added investments. A value-added investment involves buying a property, improving it in some way, and selling it at an opportune time for gain.

6. **Management** (by the owner or a property manager) – A savings account requires very little management decision making by the owner. A real estate investment on the other hand, requires substantial investment management decision-making on the part of the owner. Some examples of these decisions are the type of property to be purchased; whether to make repairs, capital improvements, or renovations; hold the real estate, sell it, or refinance it.
7. **Appreciation** of investment during holding period – Savings accounts often lose purchasing power as a result of inflation. Typically, real estate tends to appreciate over time and can be used as a hedge against inflation.
8. **Income tax** advantages – There are no income tax advantages for savings accounts. All interest is taxed directly. Real property may offer the opportunity to reduce, defer, or eliminate income taxes.
9. **Leverage** – Leverage is the borrowing of funds in hopes of earning a greater return than the cost of the borrowed funds. Investors cannot borrow money at a rate which would allow successful investment in a savings account. Investors in real estate often can borrow money at a lower rate than the yield on the real estate investment. Leverage can be positive, negative, or neutral. Positive leverage is achieved when funds are invested in property, which has a higher rate of return than the cost of the borrowed funds. Negative leverage occurs when the cost of the borrowed funds is greater than the return of the property investment. Neutral leverage occurs when the cost of the borrowed funds equals the return on the property investment.

III. Real Estate Financing

A. Types of financing for real estate purchases

1. **Cash** – The purchase of real estate is often financed entirely by the purchaser without funds provided by another party. A significant amount of commercial real estate is financed entirely by cash.
2. **Trust deed** – A legal instrument similar to a mortgage that transfers the title of a property to a trustee (sometimes referred to as a deed of trust). The borrower conveys the title to a trustee for the benefit of the lender but retains the right to use and occupy the property. Trust deeds are used to eliminate the need for a foreclosure proceeding against the borrower in the event of default.
3. **Land contract** – This agreement is also known as a contract for deed or an installment sales contract. The purchaser agrees to pay a small down payment when the contract is signed, with the balance in specified amounts over the term of the contract.
4. **Mortgages** (most common type of financing for real estate) – Two-party agreements between the borrower (mortgagor) and the lender (mortgagee). It is a written document that pledges specific real estate as a guarantee for the payment of a loan to help purchase a property. Types of mortgages are:
 - a. **First mortgage (conventional)** – A loan that is neither insured nor guaranteed by the federal government. A typical loan ratio covers 60-95 percent of the property value. A first mortgage also can be an insured mortgage when the loan is insured against loss by an agency of the federal government or a private insurer.
 - b. **Junior mortgage** – A mortgage on a property executed and recorded after a prior lien has been made. (Often called a second mortgage.) Typically covers 5 to 50 percent of the property value.
 - c. **Purchase-money mortgage** – A mortgage given by the buyer to the seller to finance a portion of the sales price of real estate. It is used in place of, or in addition to, institutional financing.

- d. Construction loan mortgage – A short-term loan made to finance new construction. Funds are advanced as construction progresses, with permanent financing arranged upon completion.
 - e. Open-end mortgage – A mortgage which allows the borrower to obtain additional funds as long as an agreed loan to value ratio is maintained or other specified terms are met.
 - f. Chattel mortgage – A mortgage only on personal property.
 - g. Package mortgage – A package mortgage covers real estate as well as personal property included with the real estate. A package mortgage may be found on apartment buildings where the stoves, dishwashers, and refrigerators are covered by the mortgage.
5. Sources of mortgage loans
- a. Commercial banks – The most common lender for commercial property financing and includes local, regional, and national banks.
 - b. Credit unions – Provide similar financing options as commercial banks, but typically allow prepayment without any penalties and flexible refinancing options.
 - c. Mortgage companies – Originate the loan and typically have a lender from the secondary market in place to purchase the loan at a later date or participate in the funding of the loan. The loans may be sold in the secondary market to insurance companies or a government agency such as Fannie Mae or Freddie Mac. Some mortgage companies have a broker that will obtain multiple quotes for the borrower to choose from.
 - d. Life insurance companies – Attracted to large development loans. Important sources of financing for multifamily and commercial properties. Also active in junior mortgage loans.
 - e. Conduit lenders – Originate commercial mortgages and hold them as investments for a short period before securitizing the loans (assigns the loan into a trust vehicle—which has several other mortgage loans) and selling them off as Commercial Mortgage-Backed Securities (CMBS) to investors.

- f. Private money lender – Financing provided by a non-institutional (non-bank) lender secured by a note and deed of trust. The private money lender may be an individual or group of investors.
6. Mortgages classified according to type of repayment structure
- a. Amortized mortgage (fixed-rate mortgage) – A permanent loan in which the sum of the principal and interest payments remains fixed throughout the term of the loan. It usually carries a higher interest rate than an adjustable-rate loan because there is more risk to the lender.
 - b. Variable or adjustable-rate mortgage (ARM) – A mortgage loan in which the interest rate is periodically adjusted in accordance with some formula or index. It typically carries a lower interest rate initially because there is less risk to the lender than a fixed-rate loan.
 - c. Straight mortgage – A short-term mortgage, usually about three years. It has no requirement for amortization. Requires level interest payments, usually monthly or quarterly; and requires that the entire unpaid principal and interest balance be paid at maturity.
 - d. Partially amortized mortgage (uses a balloon payment at end of term) – A mortgage loan which is not fully amortized at maturity and requires the total principal to be paid at the end of the mortgage term in one lump sum. The lump-sum payment made at the end of the mortgage term is referred to as a balloon payment.
 - e. Reverse mortgage – A low-interest loan, usually for senior homeowners that uses a home's equity as collateral. The loan amount is a percentage of the home's value determined by the age of the youngest homeowner. The loan does not have to be repaid until the last surviving homeowner permanently moves out of the property or passes away.
7. Effects of financing on mortgage payments (Demonstrations 1-1 to 1-4)
- a. Amount borrowed
 - b. Interest rate
 - c. Term of loan
 - d. Frequency of payments

Demonstration 1-1: Amount Borrowed Varies

	Lower Loan Ratio	Higher Loan Ratio
Amount Borrowed	\$100,000	\$150,000
Interest Rate	8.0%	8.0%
Term of Loan (years)	25	25
Partial Payment Factor	0.007718	0.007718
Monthly Payment (1)	\$772	\$1,158
Total Payment (2)	\$231,545	\$347,317
Total Interest Paid (3)	\$131,545	\$197,317

- (1) *When the amount borrowed is increased, the monthly payment increases.*
- (2) *When the amount borrowed is increased, the total payment increases.*
- (3) *When the amount borrowed is increased, the total interest paid increases.*

Demonstration 1-2: Interest Rate Varies

	Lower Interest Rate	Higher Interest Rate
Amount Borrowed	\$100,000	\$100,000
Interest Rate	6.0%	10.0%
Term of Loan (years)	25	25
Partial Payment Factor	0.006443	0.009087
Monthly Payment (1)	\$664	\$909
Total Payment (2)	\$193,290	\$272,610
Total Interest Paid (3)	\$93,290	\$172,610

- (1) *When the interest rate increases, the monthly payment increases.*
- (2) *When the interest rate increases, the total payment increases.*
- (3) *When the interest rate increases, the total interest paid increases.*

Demonstration 1-3: Payment Frequency Varies

	Monthly Payments	Annual Payments
Amount Borrowed	\$100,000	\$100,000
Interest Rate	10.0%	10.0%
Term of Loan (years)	20	20
Partial Payment Factor	0.009650	0.117460
Monthly Payment (1)	\$965	\$11,746
Total Payment (2)	\$231,605	\$234,919
Total Interest Paid (3)	\$131,605	\$134,919

- (1) *When the frequency of payments is increased, the payment decreases.*
- (2) *When the frequency of payments is increased, the total payment decreases.*
- (3) *When the frequency of payments is increased, the total interest paid decreases.*

Demonstration 1-4: Term of Loan Varies (Length of Loan)

	Shorter Term	Longer Term
Amount Borrowed	\$100,000	\$100,000
Interest Rate	8.0%	8.0%
Term of Loan (years)	15	25
Partial Payment Factor	0.009557	0.007718
Monthly Payment (1)	\$956	\$772
Total Payment (2)	\$172,017	\$231,545
Total Interest Paid (3)	\$ 72,017	\$131,545

- (1) *When the term of the loan increases, the monthly payment decreases.*
- (2) *When the term of the loan increases, the total payment increases.*
- (3) *When the term of the loan increases, the total interest paid increases.*

IV. Primary Rates

- A. The **overall yield rate** (Y_o), also known as the **discount rate**, reflects the return on investment. The rate reflects the compensation necessary to attract investors to give up liquidity, defer compensation and assume the risks of investing. It consists of four factors in the summation concept.
1. **Safe rate** – The base rate on the safest investments such as government insured investments. In the summation method it should be taken from investments having the least risk.
 2. **Risk rate** – An amount in addition to the safe rate which compensates the investor for the degree of risk in the investment. It is a component of the discount rate because the return on real estate is a desired return and may or may not be realized by the investor.
 3. **Rate for non-liquidity** – An amount in addition to the safe rate and the risk rate which compensates the investor for the time necessary to convert the real estate into cash.
 4. **Rate for management** (management of investment rate) – An amount in addition to the safe rate, risk rate, and non-liquidity rate which compensates the investor for the decision-making process to manage the real estate investment.
- B. **Recapture rate** – Provides for the return of the investment in the wasting portion of the asset. This is similar to the depreciation rate for the improvement.
- C. **Effective tax rate** (ETR) – Reflects the relationship between the real estate taxes and the value of the property.

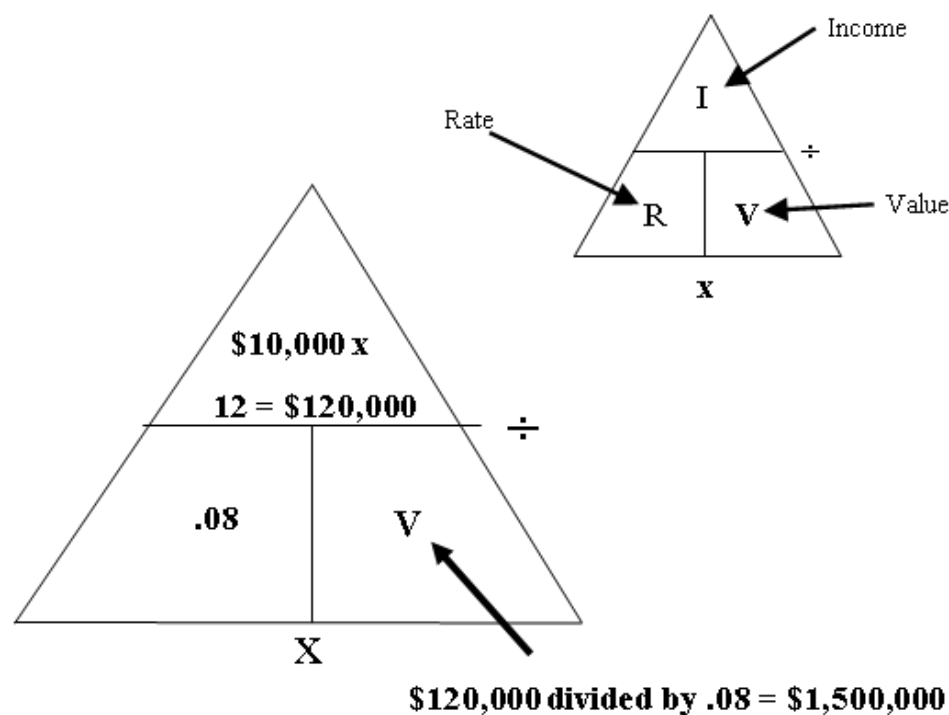
V. Generic Capitalization Formulas

A. The IRV formula (see Demonstration 1-5)

1. I (income) = Rate $\times$ Value
2. R (rate) = $\text{Income} \div \text{Value}$
3. V (value) = $\text{Income} \div \text{Rate}$
4. " I " represents Net Operating Income (NOI) in the IRV formula

Demonstration 1-5: IRV

A corner lot in a business district is used as a parking lot. The monthly gross income earned by the property is \$12,500 and the monthly expenses are \$2,500. The monthly net income is \$10,000. The owner's required rate of return for this type of property is 8% per year. What is the value of this property?

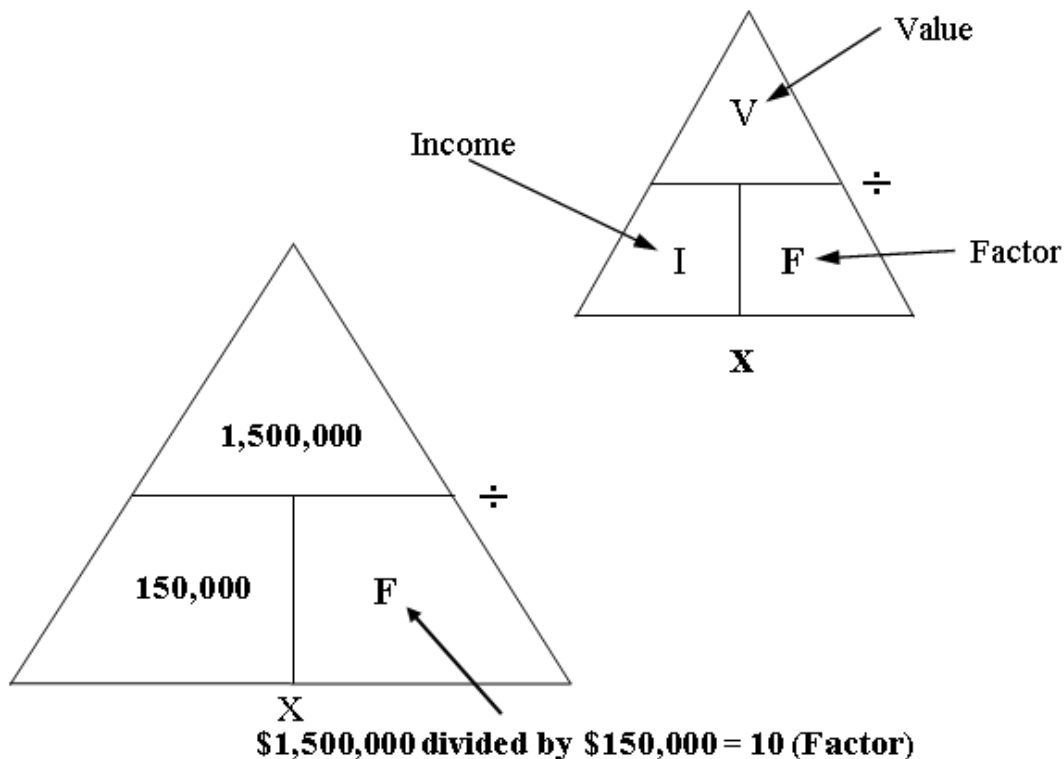


B. The **VIF formula** (see Demonstrations 1-6 and 1-7) – The VIF factor is also known as the **Gross Income Multiplier**.

1. $V (\text{value}) = \text{Income} \times \text{Factor}$
2. $I (\text{income}) = \text{Value} \div \text{Factor}$
3. $F (\text{factor}) = \text{Value} \div \text{Income}$
4. "I" represents Gross Income in the VIF formula

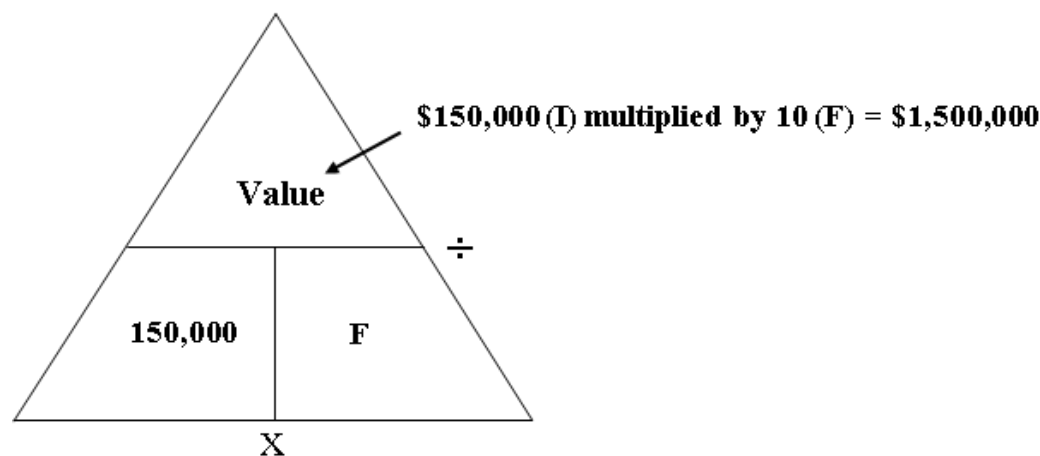
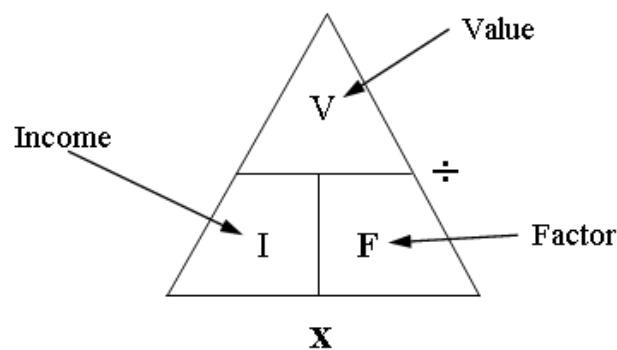
Demonstration 1-6: VIF/GIM

You derived the value of \$1,500,000 for the corner parking lot in a business district in Demonstration 1-5. The annual gross income earned is $\$12,500 \times 12 = \$150,000$. Using VIF, calculate the income factor for this property. Note: The VIF factor is also known as the Gross Income Multiplier (GIM).



Demonstration 1-7: VIF/GIM

You derived a factor of 10 that is applicable to the corner parking lot in the business district in Demonstration 1-6. You calculated the annual gross income earned of \$150,000. Using VIF, calculate value for this property. Note: The VIF factor is also known as the Gross Income Multiplier (GIM).



When using the VIF formula, the F component can also be referred to as a multiplier. A multiplier for commercial property is a gross income multiplier (GIM) and is calculated by dividing the sale price by the annual gross income (Sale Price/Annual Gross Income). A residential multiplier is called a gross rent multiplier.

REVIEW QUESTIONS – Chapter 1

1. One of three approaches to value in which the appraiser derives a value indication by converting anticipated benefits through ownership of income-producing property is the _____ approach.
2. Real estate competes with other investments for the investor's dollar. As an investor analyzes various opportunities, what will he or she consider?
 - _____
 - _____
 - _____
 - _____
 - _____
3. The economic principle of _____ states that value is created by the expectation of benefits to be derived in the future.
4. The economic principle of _____ states that a property's maximum value tends to be set by the lowest cost or price at which another property of equivalent utility can be acquired.
5. Competition is created by the potential for profits. However, competition among sellers may lead to a/an _____, which reduces prices and profits. Competition among buyers may lead to _____, which increases prices and profits to sellers.
6. The _____ rate reflects the return of the investment in the wasting asset.

7. List nine factors influencing investor decisions.

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

8. _____ leverage is achieved when funds are invested in property, which has a higher rate of return than the cost of borrowed funds.

9. The four most common methods of financing real estate are:

- _____
- _____
- _____
- _____

10. A mortgage on personal property is termed a _____ mortgage.

11. A _____ is also called a second mortgage.

12. List four types of mortgages classified according to repayment provisions.

- _____
- _____
- _____
- _____

13. A payment on the balance due of a note at the end of the loan term that is in excess of the regular payment amounts is called a _____ payment.

14. A _____ mortgage covers real estate as well as personal property included with the real estate.

15. The four variables in real estate financing that affect the mortgage payment are:

- _____
- _____
- _____
- _____

16. The _____ rate (or discount rate) reflects the return on the investment.

17. The summation concept is a theoretical procedure in developing the discount rate for a real estate investment and is comprised of the following four parts:

- _____
- _____
- _____
- _____

18. The _____ rate reflects the relationship between the real estate taxes and the value of the property.

19. To obtain value using the IRV formula, one must _____ the income by the rate.

20. To obtain value using the VIF formula, one must _____ the income by the factor.

PAGE LEFT BLANK INTENTIONALLY

Chapter 2 | Development of the Net Operating Income Estimate

APPENDIUM KNOWLEDGE AREAS



SUGGESTED READING

Source	Chapter	Pages
Property Assessment Valuation	Chapter 12	317-337

CHAPTER TOPICS AND TIMETABLE

Topic	Time	Percentage
Sources of Income Data	15 minutes	3
Basis of Income	1 hour, 30 minutes	21
Considerations When Comparing Rental Properties	20 minutes	5
Commercial Property Classifications	40 minutes	9
Types of Rent	1 hour	14
Rental Units of Comparison	30 minutes	7
Income and Expense Statements	2 hours, 30 minutes	35
Review Questions	25 minutes	6

OBJECTIVES

Upon completion of Chapter 2, the student should be able to:

- Identify the sources of income data used to develop income and expense estimates for the appraised property.
- Define the term “lease.”
- Identify and explain the various types of rents associated with income-producing properties.
- Identify the different lease types associated with the rental of income-producing properties.
- Calculate various units of comparison for rental rates and apply the appropriate unit of comparison using the income approach analysis.
- Outline an income and expense statement using potential gross income (PGI), vacancy and collection loss (V&C), effective gross income (EGI), operating expenses (OE), and net operating income (NOI) formulas.
- Define and calculate effective rent using rent concessions, percentage rent, expense stops, and tenant improvements.
- Identify the categories of income and differentiate between proper and improper expenses used in reconstructed operating statements of income-producing properties.
- Develop an accurate reconstructed operating statement for appraisal purposes.
- Calculate a vacancy and collection loss rate (V&C), an operating expense ratio (OER), and a gross income multiplier (GIM).

Chapter 2 | Development of the Net Operating Income Estimate

I. Sources of Income Data

A. Market sources of data

1. Public records (deeds, sale validation forms, tax stamps, etc.) – May create leads for buyer/seller contact information or whether the property was leased at the time of sale. Lease deeds can be recorded, but are not required, and may assist in detecting land only leases and building only leases.
2. Leases – Contracts that detail the terms and other important considerations between the tenant (lessee) and the property owner (lessor).
3. Offers, listings, opinions, and other market transactions can be a valid source of market information, if care is exercised in their use.
4. Market participants:
 - a. Commercial property managers
 - b. Tenants of comparable properties
 - c. Real estate brokers and salespeople
 - d. Sellers and purchasers of comparable properties
 - e. Income and expense surveys
 - f. Appraisers
 - g. Banks/commercial lenders
 - h. Appeal process

B. Established data banks

1. Subscriber services – Local and national organizations may sell membership or access to rent, income and/or expense information that is available through online databases. These groups include local real estate boards/Multiple Listing Services, Costar, LoopNet, Smith Travel Research (for hotels), REIS, Reonomy, Real Capital Analytics, or a group of local real estate professionals that pool information into a private database.

2. In-house databases

- a. Information obtained through various sources is stored in a private and secure database.
- b. Data can be organized by use group, location, and investment class to be applied in the income analysis.
- c. Database fields may include:
 - i. Rent information
 - ii. Vacancy and collection loss rates
 - iii. Expense data
 - iv. Sales price
 - v. Capitalization rates

C. Published sources of data

1. BOMA Experience Exchange Report: An Income/Expense Analysis for Office Buildings published by Building Owners and Managers Association.
2. Income/Expense Analysis, Apartments, Condominiums and Cooperatives published by the Institute of Real Estate Management.
3. Trends in the Hotel Industry published by CBRE.
4. Smith Travel Research provides operating reports on hotel performance by market area, market segment, or custom reports on a select group of hotels.

D. Internet/Commercial listing websites – A search with the property address and "for rent" or "for sale" may result in offering information for the property. Free commercial listing websites include:

1. Craigslist (www.craigslist.org)
2. CREXi (www.crex.com)
3. LoopNet (www.loopnet.com; see sample listing on the following pages)
4. CIMLS (www.cimls.com)
5. Commercial Brokers Association (www.commercialmls.com)
6. Brevitas (www.brevitas.com)



FOR
LEASE
AND SALE

OFFICE

10755 BURT STREET

OMAHA, NEBRASKA 68114

**OFFICE SPACE
FOR LEASE AND SALE**

SPACE AVAILABLE:
10,000 SF - 31,264 SF

BUILDING SIZE:
31,264 SF

LEASE RATE:
\$7.00 PSF, NNN

SALE PRICE:
\$2,400,000

LOCATION MAP





FEATURES:

- Nice building with abundant windows and natural light
- Very well built office building close to hotels, restaurants and other area amenities
- Centrally located at I-680 and West Dodge Road
- Underground parking available :: 32 stalls
- Built in 1991
- Building signage available
- Backup generator in place
- Additional building available next door

**FOR MORE
INFORMATION
PLEASE
CONTACT**

Seth V. Campbell
Senior Associate
402.557.6003
seth.campbell@cbre.com

Dale Scott, SIOR
Vice President, Office Brokerage
402.697.5846
dale.scott@cbre.com



Part of the CB Richard Ellis
affiliate network

AVAILABLE



OFFICE
10755 BURT STREET
OMAHA, NEBRASKA 68114

PHOTOS



SITE PLAN



**FOR MORE
INFORMATION
PLEASE
CONTACT**

Seth V. Campbell

Senior Associate

402.557.6003

seth.campbell@cbre.com

Dale Scott, SIOR

Vice President, Office Brokerage

402.697.5846

dale.scott@cbre.com



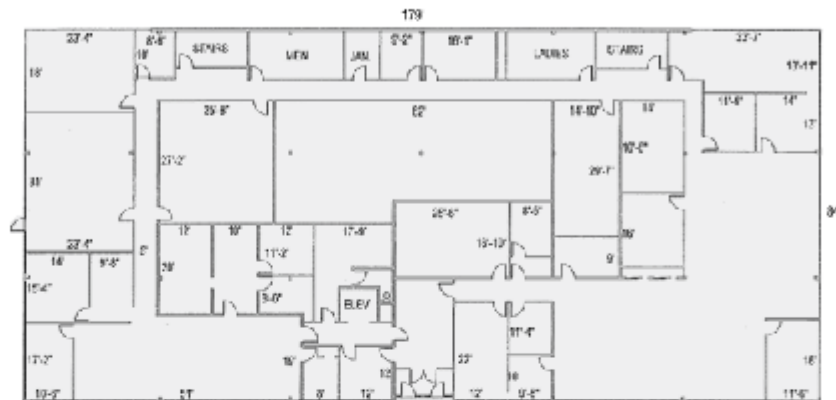
Part of the CB Richard Ellis
affiliate network

AVAILABLE

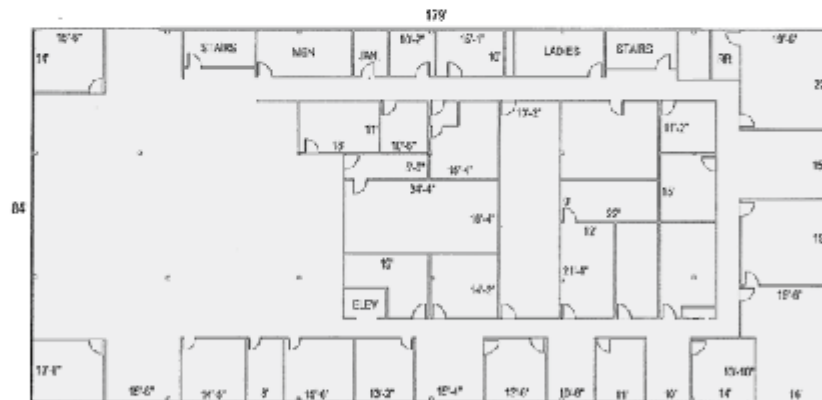


OFFICE
10755 BURT STREET
 OMAHA, NEBRASKA 68114

FIRST FLOOR



SECOND FLOOR



**FOR MORE
 INFORMATION
 PLEASE
 CONTACT**

Seth V. Campbell

Senior Associate
 402.557.6003
seth.campbell@cbre.com

Dale Scott, SIOR

Vice President, Office Brokerage
 402.697.5846
dale.scott@cbre.com

CB Richard Ellis | MEGA | 11213 Davenport Street | Suite 300 | Omaha, NE 68154

032411m ©2011 CB Richard Ellis, Inc. This information has been obtained from sources believed reliable. We have not verified it and make no guarantee, warranty or representation about it. Any projections, opinions, assumptions or estimates used are for example only and do not represent the current or future performance of the property. You and your advisors should conduct a careful, independent investigation of the property to determine to your satisfaction the suitability of the property for your needs.



Part of the CB Richard Ellis
 affiliate network

II. Basis of Income

A. Rental information

1. Rental information may be recorded on **leases** that are a matter of public record. A real estate lease is defined as a written document in which the rights to use and occupy land or structures are transferred by the owner to another for a specified period in return for a specified rent.
2. When analyzing a lease, it is important to look at the **quantity** (the amount of rent), the **quality** (the income stream), and the **durability** (the length of the lease).
3. A lease is considered both a **contract** and a **conveyance**. It has contractual obligations on the tenant to pay rent to the landlord and it may contain other promises and agreements between the landlord and the tenant. It is considered a conveyance because the landlord gives the tenant the right to occupy the property for the term specified in the lease.
4. Leases ordinarily detail the terms and other important considerations between the **tenant** and the **property owner**. Fee ownership remains with the landlord, who is said to have a leased fee interest. The tenant is said to have a leasehold interest, which allows the rights of use and occupancy under conditions specified in the lease.
 - a. In a lease fee situation, ownership remains with the landlord, who is said to have a leased fee interest. The leased fee interest reflects the value of the leases to the owner. This value is based on the cash flow to the owner from the lease in place at the time of the appraisal. Leased fee value can be higher, lower, or the same as the fee simple value depending upon the relationship between the contract rent and market rent as of the effective date of value for the property use. The relationship between fee simple and leased fee values is also affected by the capitalization rate, which can differ between the prevailing market norm and what is appropriate for credit (low risk) tenant(s) and/or long-term leases. *

***SEE ADDENDUM FOR COMPLETE AND ACCURATE DESCRIPTION OF FEE SIMPLE**

- c. The tenant is said to have a leasehold interest, which allows the rights of use and occupancy under conditions specified in the lease. If the rent amount falls below the market level, then a positive leasehold value is created, which is a benefit to the tenant. If the market rent drops below the rent amount in the lease, there is a negative leasehold value, which is a benefit to the landlord.

B. Types of rent

1. **Market rent** (also known as economic rent) is the amount of rent that a property should command in the open market. Market rent should reflect the location of the property, size of the property, supply and demand factors, and terms of lease between knowledgeable and prudent owners and tenants. Market rent is typically what is used to calculate income in the income approach.
2. **Contract rent** is the rent the tenant is paying according to the lease in effect at the time of appraisal or analysis. It may be that at the time the lease was written, the rent amount being paid was considered market rent but over the course of time, the market lease rate for similar properties may have changed and the contract rent is no longer market rent. Some other reasons why contract rent may not reflect market rent include:
 - a. Rents for specialized, built-to-suit improvements are typically based on the cost of the improvements and therefore may not be reflective of market rent.
 - b. Changes in technology have caused the improvements to become obsolete, which means the property is rented at a rate less than the typical updated property.
 - c. Competition has caused some obsolescence in the improvements, which tends to cause the property to rent for less.
 - d. The land component of cost is unreasonably high due to locating improvements at a specific site for specific business reasons.

3. **Excess rent** is the amount derived when contract rent exceeds market rent. This could be caused by a locational advantage, unusual management, uninformed tenants, or other economic factors.
4. **Deficit rent** (also known as leasehold rent or leasehold income) occurs when market rent is greater than contract rent. In effect, the tenant is receiving the amount of the difference between contract rent and market rent as income.
5. **Effective rent** is used as a common denominator to compare leases with different provisions. Effective rent = lease base rent - rent concessions. Rent concessions take the form of either free rent or extra tenant improvement allowances. The free rent is usually given for a period at the beginning of a lease. Concessions are offered when the supply or rental property exceeds demand. The purpose of analyzing effective rent is to develop market rent. Example: The annual rent spelled out in the lease is \$12,000 or \$1,000 per month plus one-month free rent. In this scenario, the effective rent would be \$916 per month (\$11,000/12) not \$1,000 per month:

$$\$12,000 - \$1,000 = \$11,000$$

$$\$11,000 \div 12 = \$916 \text{ per month (not \$1,000 per month)}$$

C. Lease terms

1. **Month-to-month leases** are short-term leases that may or may not be in written form. This type of lease provides no security for the tenant or the landlord. For the landlord, future occupancy and income are risky. For the tenant, there is no long-term guarantee to right of occupancy.
2. **Short-term leases** are generally written with the terms and provisions of the lease detailed. This type of lease is usually for a period of less than five years. This type of lease may have a fixed rental rate or be graduated which allows for adjustments to the rent during the lifetime of the lease. This type of lease typically has a clause allowing for renewal of the lease or a lease extension. Short-term leases are most commonly used in office, retail, and apartment properties. Apartment leases are typically only for a period of one year, but depending on the market, leases could go from month-to-month to one year.

3. **Long-term leases** should always be written and are very similar to the short-term lease. They typically provide for terms extending more than five years. A long-term lease may be either a fixed-rate lease or a graduate lease. Industrial, big box, and fast-food properties are examples of commercial properties that are typically leased using a long-term lease.
4. **Renewal leases** provide for one or more extensions of the lease term in the original lease document at the option of the tenant. The rent under such renewals may be predetermined or negotiated at the time of renewal.

D. Rent payment structure

1. Leases may be set up on a flat rent or a variable rent over the term of the lease. Other methods of rent payment may include:
 - a. **Graduated leases** (also known as a step-up or step-down lease) provide for changes in the lease agreement at one or more points during the lease term.
 - b. **Percentage leases** (see Demonstration 2-1) address the amount of rent received in accordance with the terms of a percentage clause in a percentage lease contract.
 - i. **Minimum base rent** – The fixed portion of the rent under the terms of a percentage lease.
 - ii. **Overage rent** – The variable portion of the rent under the terms of a percentage lease. This is the rent over and above the minimum base rent. Typically, the overage rent is based on a percentage of sales or revenue from the business conducted on the premises.

Demonstration 2-1: Percentage Lease

The ABC Company recently signed a five-year lease for 3,000 square feet of retail space in a downtown strip center. The lease calls for the ABC Company to pay \$2,700 per month plus 3% of annual gross sales above \$500,000. The previous year, the company's gross sales were \$900,000. What is the annual lease amount per square foot the company will pay based on last year's gross sales?

Minimum Rent	$\$2,700 \times 12$	\$32,400
Overage rent	$\$400,000 \times .03$	\$12,000
Total annual rent		\$44,400

$\$44,400$ (annual rent) / $3,000$ (square footage) = $\$14.80$ (rent per square foot)

E. Expense responsibilities

1. In general terms, a gross lease is typically when the landlord pays the property's operating expenses, and a net lease is where the landlord passes the property's operating expenses on to the tenant. The terms net lease and gross lease may carry different meanings in different market areas. It is more important to confirm what expenses the landlord and tenant specifically pay rather than whether the rent is called a gross or a net lease. The following types of leases address the expense responsibility of the tenant and the landlord:
 - a. **Full-service lease** – A real estate lease used predominantly for commercial office space for lease in “multi-tenant” office buildings. These types of leases almost always include the base rental rate and all the expenses including the real estate taxes. Most prevalent in mid-to-high value office buildings.
 - b. **Gross lease** – Typically, the landlord is required to pay all operating expenses associated with the real estate. However, some gross leases may call for the tenant to be responsible for utilities expenses, janitorial expenses (for office buildings), and/or include an expense stop on certain expense categories. Gross leases are common for office buildings, mixed-use commercial buildings, and/or older multi-tenant properties.

- c. **Gross industrial lease** – The landlord is required to pay insurance, real estate taxes, and structural maintenance. The tenant is responsible for utilities and property maintenance. Typically used on industrial properties, such as office/warehouse or multi-tenant flex buildings.
- d. **Modified gross lease** – The tenant and the landlord share expense. The specific expenses paid by the landlord will vary from lease to lease. An example would be where the landlord pays for utilities, real estate taxes, and structural maintenance, and the tenant pays for insurance and maintenance.
- e. **Net lease** – The tenant is required to pay all or part of the operating expenses associated with the real estate. In a single net, double net, or triple net lease the tenant pays some or all the property expenses.
 - i. Single net leases and double net leases are degrees between the gross lease and the triple net lease and may require specific expenses to be paid by either tenant or landlord. For example, a single net lease might mean that the tenant pays their own utilities while the landlord pays the remaining expenses.
 - ii. A double net lease (NN) might require that the tenant pay all the utilities, janitorial costs, and common area maintenance costs while the landlord is responsible for the remaining expenses.
 - iii. In a triple net lease (NNN), the tenant reimburses the landlord for insurance, real estate taxes, and common area maintenance (CAM) separate from the base rent, and the tenant is responsible for its utility expenses. Triple net means the rent paid is net of insurance, net of real estate taxes, and net of common area maintenance (CAM).
 - iv. In an absolute net lease, the tenant pays all expenses associated with the operation of the property and the landlord absolutely does not pay any property expenses.

III. Considerations When Comparing Rental Property Leases

A. In comparing one rental property with another, the following factors should be considered:

1. Effective date of lease
2. Location of property
3. Physical characteristics
4. Terms of the lease

B. Lease Data Characteristics

1. By analyzing the factors thoroughly, the assessor/appraiser can determine the comparability of the rental properties. Because the income approach depends upon an accurate estimate of market rent, the leases covering the subject property as well as those covering comparables in the area must be carefully analyzed.
 - a. **Lease Date**– For purposes of determining market rent, it is important when comparing rental properties, that the date of the lease agreement be recent and representative of current market conditions.
 - b. **Property Location** – It is important that properties used as a subject's rent comparables are in similar locations or exposed to the same economic influences.
 - c. **Physical characteristics of property** – Properties used as a subject's rent comparables should be similar in age, size, condition, quality, desirability, and other amenities, etc.
 - d. **Terms of lease** – It is important that properties used as a subject's rent comparables are influenced by the same or similar terms, such as who pays maintenance expenses, taxes, insurance, or other special agreements. The following considerations provide helpful guidelines for comparing leased properties:
 - i. **Date of lease** – The beginning date of the lease term is the key date for comparison and analysis purposes in the income approach. Leases negotiated several years earlier may not accurately reflect current prevailing conditions and the rents paid may not be indicative of current market rent.

- ii. **Name of the owner (landlord or lessor)** – The owner (lessor) is the party who receives the rent.
- iii. **Name of the tenant (lessee)** – The tenant (lessee) is the party who occupies the premises and pays the rent.
- iv. **Reference** – The reference indicates where the lease is recorded.
- v. **Legal description** – The legal description or other identification may include a parcel identifier and property address (situs).
- vi. **Lease term** – The lease term should state the beginning and ending dates for the tenant's occupancy. The number of months and years remaining and covered by the lease can thereby be calculated.
- vii. **Amount of rent** – The lease should contain a clause stating the amount of rent, how and when the rent is to be paid, and whether any percentages, graduations or overages are applicable.
- viii. **Owner's responsibilities** – Statements in the lease that specify the owner's responsibility for maintenance, insurance, and other expenses during the terms of the lease. Assessors need to determine whether there is any expense cap or expense stop language in the lease. With an expense cap, operating expenses are borne by the tenant to a specified level above which the landlord picks up the additional expenses. With an expense stop (see Demonstration 2-2), operating expenses are borne by the landlord to a specified level above which the tenant picks up any additional expenses. If the tenant's annual expenses exceed that limit, the tenant must reimburse the owner for the difference based on a formula specified in the lease. For example, if the lease states that the landlord will pay all expenses up to \$5.00 per square foot, but the actual expenses turn out to be \$6.50 per square foot, then the base lease rate would be adjusted upwards by \$1.50 per square foot. An expense stop or expense cap is usually expressed as a dollar amount per square foot of leased space.

Demonstration 2-2: Expense Stop

You have a tenant who has signed a lease for a new 12,000 square foot office space in a 100,000 square foot office building. The lease term is for five years and has a base rental rate of \$20.00 per square foot. The owner of the office building set the expense stop in the lease at \$7.50 per square foot. This was based on projected annual operating expenses for the building of \$750,000, including property taxes. If operating expenses were \$975,000, what would the total rent per square foot become to cover the extra expenses?

Actual expenses		\$975,000
Projected expenses		\$750,000
Overage	$\$975,000 - \$750,000$	\$225,000
Square footage		100,000
Overage per square foot	$\$225,000 / 100,000$	\$2.25
Base rent		\$20.00
Total rent per square foot	$\$2.25 + \20.00	\$22.25

(Terms of Lease, continued)

- ix. **Tenant's responsibilities** – Statements in the lease that relate to the tenant's responsibilities for insurance, utilities, and other expenses. Lease provisions may call for an escalation of rent payments due to an increase in the cost of these items.
- x. **Taxes** – The tenant may pay all taxes, a portion of the taxes (based on lease terms), or no taxes.
- xi. **Right to sublease or assign** – There should be a clause stating whether the tenant may assign or sublease the premises and under what conditions. This statement should also indicate whether an assignment or a sublease would relieve the initial tenant from future responsibility under the original lease.
- xii. **Option to renew** – This clause should give dates, terms, rent payments, and other provisions specified as the tenant's options.
- xiii. **Tenant improvements (tenant finish)** – This section of a lease will describe the tenant finish work that is planned for the leased space, or whether the space is being leased "as is." If any work is being done, the work may be defined and/or illustrated in the form of architectural drawings in the addendum of the lease. Each tenant improvement arrangement is different.
 - (i) Some leases are negotiated for the tenant to pay all the tenant finish upfront. Some leases call for the cost of the tenant finish to be split between the landlord and tenant and some leases have the landlord pay for all the tenant finish work. If there is a tenant improvement allowance, the landlord will pay for a portion of the total tenant improvement cost and the remaining balance of the cost is absorbed by the tenant. The landlord typically sets a dollar limit on the allowance that is defined in the lease.

- (ii) A tenant may negotiate for the landlord to pay for a portion or all the tenant finish work and pay back the landlord over the lease term. For example, if the landlord agrees to pay \$10,000 in tenant improvements and the lease is for five years, the annual amount that will be recovered through the rent would be \$2,000. If the building contained 4,000 square feet, then the amount added to the base rent for the tenant improvements would be \$0.50 per square foot. In some cases, the landlord may charge an interest rate in the repayment of the tenant improvements and amortize it over the lease term.
- (iii) The appraiser should consider whether tenant improvements and corresponding rent include any personal property that is removable as well as ownership of the tenant improvements at the end of the lease. The lease may also address who is responsible for any increases in taxes caused by the tenant improvements.
- (iv) Tenant improvement (see Demonstration 2-3) expenses vary based on the extent of work being done, use (retail versus office), and local market norms. Market conditions will influence how much tenant allowance is made and the allowance may change over time. For example, if vacancy rates are high and property owners are competing for tenants, landlords may have to offer a greater tenant allowance to entice a renter to sign a lease. Conversely, a market where vacancy rates are low and tenant demand is high puts the advantage in the hands of landlords, making it less likely for landlords to pay for the tenant finish. The cost of tenant improvements may range from \$5 per square foot for basic floor and paint finish to \$200 per square foot to build out an unfinished (shell) building. Given these variances, it is difficult to determine a standard tenant improvement expense in mass appraisal. However, stratifying data by investment class and use groups can assist in this process. Examples of tenant finish for various commercial uses include:

Office buildings – Typically, the landlord will set an allowance for these types (repainting, new carpet, cleaning, etc.) of tenant improvements. If the tenant wants more, it is usually the responsibility of the tenant to pay the additional cost. If the landlord pays for the tenant improvements, it is customary for those costs to be recovered through lease payments. In this case, the landlord will simply divide the total cost of the tenant improvements by the number of payments to find the total amount per pay period. This amount is added to each payment (whether monthly or annual). Next, calculate the amount per square foot by figuring the annualized payment amount (if not done in the previous step) and divide that amount by the square footage leased. This calculation is helpful to use when developing the income approach for commercial buildings because the adjustment is typically made on an annual, per square foot basis.

Retail buildings – It is common in retail leases to require the tenant to pay for any tenant improvements. If market conditions and/or lease arrangement allow for the landlord to be reimbursed for the tenant improvements, the rent adjustment is calculated in a similar fashion to the method used for office buildings.

Industrial buildings – Industrial buildings are almost always leased “as is.” The tenant typically makes tenant improvements at their own cost upon landlord approval.

Demonstration 2-3: Tenant Improvements

The subject property is a leased space that received tenant improvements in the amount of \$25,000. The lease is a five-year lease with payments made monthly. The net leasable area is 2,000 square feet. The annual lease rate before tenant improvements was \$9.50 per square foot. What is the annualized amount of tenant improvements per square foot over the term of the lease? What is the total rent per square foot, including the reimbursement of tenant improvements to the landlord?

Tenant improvements (TI)		\$25,000
Lease term		5 years
Annualized TI	$\$25,000 / 5 \text{ years}$	\$5,000 per year
Square feet (SF) leased		2,000 SF
Annualized TI per SF	$\$5,000 \text{ per year} / 2,000 \text{ SF}$	\$2.50 per SF
Base rent		\$9.50 per SF
Total rent per SF	$\$2.50 + \9.50	\$12.00 per SF

(Terms of Lease, continued)

- xiv. **Security** – This provision should state the amount of any damage deposit required in addition to advance rent. It should also include a provision for damage claims resulting from tenant abuse of the property.
- xv. **Termination** – This clause should indicate what conditions allow an owner or tenant to terminate the lease.
- xvi. **Special provisions** – This clause should include any other conditions or considerations that may affect the tenant or the owner during the term of lease.

IV. Investment Class

- A. **Commercial property** can be stratified into homogenous groups, or investment classes, that allow the appraiser to organize income and expense data for analysis. No formal international standard exists for classifying a building, but investment classes consider physical features (effective age, condition, quality, amenities), location (access, visibility), and economic characteristics (rent levels, occupancy, management).
- B. One of the most important things to consider regarding investment classifications is that property should be viewed in context and relative to other properties of similar use within the same submarket. Criteria for investment class may be different for an apartment building than a retail center. A Class A building in one submarket may not be a Class A building in another submarket. General characteristics of the different building classes are:
 - 1. Investment Class A - These buildings typically have an actual or effective age less than 10 years, are in good condition for the age (well-maintained/no deferred maintenance), a contemporary design, and/or are constructed with above average-quality construction materials. The site may have site access with two or more vehicular points, corner location, and/or high visibility frontage. The property may be in a corridor of on-going commercial growth, high density of commercial development, and/or high traffic counts for market area. These characteristics allow the property to attract the highest quality (credit) tenants and typically command the highest rents in the submarket.

2. Investment Class B - These buildings typically have an actual or effective age between 10 and 30 years, are in average condition, and/or constructed with average-quality materials. Location, access, and visibility features may be inferior to a Class A property. The building may need replacement of short-lived items and may be targeted by value-added investors who perform renovations that put the property back to a Class A status. Class B buildings are still functional and have been maintained.
3. Investment Class C - These buildings typically have an effective age over 30 years, fair or poor condition for the age (effective age is at or older than actual age), may have deferred maintenance, and/or functional obsolescence may be present. A 10 to 30-year-old building located in a declining submarket area or on a site that is far from major access points and complementary land uses may be considered an Investment Class C property. Class C buildings tend to have the lowest rental rates, take the longest to lease, and are often targeted as re-development opportunities.

Demonstration 2-4: Effective Rent Analysis

The lease on an office building is for five years with a base rate of \$20.00 per square foot. The landlord is offering a rent concession of three months free rent. What would be the effective rent amount per square foot?

Rent concession term		3 months
Lease term		5 years (60 months)
Rent concession percentage	3 months / 60 months	0.05
Base rate		\$20.00 / square foot (SF)
Rent concession per SF	$\$20.00 \times 0.05$	\$1.00 per SF
Effective rent	$\$20.00 \text{ SF} - \1.00 SF	\$19.00 SF

Demonstration 2-5: Lease and Rent Analysis

A review of the subject property's lease shows the rent stated in the lease is \$20.00 per square foot per year, with the landlord required to pay all expenses except for utilities, maintenance, and taxes. Comparable properties in the market are currently being rented at \$23.00 per square foot per year. Based on this analysis of the lease and comparable rentable properties, what type of lease and rent would this be:

- Lease: Net, net lease
- Rent: Deficit rent

Exercise 2-1: Lease and Rent Analysis

Bob recently leased a 1,500 square foot space at a local shopping center. He signed a five-year lease which requires him to pay \$50,000 in rent over the life of the lease. Bob's lease is typical for new rental space in the area. Jeff, an associate of Bob, leased retail space at the same shopping center two years ago. He has three years left on his current lease agreement. Jeff has 2,000 square feet of leased area which he rented for \$4.00 a square foot two years ago. His lease calls for annual rent adjustment of 2.5% per year. He also pays the landlord an additional three percent of all his gross sales above \$500,000 per year. Last year, Jeff had \$800,000 in gross sales.

1. What is the annual rent per square foot, that Bob pays? What type of rent is it?
2. What type of rent does Jeff pay?
3. How much rent did Jeff have to pay the landlord last year?

PAGE LEFT BLANK INTENTIONALLY

Exercise 2-2: Lease Analysis

You were assigned the responsibility of analyzing several leases in a local mini mall. After reviewing several of the leases, you determined the minimum base rent for the mall is \$5.00 per square foot per year. The tenants also pay an overage rent of 3 percent of all sales over \$100,000.

Your subject property is a 2,500 square foot retail space in a comparable mini mall. Your subject's sales in the past year amounted to \$475,000. Based on the comparable information, what was the annual rent of your subject property and what type of lease is the mall owner utilizing?

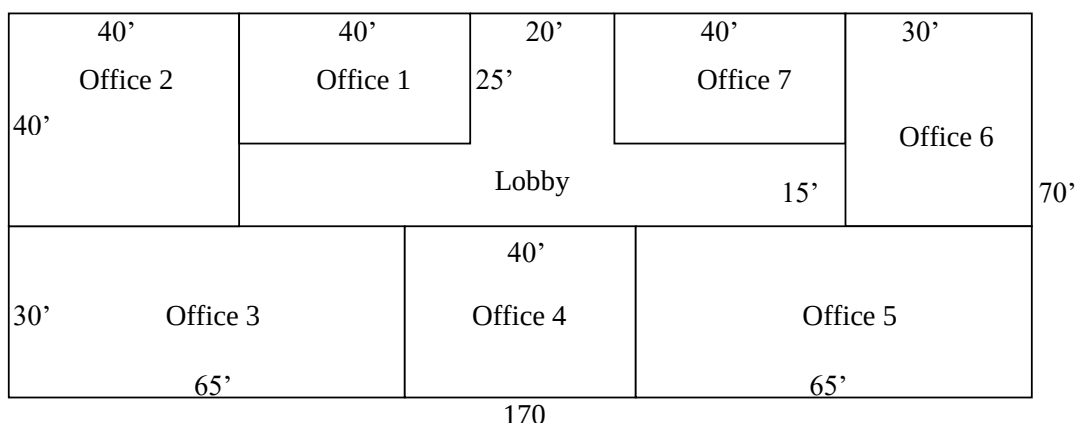
PAGE LEFT BLANK INTENTIONALLY

V. Rental Units of Comparison

- A. **Rent per unit** – Apartment buildings, self-storage facilities, and condos.
- B. **Rent per room** – Apartment buildings, motels, and hotels.
- C. **Rent per space** – Mobile home parks, travel trailer parks, and parking garages.
- D. **Rent per square foot** – Shopping centers, retail stores, office buildings, and warehouses are typically analyzed on an annual basis (rent per year per square foot). Apartment units are typically analyzed on a monthly rent per square foot basis.
 - 1. **Gross leasable area (GLA)** – Includes the entire area of the building.
 - 2. **Net leasable area (NLA)** – Includes only the floor area occupied by the tenant.
 - 3. The difference between GLA and NLA is the common space. When leases are compared, the assessor/appraiser must know whether rent is based on GLA or NLA, and what method is used to determine the NLA.

Demonstration 2-6: Gross and Net Leasable Area

When using the square foot unit of comparison on office buildings and shopping centers, care must be exercised in the comparison process. Some leases refer to GLA, but other leases are negotiated based on NLA. The GLA includes all common areas, such as halls, restrooms, and vestibules. The NLA includes only the floor area occupied by the tenant. The below floor plan of a one-story office building will be used to demonstrate the proper procedures in developing GLA and NLA units of comparison. Determine GLA and net NLA.



Determine gross leasable area (GLA) and net leasable area (NLA).

GLA: Calculate the total area of the building: $70' \times 170' = 11,900$ SF

NLA: Calculate the building area only using floor space occupied by the tenant

Office 1	40 x 25 =	1,000
Office 2	40 x 40 =	1,600
Office 3	30 x 65 =	1,950
Office 4	40 x 30 =	1,200
Office 5	30 x 65 =	1,950
Office 6	30 x 40 =	1,200
Office 7	40 x 25 =	1,000
	NLA =	<u>9,900</u> square feet

GLA:

The building contains 11,900 square feet and is leased on a basis of \$25.00 per square foot of GLA per year. Calculate the annual rental income generated by this property.

Number of square feet (GLA)	11,900
Annual rent per square foot	\$25.00
Annual rental income	\$297,500

NLA:

The offices of the same building are leased to seven separate tenants at a rate of \$30.00 per square foot per year. Calculate the annual rental income generated by this property.

Number of square feet (NLA)	9,900
Annual rent per square foot	\$30.00
Annual rental income	\$297,000

In using the NLA rent per square foot, only the area leased by the tenant can be used to develop the annual rent. If the NLA rent of \$30.00 per square foot is multiplied by the gross area of the building, the indicated potential gross income would be \$357,000 and would result in a substantial error in the value estimate utilized in the capitalization process. Only one type of rent may be used when developing rent per square foot as a unit of comparison. It is necessary that all rents be expressed in terms of either gross leasable area or net leasable area.

- E. Selecting Units of Comparison - Some uses will have several units of comparison to consider in the analysis. For example, apartments can be analyzed on the following: **rent per unit**, **rent per room**, and **rent per square foot**. The appropriate unit of comparison will have the lowest percentage spread of the data set analyzed.

Demonstration 2-7: Selecting Units of Comparison

The subject of the appraisal is an apartment building with a mixture of two-bedroom and three-bedroom units. Based on the following market data, what is the most appropriate unit of comparison - rent per unit or rent per square foot?

Step 1: Calculate the unit of comparison for each comparable.

Comparable	2-Bedroom Units		Calculation	Rent/SF	3-Bedroom Units		Calculation	Rent/SF
	Unit size	Monthly Rent			Unit size	Monthly Rent		
Rental 1	600	\$800	$\$800 \div 600$	\$1.33	1,200	\$1,500	$\$1,500 \div 1,200$	\$1.25
Rental 2	575	\$750	$\$750 \div 575$	\$1.30	1,175	\$1,400	$\$1,400 \div 1,175$	\$1.19
Rental 3	625	\$825	$\$825 \div 625$	\$1.32	1,260	\$1,525	$\$1,525 \div 1,260$	\$1.21
Rental 4	615	\$850	$\$850 \div 615$	\$1.38	1,250	\$1,575	$\$1,575 \div 1,250$	\$1.26

Step 2: Calculate the spread between the high and low indications for each unit of comparison and obtain the percentage spread by dividing the spread by the lowest figure in the range.

Comparable	2-Bedroom Units			3-Bedroom Units		
	Unit size	Rent	Rent/SF	Unit size	Rent	Rent/SF
Rental 1	600	\$800	\$1.33	1,200	\$1,500	\$1.25
Rental 2	575	\$750	\$1.30	1,175	\$1,400	\$1.19
Rental 3	625	\$825	\$1.32	1,260	\$1,525	\$1.21
Rental 4	615	\$850	\$1.38	1,250	\$1,575	\$1.26
	High	\$850	\$1.38	High	\$1,575	\$1.26
	-Low	<u>-\$750</u>	<u>-\$1.30</u>	-Low	<u>-\$1,400</u>	<u>-\$1.19</u>
	Spread	\$100	\$0.08	Spread	\$175	\$0.07
	$\div$ Low	<u>$\div \\$750$</u>	<u>$\div \\$1.30$</u>		<u>$\div \\$1,400$</u>	<u>$\div \\$1.19$</u>
Percentage Spread		13%	6%		13%	6%

Demonstration 2-7: Selecting Units of Comparison, continued

Step 3: Select the lowest percentage spread

The amount of spread between the rent per unit is a little more than twice the amount of the spread for rent per square foot. Thus, based on these data, a rent per square foot would be the best choice for the analysis.

Exercise 2-3: Units of Comparison – Rent per Square Foot

You are developing a rent per square foot as a unit of comparison for small office buildings from an analysis of the market. You have obtained the following information:

A comparable sale has been identified that is similar to the subject in terms of age, size, construction, condition, and location. It is a two-story office building and has a gross leasable area (GLA) of 16,000 square feet. The common area contains 3,000 square feet and includes all hallways, elevators, stairs, and includes the reception area on the ground floor. The building rents for \$380,000 per year.

Analyze the rent schedule and develop rent per square foot as a unit of comparison, on both a gross square footage and net square footage basis.

PAGE LEFT BLANK INTENTIONALLY

Exercise 2-4: Units of Comparison – Apartment Unit vs. SQFT

Five rental properties have been located in the neighborhood in which you are appraising apartment complexes. All five properties are comparable in age, condition, and location.

Analyze the rental data to determine the range of values on a per unit and per square foot basis for each type of unit. Discuss the merits of both approaches. Which method is the better choice for determining the monthly potential gross income of comparable buildings in this neighborhood?

	One Bedroom		Two Bedroom	
	Unit size	Monthly Rent	Unit size	Monthly Rent
Rental 1	575	\$820	810	\$890
Rental 2	550	\$740	840	\$940
Rental 3	580	\$840	850	\$1,000
Rental 4	535	\$750	775	\$870
Rental 5	525	\$720	780	\$860

PAGE LEFT BLANK INTENTIONALLY

Exercise 2-5: Development of Market Rental Estimate

You have been asked to appraise The Villas at Camelback Crossing. This 500-unit property is 505,000 total square feet, five years old and has two swimming pools and two clubhouses. Covered parking is provided for every unit. The breakdown of the complex is listed in the table below. Both the aerial photograph and front photo of the property have been provided to assist you in valuing the subject.

You have researched the area and found five comparable apartment complexes similar to the subject. Analyze the rental information on the following page to determine the appropriate unit of comparison. Pick which of the comparable rentals is the best indication of market rent for the subject. Apply the market rent to the subject's unit mix to arrive at a total annual rent estimate for the property.



Villas at Camelback Crossing					
Floor Plan	Bed	Bath	SQFT Per Unit	Number of Units	Total SQFT
1 BR	1	1	850	300	255,000
2 BR	2	1.5	1,250	200	250,000
Total				500	505,000

Rental Information on next 2 pages.

Alexan Palm Valley							
Floor Plan	Bed	Bath	SQFT	Rent	Parking	Unit Count	Rent per SQFT
1 BR	1	1	760	\$840	Uncovered	150	1.11
2 BR	2	2	1,125	\$1,080	Uncovered	150	0.96
Total	282,750			300			

Amenities: One year old and has one pool and one clubhouse

Alexan Paradise Lane							
Floor Plan	Bed	Bath	SQFT	Rent	Parking	Unit Count	Rent per SQFT
1 BR	1	1	760	\$850	Covered	150	1.12
2 BR	2	2	1,250	\$1,320	Covered	90	1.06
Total	226,500			240			

Amenities: Brand new and has one pool and one clubhouse, covered parking is included

Aventura Apartments							
Floor Plan	Bed	Bath	SQFT	Rent	Parking	Unit Count	Rent per SQFT
1 BR	1	1	850	\$925	Covered	300	1.09
2 BR	2	1.5	1,250	\$1,200	Covered	200	0.96
Total	505,000			500			

Amenities: Five years old and has two pools and two clubhouses, covered parking is included

Rock Canyon							
Floor Plan	Bed	Bath	SQFT	Rent	Parking	Unit Count	Rent per SQFT
1 BR	1	1	850	\$895	Uncovered	200	1.05
2 BR	2	2	1,025	\$985	Uncovered	100	0.96
Total	277,500			300			

Amenities: Five years old and has two pools and one clubhouse

Sage Stone at Arrowhead							
Floor Plan	Bed	Bath	SQFT	Rent	Parking	Unit Count	Rent per SQFT
1 BR	1	1	700	\$770	Covered	210	1.10
2 BR	2	1.5	1,050	\$990	Covered	100	0.94
Total	256,000			310			

Amenities: Five years old and has two pools and one clubhouse, covered parking is included

PAGE LEFT BLANK INTENTIONALLY

F. **Gross Income Multiplier (GIM)** defined - the relationship between gross income and the value of the property, which is generally represented by its sale price. The income is the annual gross income. The GIM can be developed either on a potential gross income basis or an effective gross income basis. If developed from effective gross income, it is called the effective gross income multiplier (EGIM). Either way is acceptable if it is used in the same way it was developed.

1. A GIM is generally used with commercial and apartment properties.
2. The GIM is calculated by dividing the sale price (value) by the annual gross income. For example, assume a sale price of \$2,400,000 for an apartment that has an annual gross income of \$279,000.

Sale Price	\$2,400,000
Annual Gross Income	÷\$279,000
GIM	8.60

3. The GIM can also be used to develop a value opinion for the subject property simply by multiplying the subject's gross income by the GIM.

VI. Income and Expense Statement

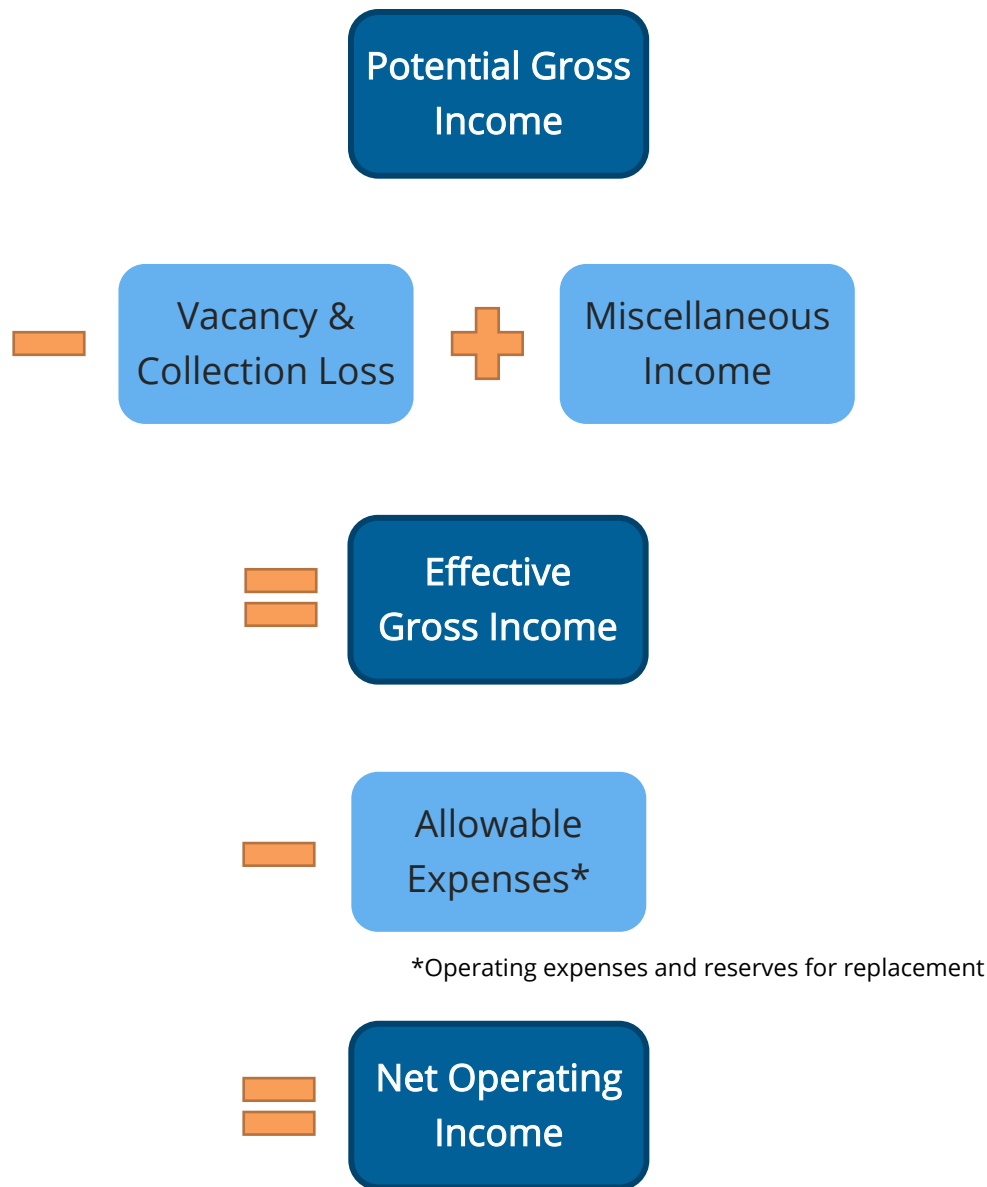
- A. **Reconstructed from owner's statement** - This process expresses the earning capacity of a property. Often the appraiser will use the information from the owner's operating statement and reconstruct an income statement for the property, based on the current market rents and typical proper expenses for one year. In this process the assessor/appraiser should analyze at least two- or three-years' worth of income and expense data for the subject property.
- B. **Developed from rent comparables in the area** - The rent that a property commands in the market depends on many different factors. The assessor/appraiser must examine all the terms and conditions surrounding the sale of a property when the sales comparison approach to value is used. All the circumstances surrounding the comparable rentals used as a basis for reestablishing market rent for the subject property must be examined in the income approach to value.

C. Format of income and expense statement:

1. Potential gross income estimate (PGI) [also referred to as gross income]
 - a. Potential gross income is the annual economic rent for the property at 100% occupancy.
 - b. It includes the market rent from all scheduled sources.
 - c. When analyzing the market some factors to keep in mind when searching for comparables are rent concessions and expense reimbursements.
 - i. **Rent concessions** – Reductions in the rent as an enticement to concessions on the potential gross are a flat percentage of the potential gross income and a reduction in market rent rate to an effective rent rate.

(section continues later in this chapter)

Illustration 2-1: Income and Expense Statement



Demonstration 2-8: Estimating PGI

A retail store that has 3,000 square feet rents for \$6.25 per square foot per month. The potential gross income can be computed as follows:

Square Feet	3,000
Rent per SQFT	× \$6.25
Time frame (months)	× 12
PGI	\$225,000

Exercise 2-6: Developing a Potential Gross Income Statement

You are appraising an apartment building with three levels. The first level contains garden level apartments with 4 one-bedroom units, 2 two-bedroom units and 2 three-bedroom units. The second level contains 6 one-bedroom units and 4 two-bedroom units. The third level contains 4 two-bedroom units and 4 three-bedroom units.

Market analysis provides the following:

Level	Beds	Market Rent
First	1	\$700/mo.
First	2	\$800/mo.
First	3	\$900/mo.

Units located on the second and third levels rent for 25% more than those on the first level. The janitor lives in a one-bedroom unit on the first level, which is rent free and a part of the salary. The owners of the building occupy a three-bedroom unit on the third level and pay no rent.

Develop a Potential Gross Income (PGI) estimate for this apartment building.

PAGE LEFT BLANK INTENTIONALLY

Demonstration 2-9: Rent Concessions

The tenant was given three months free rent on a five-year lease. The full-service rent in the lease is \$20.00 per square foot. The net leasable area is 5,000 square feet. The rent concession can be calculated using two methods:

Method #1		
Free Rent	3 months / 60 months	0.05 or 5%
PGI	5,000 SQFT x \$20.00	\$100,000
Rent Concession	\$100,000 x 0.05	\$5,000
Adjusted PGI	\$100,000 - \$5,000	\$95,000

Method #2		
Free Rent	3 months / 60 months	0.05 or 5%
Rent Concession	0.05 x \$20.00	\$1.00
Effective Rent Rate	\$20.00 - \$1.00	\$19.00
Adjusted PGI	5,000 SQFT x \$19.00	\$95,000

(Income and Expense Statements, continued)

- i. **Expense reimbursements** – The amount that the tenant is required to pay the landlord for additional expenses. These reimbursements tend to show up as income in the operating statement. The expense portion of the operating statement will probably reflect all expenses paid by the landlord, including that amount reimbursed. The assessor/appraiser needs to use care when using this information when analyzing a specific property's income and expense data. Two examples of typical expense reimbursements are:
 - (i) **Common area maintenance (CAM)** – An additional annual charge often assessed to tenants for maintenance of the property's common area such as entryways, hallways, or bathrooms. However, CAM charges can also cover other expenses such as insurance, utilities, etc. Typically, the more square feet a tenant leases, the greater the percentage the tenant pays. CAM fees can either be a variable fee or a flat fixed amount.
 - (ii) **Tenant Improvements (TI)** – These are capital expenditures to accommodate specific needs of a tenant that can be annualized and recovered by the landlord through the rent payment. Therefore, it can be considered a proper expense. For *first generation properties* (brand new space) the TI allowance is often insufficient to complete construction even with the most inexpensive quality of components. Construction expenses in excess of the TI allowance must be paid by the tenant. For *second-generation properties*, the typical TI may just cover the costs to replace worn carpets and wallpaper, paint walls, and move a minor number of walls and doors.

Demonstration 2-10: Determining Potential Gross Income (PGI)

The subject property contains 5,000 square feet and is leased on a triple net lease basis at \$20 per square foot per year. Reimbursement for taxes, insurance, and common area maintenance (CAM) expenses are reimbursed at \$7 per square foot annually. What is the Potential Gross Income (PGI) for the subject property?

	SQFT	x	\$/SQFT	=	Annual
Rent Income	5,000	x	\$20	=	\$100,000
Reimbursement Income	5,000	x	\$7	=	\$35,000
Total Potential Gross Income					\$135,000

Demonstration 2-11: Tenant Improvements

The subject property has 12,000 square feet of net leasable area and it is currently vacant. Before the former tenant moved out the property was renting for \$20.00 per square foot. It is anticipated that the tenant improvement costs needed to attract a new tenant for a five-year lease will be \$200,000. What would the rent be after making the tenant improvements, if you intend to recover the capital cost during the five-year lease?

TI per SQFT	$\$200,000/12,000$	\$16.67
TI per SQFT per Year	$\$16.67/5 \text{ years}$	\$3.33
Rent per SQFT for TI Cost Recovery	$\$20.00 + \3.33	\$23.33

(Format of Income and Expense Statement, continued)

2. Vacancy and collection loss (derived from market)
 - a. Typically, all properties will experience some amount of vacancy during their economic life. Vacancy formulas:
 - i. $\text{Vacant units} / \text{Total units} = \text{Vacancy Rate}$
 - ii. $\text{Vacant square footage} / \text{Total square footage} = \text{Vacancy Rate}$
 - b. During the economic life of a property, there will be a loss of rental income because of the failure of tenants to pay the rent. Collection loss formula:
 - i. $\text{Billable rent} - \text{Collected rent} = \text{Uncollected rent}$
 - ii. $\text{Uncollected rent} / \text{Billable rent} = \text{Collection loss rate}$
 - c. The amount that is allowed for vacancy and collection loss must be based on an analysis of properties that are similar to the subject.
 - d. In the reconstruction of the income and expense statement vacancy and collection loss are a percentage of potential gross income. Typically, in this process are added together into one percentage.

Illustration 2-2: Vacancy Loss

The following is an example of developing vacancy rates for apartments.

Complex	Units	Occupied	Vacant	Vacant ÷ Total Units	Vacancy Rate
Rosewood	142	132	10	$10 \div 142$	0.0704
Applewood	144	134	10	$10 \div 144$	0.0694
Tanglewood	128	119	9	$9 \div 128$	0.0703

The indicated vacancy rate is **0.07** or **7%**.

Illustration 2-3: Collection Loss

The following is an example of developing a rate for collection loss.

Complex	Billable	Collected	Collection Loss	Loss ÷ Billable	Rate
Rosewood	\$153,000	\$150,000	\$3,000	$\$3,000 \div \$153,000$	0.0196
Applewood	\$160,000	\$156,800	\$3,200	$\$3,200 \div \$160,00$	0.0200
Tanglewood	\$132,000	\$129,300	\$2,700	$\$2,700 \div \$132,000$	0.0205

The indicated rate for collection loss is **0.0200** or **2%**.

The indicated rate for vacancy and collection loss is **9% (7% + 2%)**.

Exercise 2-7: Income & Expense Statement - Vacancy Rate

You are reconstructing an income and expense statement for an apartment complex containing 72 units. You have found four apartment complexes having similar characteristics and amenities, located in the same neighborhood as that of your subject.

Apartment	Number of Units	Occupied Units
Briargate	80	76
Creekside	60	57
Beachfront	78	74
Lakeside	62	59

What vacancy rate would be appropriate for your reconstructed income and expense statement?

NOTE: remember to round the answer to 2 decimal places

Exercise 2-8: Income & Expense Statement - Collection Loss Rate

You are reconstructing an income and expense statement for an apartment complex containing 120 units and need to determine a rate for collection loss. You have found four apartment complexes with rental information, having similar characteristics and amenities, located in the same neighborhood as your subject.

Apartment	Number of Units	Billable Rent	Rents Collected
Peachtree	108	\$984,900	\$965,100
Applewood	120	\$1,094,400	\$1,072,600
Elderberry	110	\$1,056,000	\$1,035,200
Brentwood	124	\$1,130,800	\$1,107,900

What collection loss rate will you use in your reconstructed income and expense statement?

PAGE LEFT BLANK INTENTIONALLY

Exercise 2-9: Income & Expense Statement - Vacancy & Collection Loss Rate

You are reconstructing an income and expense statement for a 180-unit apartment complex located just south of the central business district. The complex is eight years old and has typical amenities and management.

The following five apartment complexes are similar to the subject in management, age, condition, etc.

Apartment	Number of Units	Occupied Units	Billable Rent	Rent Collected
Complex A	210	199	\$1,925,300	\$1,886,800
Complex B	180	171	\$1,744,200	\$1,709,500
Complex C	196	186	\$1,899,200	\$1,861,200
Complex D	200	190	\$1,938,000	\$1,898,000
Complex E	220	209	\$2,131,800	\$2,089,300

Use the above data to determine what rate you will use for vacancy and collection loss in your reconstructed income and expense statement.

PAGE LEFT BLANK INTENTIONALLY

3. Miscellaneous income

- a. Miscellaneous income may come from several sources other than the scheduled rents.
- b. Miscellaneous income covers all income generated by the operation of the real property that is not derived directly from scheduled rental of space.
- c. Examples of miscellaneous income include parking fees, vending machines, coin operated laundries, application fees, pet rent, late fees, forfeited deposits, and cable/phone/data service plans. This income may or may not be attributable to the real property. The appraiser must consider jurisdiction guidelines and statutes in determining what miscellaneous income can be included in the analysis and attributed to the real property being appraised.
- d. Effective gross income is obtained **after** determining the potential gross income for the property.
- e. EGI is PGI less vacancy and collection loss plus appropriate miscellaneous income.

$$\text{EGI} = \text{PGI} - (\text{Vacancy} + \text{Collection Loss}) + \text{Appropriate Misc. Income}$$

Demonstration 2-12: Calculating EGI

A retail building rents for \$10,400 per month and the vacancy and collection loss is estimated at 8%. The property also generates \$625 per month in miscellaneous income. The effective gross income would be calculated as follows:

Potential gross income	$\$10,400 \times 12$	\$124,800
Less vacancy and collection loss	$0.08 \times \$124,800$	-\$9,984
Add miscellaneous income	$\$625 \times 12$	<u>+\$7,500</u>
Effective gross income		\$122,316

4. Proper expenses

- a. Expenses for real estate, which are considered in the income approach to value, can be classified:
 - i. Fixed expenses – an expense that does not vary by rate of occupancy.
 - ii. Variable expenses – expenses that vary based on the rate of occupancy.
 - iii. Reserves for replacement – annual charges for items that have relatively short lives (short-lived items) and that must be replaced before the end of the lease period or before the improvement reaches the end of its useful life.

The sum of these expenses is referred to as **operating expenses**. Operating expenses are ordinary and typical expenses that are necessary to keep the property functional and rented competitively with other properties in the area. Operating expenses can be expressed as either individual expense amounts or by an operating expense ratio.

The formula to calculate an operating expense ratio (OER) is:

$$\text{Total expenses} \div \text{Effective gross income} = \text{Operating expense ratio}$$

Demonstration 2-13: Calculating an OER

You have reconstructed the income and expenses on the subject property and have determined the following:

	SQFT		\$/SQFT		Annual
Rent Income	20,000	×	\$25.00	=	\$500,000
Reimbursement Income	20,000	×	\$8.00	=	\$160,000
Potential Gross Income					\$660,000
Less: Vacancy and Collection Loss	10%				-\$66,000
Add Miscellaneous Income					<u>+\$15,000</u>
Effective Gross Income					\$609,000
Less: Operating Expenses	20,000	×	\$9.12	=	<u>-\$182,400</u>
Net Operating Income					\$426,600

The market data indicates expenses of \$9.12 per square foot, or \$182,400. Based on this reconstruction the operating expense ratio (OER) would be:

$$\$182,400 \div \$609,000 = .30 \text{ or } 30\%$$

- b. Operating expenses vary from property to property, depending on the type of occupancy, use type, and quality of management.
- c. In analyzing the operating expenses for a property, the operating statements from comparable properties must be reviewed and the following considered:
 - i. Does the expense amount appear to be typical for the property in question, and is the amount substantiated by expense statements of comparable properties?
 - ii. Do the expenses tend to appear infrequently?
 - iii. Do the expenses appear to indicate typical management?
 - iv. Do the expenses indicate a specific weakness of the property in question?
 - v. Are the various reported expenses consistent as they relate to each other (maintenance, age, and reserves for replacement)?
 - vi. Is the ratio of expenses to effective gross income comparable to those for competitive properties?
- d. Examples of **proper** operating expenses are:
 - i. **Management** – A percentage of effective gross income. Included in management fees may be accounting expenses, office expenses, rent collections, legal fees, advertising, lease commissions, etc.
 - ii. **Insurance** – May need to be prorated if the insurance policy is more than one year.
 - iii. **Utilities** – Would include items like gas, water, electricity, sewer charge, telephone, and trash removal.
 - iv. **Maintenance** – These would be normal maintenance expenses generated by the physical use of the property.
 - v. **Employee wages and fringe benefits** –Salaries and fringe benefits for employees necessary to maintain the property and to provide the operational activities required to keep the property rented.
 - vi. **Grounds keeping** (e.g., yard care and snow removal).

- vii. **Miscellaneous expenses** – These items should be small ones that do not justify listing and amount to only a small percentage of the effective gross income.
 - viii. **Reserves for replacement** – Expense items in this category are similar to items that are depreciated in the cost approach as short-lived features.
- e. Reserves for replacement - are considered short-lived items.
- i. Short-lived items are items which are expected to have a remaining useful life less than the remaining useful life of the property
 - (i) Some examples are carpets, drapes, heating, air conditioning, ranges, refrigerators, water heaters, and paving, etc.
 - ii. The amount of expense for short-lived items is computed by the following steps:
 - (i) Estimate the useful life of the item.
 - (ii) Estimate the replacement cost new.
 - (iii) Calculate the percentage of reserve per year by dividing 100 percent by the useful life of the item.
 - (iv) Multiply the replacement cost new by the percentage of reserve per year to arrive at the annual charge.

Demonstration 2-14: Replacement Reserves

A small retail store has a remaining useful life of 40 years. The roof cover has a useful life of 20 years and the cost of the roof cover is \$12,000. The annual percentage can be calculated as follows:

$$100 \text{ percent} \div 20 \text{ years} = 5 \text{ percent per year}$$

The annual percentage is then multiplied by the cost of the roof to determine the reserve for replacement amount.

$$\$12,000 \times .05 = \$600 \text{ (annual reserve for the roof)}$$

i. **Improper Operating Expenses**

(i) **Depreciation**

1. Depreciation (or appreciation) will be considered in the income approach as recapture and handled as part of the capitalization rate.
2. Depreciation found on operating statements for income tax purposes will generally not be the same as the recapture provision in the capitalization rate.

(ii) **Debt service** (principal and interest on mortgage)

1. Most appraisals whether single property or mass appraisal are done under the assumption that the property is free and clear of any debt or liens. The amount or presence of debt will vary from property to property.
2. Principal and interest payments are financing expenses that are part of the cost of capital, if debt is used to acquire the property. Debt is the cost of capital component for the discount rate and reflected in the capitalization rate, not the net operating income.

(iii) **Owner's personal expenses** (income taxes, etc.)

1. Any business expenses not necessary for generation of income to the property are not valid operating expenses of the property.
2. The income tax shown on the owner's personal income, not the value of the property is considered an improper expense.

(iv) **Capital improvements**

1. Capital improvements are not items of annual expense. Typically, capital improvements will result in either increases in property value, economic life of the property, or income to the property.
2. For example, the addition of a swimming pool to an apartment complex would not be considered a normal annual operating expense of the property.

(v) Franchise fees and special corporation costs

1. A franchise is usually an exclusive right to furnish public services or to sell a product in a certain community.
2. Normally franchise fees, special corporation costs, and items of this nature are charges against the owner of the property or the business, rather than the property itself, thus would not normally be considered operating expenses of the real estate.
3. Franchise fees should typically be excluded for fast-food or other retail uses where the value of the real estate is not tied to the performance of the business.
4. Franchise fees can be a proper expense for hotel properties and included in the expense analysis when the daily rates and market data used to develop the effective gross income estimate includes franchised hotels. In such cases, the value indication may include intangible and personal property value that would need to be removed so the real property value can be estimated. The income and expense analysis should be consistent with the capitalization rates derived from comparable franchised hotel sales. Capitalization rates are addressed in Chapter 3.

(vi) **Real Estate Taxes** - are a valid operating expense of property.

However, because the ad valorem tax appraiser is appraising the property for the ultimate purpose of determining the amount of ad valorem taxes, the appraiser must not make a deduction for real estate taxes. It is not possible to know the proper amount of real estate taxes until the amount of the assessed value is known. Thus, to make a deduction for real estate taxes, the appraiser would have to pre-suppose the amount of value in order to compute the amount of real estate taxes. The ad valorem tax appraiser circumvents this problem by making no deduction for real estate taxes from the income and instead the appraiser includes the effective tax rate as a component of the capitalization rate for the property (this is very similar to property depreciation/appreciation discussed previously).

1. Proper as an expense under certain conditions, such as appraising property for other than ad valorem tax purposes.
2. Not proper as an expense, under certain conditions, such as appraising property for ad valorem tax purposes.

Demonstration 2-15: Reconstructing an Income and Expense Statement

You have been given the owner's operating statement for a 60-unit apartment complex. The statement has been prepared by the owner's accountant for income tax purposes and you will need to reconstruct it for appraisal purposes.

Income and Expense Statement -

Peachtree Apartments

Income:	
Apartment rents	\$581,400
Laundry facilities	\$7,500
Gross income	\$588,900
Expenses:	
Real estate taxes	\$45,450
Insurance	\$30,600
Salaries	\$34,500
Fringe benefits	\$9,650
Painting 10 units	\$20,000
Utilities	\$73,100
Ground maintenance	\$18,500
Advertising	\$4,800
Depreciation	\$195,000
Debt service (principal & interest)	\$198,400
Replace 5 refrigerators	\$4,000
Replace 8 stoves	\$5,600
Replace 10 Water heaters	\$6,000
Total expenses	\$645,600

Analysis of the market indicates a potential gross income (PGI) of \$630,000, and a rate of 5% for vacancy and collection loss. The owner's miscellaneous income is typical, and the allowable expenses are similar to comparable complexes, thus requiring no adjustment. Also, expenses for items that are considered reserves for replacement are comparable to similar complexes. The owner did not include any allowance for management. However, a management fee of 5% of the effective gross income is an appropriate adjustment.

The stoves and refrigerators have a 15-year life, while the water heaters have a 10-year life. There is one stove, refrigerator, and water heater in each apartment. The roof cover for the complex has a 20-year life and would cost \$60,000 to replace. All units should be repainted on a five-year cycle. The floor covering in each unit lasts 9 years and costs \$1,200 per unit to replace.

Using market data and the owner's allowable expenses you can now reconstruct the income and expense statement.

Reconstructed Income and Expense Statement - Peachtree Apartments

Potential Gross Income			\$630,000	100.0%
Vacancy and Collection Loss			-\$31,500	5.0%
Miscellaneous Income			+ \$7,500	
Effective Gross Income			\$606,000	
Operating Expenses				
Management	\$30,300	5.0%		
Insurance	\$30,600	5.05%		
Salaries	\$34,500	5.69%		
Fringe benefits	\$9,650	1.59%		
Utilities	\$73,100	12.06%		
Ground maintenance	\$18,500	3.05%		
Advertising	\$4,800	0.79%		
Reserves for Replacements				
Refrigerators $[(800 \times 60) \div 15]$	3,200	0.53%		
Stoves $[(700 \times 60) \div 15]$	2,800	0.46%		
Water heaters $[(600 \times 60) \div 10]$	3,600	0.59%		
Painting $[(2,000 \times 60) \div 5]$	24,000	3.96%		
Floor cover $[(1,200 \times 60) \div 9]$	8,000	1.32%		
Roof cover $(60,000 \div 20)$	3,000	0.50%		
Total Expenses:			-\$246,050	40.60%
Net Operating Income:			\$359,950	59.40%

To assist the appraiser in comparing expenses with those of other properties and assist in developing benchmarks for mass appraisal of income-producing properties, expense ratios have been provided.

Analysis of the Reconstructed Income and Expense Statement

Potential Gross Income: This is market rent or economic rent as determined from an analysis of similar properties. The owner's statement reflected the rents collected not potential gross income (PGI). PGI is market rent at 100% occupancy. If there is a difference between market rent and contract rent, market rent must be used to develop PGI.

Vacancy and Collection Loss: It is highly unlikely a property will remain fully rented, or that all the rents will be collected, during its economic life. Thus, a vacancy factor and/or a factor for collection loss must be determined by a study of other properties, similar to the subject being appraised. The owner reported his rents collected but did not provide any information on vacancies or collection loss amounts. Since the purpose of reconstructing the income and expense statement is to reflect typical income and expenses compared to similar properties, an adjustment for vacancy and collection loss is appropriate. Market analysis indicated an adjustment of 5% was typical, so this was the amount that was applied in reconstructing the owner's statement.

Miscellaneous Income: The owner did have miscellaneous income and listed it separately on the statement as income from his coin-operated laundry facilities. Similar units with coin-operated laundry facilities reported similar incomes, thus the owner's listed amount was considered typical.

When considering miscellaneous income and analyzing owner's operating statements for various use-types of properties, the appraiser must decide whether the miscellaneous income is attributable to real estate or if it might be a part of the business income.

Proper Expenses:

Management: Management fees are a proper expense for this type of property even though the owner did not list an expense for management. A review of similar apartment complexes indicated a fee of 5% of effective gross income was appropriate for apartment complexes in this area.

Insurance: Insurance is a proper expense. However, when analyzing insurance expenses, the premium should reflect the cost for one year. If the premium is for something more or less than one year, it must be adjusted to reflect an annual premium.

Salaries and fringe benefits: Salaries and fringe benefits necessary to maintain the rental income stream to the property are a proper expense. The owner's expense for salaries and fringe benefits was considered typical and the entire amount was allowed.

Utilities: Utilities paid by the landlord are considered a proper expense.

Ground maintenance: Maintaining the grounds, which includes lawn care is a proper expense in the operation of an apartment complex.

Advertising: Advertising is a proper expense. However, a comparison to similar properties should be made to determine if the amount on the owner's statement is too high or too low.

Refrigerators: This item should be considered as a reserve for replacement. The proper allowance can be determined by first multiplying the cost of one refrigerator by the total number of units. Next, divide this amount by the total life for the refrigerators.

Stoves: This item should be considered as a reserve for replacement. The proper allowance is calculated in the same manner as the calculation for refrigerators.

Water Heaters: Water heaters should be considered as a reserve for replacement and is treated the same as refrigerators and stoves.

Painting: Painting should be considered as a reserve for replacement. The proper allowance is determined by multiplying the cost to paint one unit by the total number of units in the complex and then dividing the result by the cycle for painting. In this statement, painting is on a five-year cycle.

Floor Cover: This item was not included in the owner's operating statement. However, market analysis indicates floor coverings last nine years and the cost to replace the floor cover in an individual unit is \$1,200. The proper allowance is calculated in the same manner as the calculations for refrigerators, stoves, and water heaters.

Roof Cover: This is another item that did not appear on the owner's statement. However, it should be considered as a reserve for replacement. The proper allowance is calculated by dividing the cost to replace the roof cover by its total life.

Improper Expenses:

These items were listed as expenses on the owner's statement. However, they are not proper expenses and should not be used in developing an estimate of net operating income. Thus, they were omitted from the reconstructed income and expense statement.

Income Taxes: Income taxes are not a legitimate operating expense, because the tax is based on the personal income of the owner.

Depreciation: Depreciation is recognized in the income approach by the recapture rate which is a component of the overall rate or the building capitalization rate.

Debt Service: This includes principal and interest of the mortgage. These items are considered in the capitalization rate in the form of a discount rate.

Real Estate Taxes: Real estate taxes are an allowable expense. However, since we are appraising for ad valorem tax purposes, we do not deduct the taxes as an operating expense but allow them as a component of the capitalization rate.

Demonstration 2-16: Reconstruction of Income and Expense with CAM Expenses/Reimbursements

You have been given an owner's operating statement for an office complex covering the last eight months. The property is 100 percent occupied. The vacancy and collection loss for this type of property is 7 percent in the local market. The leases provided by the owner indicate that the tenants are responsible for their portion of the property's expenses for insurance, real estate taxes and common area maintenance (CAM) expenses (triple net lease). The tenants reimburse the landlord monthly for these three expenses. Based on your review of the market, the contract rents submitted by the property owner are at market rates and there are no rent collection issues. Other income to the landlord is generated by ten covered parking spaces that are 100% leased at a market price. The expense amounts are representative of market data and the fire sprinkler system service is based on an annual amount. Your assignment is to estimate the fair market value property for property tax purposes, so the first step is to determine the appropriate net operating income for the analysis.

Income and Expense Statement

ABC Office Complex

Item	Owner's statement		Reconstructed Statement
Company A rent	\$382,400	$(382,400 \div 8) \times 12 =$	\$573,600
Company B rent	\$62,000	$(62,000 \div 8) \times 12 =$	\$93,000
Company C rent	\$137,600	$(137,600 \div 8) \times 12 =$	\$206,400
Expense reimbursements	\$0	$27,000 + 87,300 =$	\$114,300
Other Income	\$5,600	$(5,600 \div 8) \times 12 =$	\$8,400
Potential Gross Income	\$587,600		\$995,700
Less: Vacancy & Collection Loss	\$0	7%	-\$69,699
Misc. Income			included above
Effective Gross Income	\$587,600		\$926,001

Item	Owner's Statement		Reconstructed Statement
Expenses:			
Insurance	\$18,000	$(18,000 \div 8) \times 12 =$	\$27,000
CAM	\$58,200	$(58,200 \div 8) \times 12 =$	\$87,300
Real Estate Taxes	\$232,800	Exclude for Assessment	\$0
Management Fees - Property	\$29,380	$(29,380 \div 8) \times 12 =$	\$44,070
Management Fees - Assets	\$47,008	Improper Expense	\$0
Roof Maintenance	\$2,400	$(2,400 \div 8) \times 12 =$	\$3,600
Fire Sprinkler System Service	\$1,360	$(1,360 \div 8) \times 12 =$	\$2,040
Miscellaneous Expenses	\$1,200	$(1,200 \div 8) \times 12 =$	\$1,800
Real Estate Tax Consultant Fees	\$1,650	Improper Expense	\$0
Depreciation	\$105,000	Improper Expense	\$0
Debt Service	\$276,830	Improper Expense	\$0
Total expenses:	\$773,828		\$165,810
Net Operating Income:	-\$186,228		\$760,191

Demonstration 2-16: Reconstruction of Income and Expense with CAM Expenses/Reimbursements (continued): ANALYSIS AND EXPLANATION

The owner has provided a statement of eight months of operations, while the appraisal analysis is done from the annual perspective. A majority of the income and expense reconstruction involves prorating the information provided by the owner by dividing it by eight months, then multiplying it by 12 months.

Potential Gross Income: Potential Gross Income (PGI) is market rent at 100% occupancy. If there is a difference between market rent and contract rent, market rent must be used to develop PGI. The contract rents are at market and since there are no collection issues, the only adjustment is to prorate the figures to an annual perspective to include the property's PGI.

Usually, the building rent would only be included in the PGI, but the expense reimbursements and other income for the garage are also subject to vacancy and collection losses, so they need to be included in the PGI as well. In other words, the landlord will not receive reimbursement income when the property is vacant and not earning rent.

Miscellaneous Income: Miscellaneous income is included in the PGI. The owner's statement includes the other income attributed to the covered parking, but it does not report the expense reimbursements for insurance, real estate taxes and common area maintenance (CAM) charges to the tenant.

Under a triple net lease, it is common to have the reimbursed expenses billed to the tenant by the landlord for insurance, taxes, and CAM charges. However, they may not show up in the owner's statement if the tenant pays their pro-rata share directly to a management company or the respective insurance company and taxing jurisdiction.

The reconstructed statement utilizes the actual expense items provided for insurance and CAM charges. Since the analysis is being done for assessment purposes, real estate taxes are excluded. The analysis requires using the prorated amount for the tenant's share of insurance and CAM expenses.

The reconstructed statement also includes the other prorated income that was reported for the covered parking. The covered parking space was reportedly full

and receiving market rent, so the other income is adjusted to an annual perspective.

Vacancy and Collection Loss: Most properties will not be 100% occupied through their economic life without any collection losses. Although the subject is 100% occupied without any collection loss issues, the appraisal analysis is done from the market perspective. The local market is experiencing a vacancy and collection loss rate of 7%, which is applied in the analysis.

Proper Expenses

Insurance – Insurance is a proper expense. However, when analyzing insurance expenses, the premium should reflect the cost for one year. If the premium is for something more or less than one year, it must be adjusted to reflect an annual premium. The insurance expense for eight months has been adjusted to an annual perspective.

CAM – Common area maintenance (CAM) is a proper expense. For the office complex in this example, it reflects keeping up the shared space on the property such as the following: pest control, snow removal, landscaping, lawn care and trash removal, as well as janitorial services and utilities for the common areas. The stabilized statement should reflect the cost for one year, but only eight months of CAM expenses was provided by the owner. The CAM expense has been adjusted to an annual perspective.

Management fees (property) – Management fees are a proper expense for this type of property. With multiple tenants and common areas to oversee, a management company is needed to supervise the property. The eight months of property management fee has been prorated to an annual perspective.

Roof maintenance – Under a triple net lease, the landlord is typically responsible for the structure and roof of the property. The expense information provided by the owner includes a roof maintenance contract to a third-party company. These types of contracts typically include a monthly roof inspection and to address minor issues as needed. The expense information was provided for eight months and is prorated to one year.

Fire Sprinkler System Service – Buildings that feature a fire sprinkler system typically require a maintenance service contract to ensure the system is

working properly in conjunction with the fire alarm system. This expense is billed annually, so no adjustment was made in the analysis.

Miscellaneous Expenses – Most property expenses include miscellaneous expenses that do not fall into other categories. These could be administrative expenses such as professional fees (accounting, legal, etc.), income tax preparation, LLC filing, or other minor expense items. This expense was provided for eight months and is adjusted to an annual perspective.

Improper Expenses

Real Estate Taxes – The purpose of the appraisal is for tax assessment purposes, so real estate taxes are excluded as an expense item.

Management Fees (Assets) – Asset management fees are expenses charged by the investment management firm associated with a multiple property or portfolio of properties under one entity. An example might be a Real Estate Investment Trust (REIT) or privately held investment group that manages multiple properties for a group of private investors to make decisions on when to buy and sell properties with the fund.

The Asset Management Fee is determined by the nature of the ownership and not the market. As a result, it is not a proper expense item for this type of property and excluded from the analysis.

Real Estate Tax Consultant Fees – The actual expenses include an expense paid to a real estate tax consultant to assist the owner in reducing its tax liability. This expense is not required for the office use and is excluded from the analysis.

Depreciation – Depreciation on the owner's expense is a function of when the asset was acquired and what is reported to the federal or state income tax agencies. This expense varies from owner to owner and is not required for the office use.

Debt Service – This includes principal and interest of the mortgage for the monthly payments to the owner's lender. Debt varies from property to property. Some properties are owned free and clear of debt, while others may have millions of dollars in debt. As a result, debt-service is excluded from the analysis.

Exercise 2-10: Reconstruction of an Income Statement

You have been given an apartment owner's operating statement to analyze for a 28-unit apartment complex. The figures are validated by check stubs and you believe them to be accurate. You are required to determine which figures to use as stated, which figures to pro-rate, and which figures to totally eliminate.

Mark an "X" in the appropriate column to indicate what should be done with each item of expenses.

Expense Item:	Use as Stated	Pro-Rate	Eliminate
Management fee			
Repairs			
Miscellaneous			
Utilities			
Interest on mortgage			
Principal on mortgage			
New roof			
Insurance fire (3-year policy)			
Insurance liability (1-year)			
Employee's health policy (1-year)			
Janitor's salary			
Painting exterior			
Purchase of 3 new refrigerators			
Purchase of 4 range/ovens			
Supplies			
Corporate income taxes			
Red Cross donation			
Carpet replacement			
Redecorate 7 apartment units			
Real estate taxes			

Exercise 2-11: Reconstruction of an Income Statement

Your assignment is to appraise a 30-unit apartment building. You have the owner's operating statement prepared by the accountant. After careful analysis, you decide that all items are essentially correct. However, based on data you have on file the owner did not include certain items that normally occur in the market. The owner did not include an allowance for vacancy and collection loss, which is 4% using a market study you recently conducted. Also, the owner manages the property and did not include a management expense. An analysis of comparable data indicates that a typical management expense for this type of property is 5% of EGI. Painting and decorating should last 3 years.

Income or Expense Items	Owner's Statement	Reconstructed Statement
Gross Income	\$365,000	
Vacancy & Collection Loss	0	
Effective Gross Income	\$365,000	
Expenses:		
Management		
Employees' salaries	\$34,022	
Employees' benefits	\$2,160	
Insurance	\$7,000	
Natural gas	\$13,445	
Painting & decorating 5 units	\$6,000	
Repairs	\$8,425	
Supplies	\$2,040	
Electricity	\$4,850	
Water	\$1,680	
Real Estate Taxes	\$32,550	
Depreciation	\$56,000	
Interest on mortgage	\$120,650	

Expenses:		
Legal & accounting fees	\$4,800	
Principal on mortgage	\$19,100	
Miscellaneous expenses	\$3,150	
Total Expenses	\$315,872	
Net Operating Income	\$49,128	

Reconstruct the operating statement, concluding with your own estimate of net operating income and then calculate the operating expense ratio.

Exercise 2-12: Reconstruction of an Income Statement

You are appraising a 12-year-old apartment complex and have been given the owner's income and expense statement for last year. This statement shows the actual amounts received and paid out as shown below:

Income or Expense Items	Owner's Statement	Reconstructed Statement
Gross Income	\$697,500	
Vacancy & Collection Loss	0	
Miscellaneous Income (annual)	\$2,750	
Effective Gross Income	\$700,250	
Expenses:		
Management		
Interest on mortgage	\$168,238	
Insurance	\$4,500	
Real Estate Taxes	\$112,500	
Janitor's salary	\$36,000	
Water	\$11,000	
Electricity	\$16,000	
Natural gas	\$18,000	
Miscellaneous repairs	\$11,250	
Supplies	\$3,375	
Principal on mortgage	\$70,000	
Exterior repairs	\$31,250	
Depreciation	\$72,000	

<i>Expenses (cont.):</i>		
Building addition	\$120,000	
6 Refrigerators	\$10,500	
8 Ranges	\$12,000	
2 Water heaters	\$4,500	
Roof cover		
Floor cover		
Redecorating		
Total Expenses	\$701,113	
Net Operating Income	\$(863)	

The apartment complex consists of five buildings with 10 units per building. Each building contains two efficiency units, six 1-bedroom units, and two 2-bedroom units. One water heater serves each building, and each unit contains a range and refrigerator.

An analysis of similar apartment complexes provides an indication of monthly market rent as follows:

Unit Type	Market Rent
Efficiency	\$1,000
1 bedroom	\$1,250
2 bedrooms	\$1,500
Miscellaneous income: \$2,750 (annual)	

Your analysis also indicates a vacancy and collection loss of 5% and a management allowance of 4% of the effective gross income for the subject property. Proper operating expenses are typical, needing no adjustment. Reserves for replacement include water heaters, ranges and refrigerators, roof cover, floor cover, and redecorating. The costs shown on the owner's statement for ranges, refrigerators, and water heaters are representative of the market.

Further analysis indicates a 10-year life for ranges and refrigerators, and a 5-year life for water heaters. Replacement of the roof cover for the entire complex costs \$45,000 and has a 12-year economic life. Replacement of floor covers for the entire complex costs \$37,500 and has an economic life of five years. Each apartment unit requires redecorating, on average, every five years. The cost to redecorate each unit is \$3,750.

Reconstruct the previous statement and develop it into a proper income and expense statement.

PAGE LEFT BLANK INTENTIONALLY

REVIEW QUESTIONS – Chapter 2

1. In addition to sales and leases, _____ and _____ can be valid sources of market information, if the appraiser exercises care.
2. List five sources of rental income data:
 - _____
 - _____
 - _____
 - _____
 - _____
3. Another name for market rent is _____ rent.
4. A lease that provides for one or more extensions of the lease term in the original lease document at the option of the tenant is a _____ lease.
5. The fixed portion of the rent under the terms of a percentage lease is known as _____.
6. When contract rent exceeds economic rent, the difference is known as _____ rent.
7. The rent that is over and above a guaranteed minimum base rent is referred to as _____ rent.
8. A contract that requires a fixed minimum base rent and a variable rent based on volume of business, sales, productivity, or use of the property by the tenant is called a _____ lease.
9. A lease in which the landlord is required to pay all operating expenses associated with the real estate is known as a _____ lease.

10. A lease in which the tenant is required to pay all operating expenses associated with the real estate is known as a _____ lease.

11. List four factors that must be considered when comparing rental properties.

- _____
- _____
- _____
- _____

12. List five common units of comparison when comparing rental properties.

- _____
- _____
- _____
- _____
- _____

13. The total leasable area, including common areas is known as _____.

14. The _____ is the potential gross income, less vacancy and collection loss, plus appropriate miscellaneous income.

15. Income from vending machines and coin-operated laundries is/are considered examples of _____ income.

16. The first step in reconstructing the income and expense statement is to determine the _____.

17. Which of the following would be considered a proper expense in the reconstructed operating expense statement?
- Salaries
 - Management
 - Utilities
 - All of the above
18. Expense amounts for short-lived items such as carpet, drapes, and water heaters are considered _____.
19. When reconstructing an income/expense statement, depreciation and debt service would be considered a/an _____ expense.
20. The income remaining after developing effective gross income and allowing for operating expenses and reserves for replacement is known as _____.
21. _____ take the form of either free rent or extra tenant improvement allowances.
22. _____ tends to be less than fee simple value because leases are typically set at market level when written, but usually fall below market levels as time passes.
23. _____ buildings are generally a little older but still have good quality management and tenants.
24. _____ rent is used as a common denominator to compare leases with different provisions. This rent is the lease base rent less rent concessions.
25. The formula to calculate an operating expense ratio (OER) is:
_____.

PAGE LEFT BLANK INTENTIONALLY

Chapter 3 | Development of Capitalization Rates

APPENDIUM KNOWLEDGE AREAS



SUGGESTED READING

Source	Chapter	Pages
Property Assessment Valuation	Chapter 13	339-361

CHAPTER TOPICS AND TIMETABLE

Topic	Time	Percentage
Introduction to Relationship of Capitalization Rates & Value Estimate	30 minutes	7
Overall Capitalization Rate (R_o) or (OAR)	15 minutes	4
Five methods of Developing an Overall Capitalization Rate (R_o)	3 hours, 30 minutes	49
Capitalization Rate Individual Components	1 hour, 35 minutes	22
Review Questions	35 minutes	8
Quiz #1	45 minutes	10

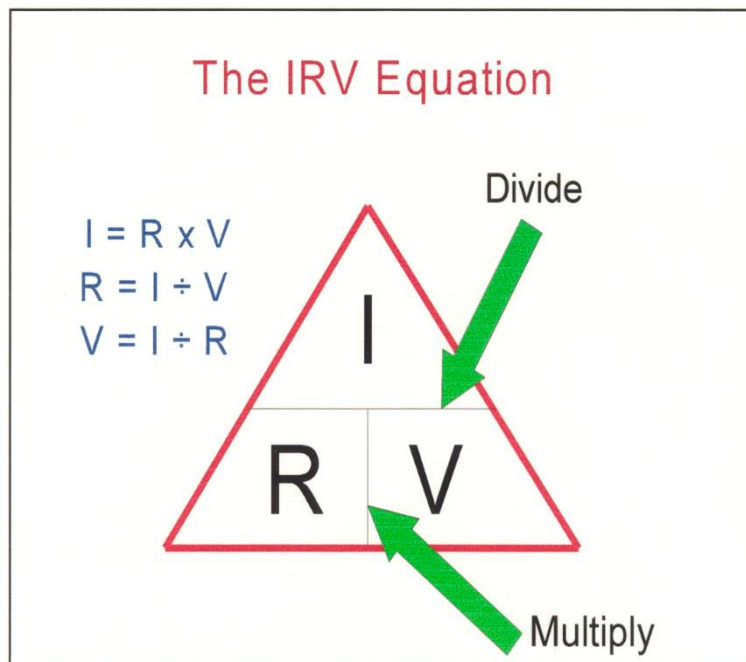
OBJECTIVES

Upon completion of Chapter 3, the student should be able to:

- Express the relationship between income and value as a capitalization rate (R_o).
 - Apply the IRV and VIF formulas in the valuation of income-producing properties.
 - Identify and properly apply the five methods of developing an overall capitalization rate (R_o).
 - Define and calculate recapture rate.
 - Calculate land-to-improvement (building) ratios, expense ratios, and remaining economic lives and use to determine comparability to the subject property appraised.
 - Calculate an effective tax rate (ETR) using the EAT equation.
 - Appropriately "load" a capitalization rate (R_o) with an effective tax rate (ETR)
 - Describe the three parts of a capitalization rate.
 - Identify the four components of a yield rate.
 - Develop an overall yield rate (Y_o) using the band-of-investment technique
- Introduction.

Chapter 3 | Development of Capitalization Rates

- I. Introduction to Relationship of Capitalization Rates and the Value Estimate
 - A. Interest rates, yield rates, capitalization rates, and gross income multipliers are all considered relationships between income and value. For example, a savings account of \$100 earning an interest rate of 6% produces interest income of \$6 annually. An overall capitalization rate of 10% shows the relationship between an annual income of \$10,000 and market value of \$100,000, and a gross income multiplier of 5 can be used to demonstrate the relationship between a gross income of \$50,000 and market value of \$250,000.
 - B. No unit of comparison from market information has validity unless it relates to the subject property in the same way that it relates to market comparables from which it was derived.
 - C. IRV equation
$$I = R \times V$$
$$R = I \div V$$
$$V = I \div R$$

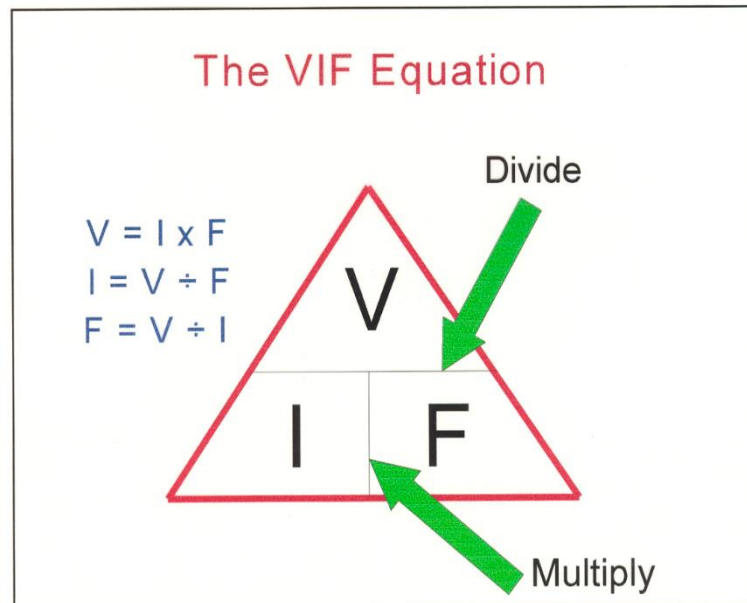


D. VIF equation

$$V = I \times F$$

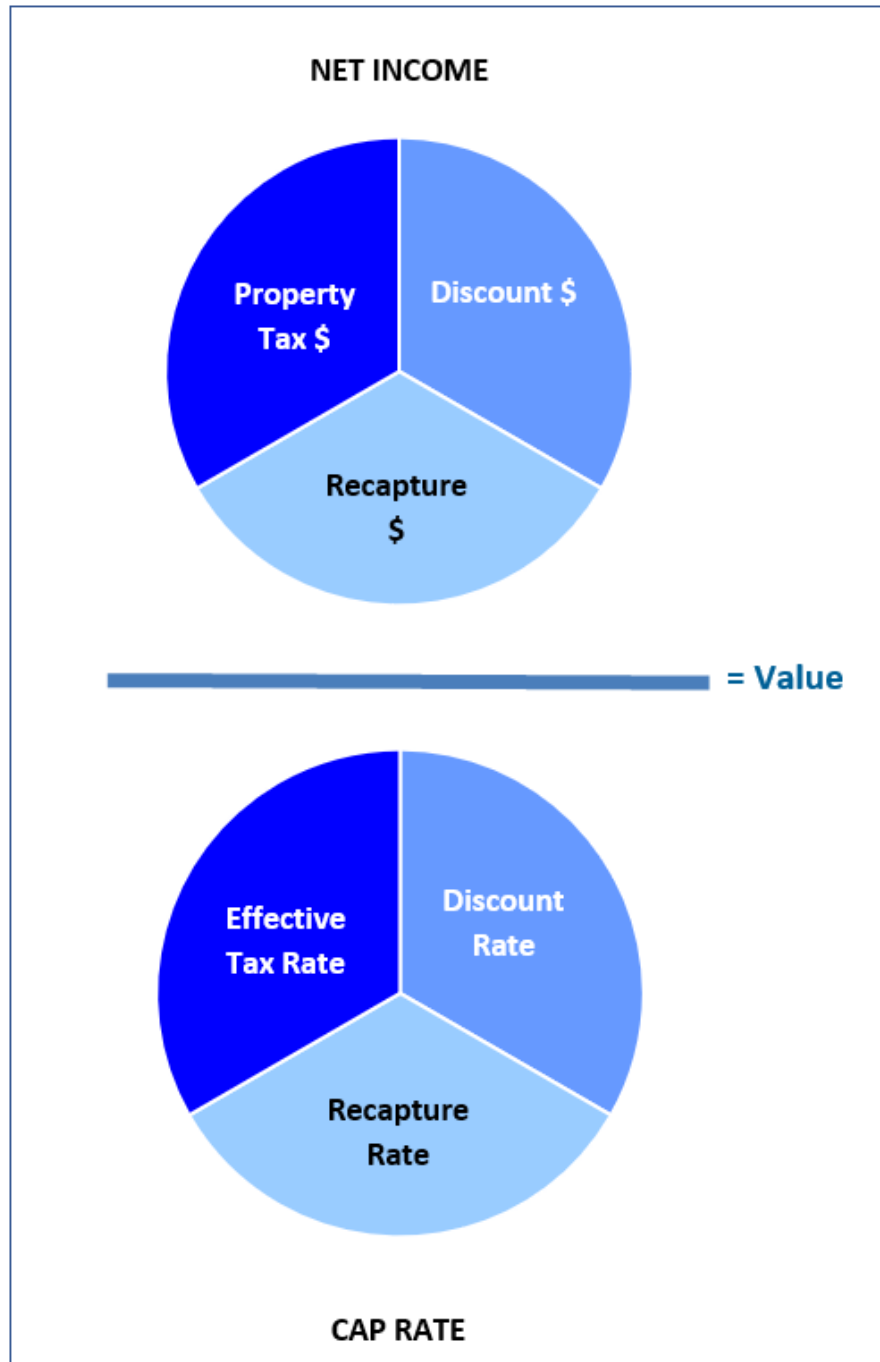
$$I = V \div F$$

$$F = V \div I$$



E. Income expressed as percentage of an unknown value

1. Income expressed as a percentage of an unknown value - net operating income when expressed as a certain percentage of value is called the overall capitalization rate.
2. In appraising a property, divide the percentage (known as the overall capitalization rate) into the property's net operating income, to obtain value. Mathematically, if income is 10% of value of the property, then dividing the income by 10% will indicate the total value of the property.
3. Income and capitalization rates
 - a. Income must match rate
 - b. Rate must match income
 - c. No exceptions to this rule



II. Overall Capitalization Rate (Ro) or (OAR)

- A. Expresses the relationship between net operating income and market value of property. The overall capitalization rate represents the percentage that net operating income is in relation to total property value.
- B. Used in direct capitalization. The components of the overall capitalization rate are:
 - 1. the overall yield rate
 - 2. recapture rate
 - 3. effective tax rate (ETR).
- C. It is essential that the overall capitalization rate be matched with the net operating income from which it is derived. Failure to do so can substantially under- or overstate estimates of market value.
- D. Theoretically composed of the 'return on' rate (or overall yield rate or property discount rate) and a portion of the 'return of' (or recapture rate) for the improvements. While all the component parts are included in the overall capitalization rate, there is not necessarily any separate identification of each of the rates included within the overall capitalization rate.
- E. Does not distinguish between land rate and improvement rate components. The overall capitalization rate (R_O) is a weighted average of the land and improvement capitalization rates used in straight-line capitalization. While all the component parts (land capitalization rate (R_L) and improvement capitalization rate (R_B)) are included in the overall capitalization rate, there is not necessarily any separate identification of each of the rates included within the overall capitalization rate.

III. Five Methods of Developing an Overall Capitalization Rate (R_o)

- A. There are five methods which can be used to derive an overall capitalization rate (R_o). All five methods should yield very similar overall capitalization rates, provided valid market information is derived and applied correctly and consistently.

The standard overall capitalization rate used in most appraisal situations, not for ad valorem taxation, does not have an ETR component and is referred to as an unloaded capitalization rate. The overall capitalization rate that is commonly used in ad valorem property tax appraisals does have an ETR component and is referred to as a loaded capitalization rate.

When property is valued for ad valorem tax purposes, property taxes should not be considered an expense item. Any deduction from gross income directly affects the indicated property value through the income approach, only typical and reasonable expenses can be used. It might be questioned how a typical and reasonable figure for taxes can be found when taxes are usually based on the property value itself.

Furthermore, taxes are assessed annually on the basis of the property value, the level of assessment, and the current tax rate or mill rate.

- B. The five methods are:

1. **Derivation from comparable sales**
2. **Band of investment technique (mortgage & equity components)**
3. **Net income ratio method**
4. **Debt coverage ratio method**
5. **Band of investment technique (land & improvement components)**

C. Derivation from comparable sales

1. Uses IRV equation
2. Known components of the IRV equation are:
 - a. Net operating income (NOI)
 - b. Property value (sale price)
3. Net operating income divided by the property value equals overall capitalization rate (R_o).

$$\text{NOI} \div \text{Sale Price (Value)} = \text{Overall Cap Rate (} R_o \text{)}$$

4. Derivation from comparable sales is the preferred method to obtain a capitalization rate (R_o) in the Income Approach because it brings actual market rates into the analysis. Similar to sale data selection in the Sales Comparison Approach, comparability and similar use are important factors in arriving at a reliable capitalization rate (R_o) indication. Considerations for comparability include:
 - a. Comparable types of property, with similar remaining economic lives, operating expense ratios, physical condition, and ratios of land-to-improvements as proportions of total property value
 - b. Comparable income streams with similar characteristics of risk, timing, stability, and income projection pattern
 - c. Comparable terms and types of financing
 - d. Comparable types of buyers with buying motivations similar to those of the most probable type of buyer
 - e. Comparable terms of sale
 - f. Comparable market conditions at the time of sale and the time of appraisal.
5. The following key characteristics of comparable sales used in the analysis should be similar to the subject property appraised:
 - a. Land-to-improvement (building) ratios
 - b. Income and expense ratios
 - c. Remaining economic lives (improvements)

Demonstration 3-1: Development of an Overall Capitalization Rate (R_o) Derivation of Comparable Sales

You are appraising a retail commercial property. In an effort to derive a proper overall capitalization rate (R_o) from market sales, you have found a similar retail commercial property, which recently sold for \$900,000 and has an effective gross income of \$200,000. Total operating expenses, including reserves for replacement, for the comparable sale amount to \$96,500 annually. What is the indicated overall capitalization rate (R_o) for the comparable sale property?

Effective Gross Income	\$200,000
Total Operating Expenses & Reserves	<u>-\$96,500</u>
Net Operating Income	\$103,500

Net Operating Income	÷	Property Value	=	Overall Capitalization Rate (R_o)
\$103,500	÷	\$900,000	=	11.5%

Illustration 3-1: Development of an R_o By Market Comparison Method

The following example demonstrates that even though two seemingly comparable properties may have the same sales price, the ratio of land-to-improvement (building) will affect the overall rate of the property even when the discount rate and the recapture rate are the same.

Item	Sale #1	Sale #2
Sale Price	\$800,000	\$800,000
Land Value	\$400,000	\$200,000
Improvement Value	\$400,000	\$600,000
Property Discount Rate (Y_o)	10%	10%
Recapture Rate (20 yrs)	5%	5%
Return for Land and Improvement	\$80,000	\$80,000
Recapture dollars (Depreciation)	\$20,000	\$30,000
Total Net Operating Income	\$100,000	\$110,000
Overall Capitalization Rate	12.50%	13.75%

Illustration 3-2: Development of an R_o By Market Comparison Method

In addition to the need for land-to-improvement (building) ratios to be comparable, the income and expense ratios of the comparable sales must be similar to ratios for the subject property being appraised. The following example demonstrates how two sale properties with comparable gross incomes and resulting identical gross income multipliers can have significantly different overall capitalization rates, due to differences in expense ratios.

Item	Sale #1	Sale #2
Gross Income	\$200,000	\$200,000
Expenses	\$60,000	\$80,000
Expense Ratio	30%	40%
Net Operating Income	\$140,000	\$120,000
Sale Price	\$1,240,000	\$1,240,000
Gross Income Multiplier	6.2	6.2
Recapture dollars (Depreciation)	\$20,000	\$20,000
Overall Capitalization Rate	11.3%	9.7%

Illustration 3-3: Development of an R_o From Comparable Sales

Another requirement which must be met by the comparable sales properties and the subject property is they must have similar remaining economic lives. The example below shows how a change in the recapture rate (reciprocal of remaining economic life) can affect the overall rate.

Property Components (Assume 80% Improvement and 20% Land)		
Item	Sale #1	Sale #2
Estimated Remaining Economic Life	25 years	50 years
Annual Recapture Rate	4.0%	2.0%
Property Discount Rate (Y_o)	9.0%	9.0%
Improvement Cap Rate (R_B)	13.0%	11.0%
Improvement Component of Overall Cap Rate	10.4%	8.8%
Land Component of Overall Cap Rate	1.8%	1.8%
Overall Cap Rate (R_o)	12.2%	10.6%

Sale #1					
Capital Investment	Investment %		Cap Rate		Product
Land	0.20	X	0.09	=	0.018
Improvement	0.80	X	0.13	=	0.104
Totals	1.00	R_o		=	0.122

Sale #2					
Capital Investment	Investment %		Cap Rate		Product
Land	0.20	X	0.09	=	0.018
Improvement	0.80	X	0.11	=	0.088
Totals	1.00	R_o		=	0.106

From the preceding examples, we can see that the derivation of overall rates from properties that are not very similar to the subject can result in an invalid overall rate when applied to the subject. Derivation of overall rates from comparable sales requires highly comparable, although not necessarily perfect data. It is imperative that the properties used as comparables be comparable to the subject in terms of physical, locational, financial, and investment characteristics as possible.

Exercise 3-1: Development of an R_o Derivation from Comparable Sales

You are appraising an apartment complex and want to develop an overall rate from the market. You find six sales of similar apartment complexes. The six comparables are as follows:

Sale No.	Sale Price	Remaining Economic Life	Land-to-Improvement Ratio	Expense Ratio	Net Operating Income
1	\$760,000	18	1:4	34%	\$64,600
2	\$780,000	20	1:5	27%	\$66,300
3	\$720,000	20	1:4	30%	\$64,800
4	\$712,000	20	1:4	30%	\$64,000
5	\$744,000	20	1:4	30%	\$67,000
6	\$750,000	22	1:5	30%	\$82,500

The subject property has a remaining economic life of 20 years, a land-to-improvement ratio of 1:4 and an expense ratio of 30%. The six comparables and the subject property are of similar construction and condition.

What is the indicated overall capitalization rate by market comparison, based on the six comparable sales?

Exercise 3-2: Development of an R_o Derivation from Comparable Sales

You have been asked to appraise an apartment complex in your small town. You have only been able to find three sales of similar properties to the subject, all of which have remaining economic lives of 25 years. The subject is a 25-unit complex that has a 44% operating expense ratio and a remaining economic life of 25 years. The land-to-improvement ratio for the subject is 1:4. From the three comparable sales shown in the grid below, derive the overall capitalization rates for each and select the one that you determine is most appropriate for appraising the subject property.

	Comparable Sales		
	Sale 1	Sale 2	Sale 3
Sale Price	\$2,100,000	\$2,025,000	\$2,550,000
No. of Units	25	24	30
Potential Gross Income	\$410,000	\$360,000	\$450,000
V&C Loss	\$18,450	\$16,200	\$20,250
Operating Expenses	\$187,944	\$151,272	\$171,900
Land Value	\$420,000	\$405,000	\$637,500

PAGE LEFT BLANK INTENTIONALLY

Exercise 3-3: Development of an R_o Derivation from Comparable Sales

You are deriving an overall capitalization rate (R_o) from a market transaction which occurred recently for a sale price of \$18,000,000. You have been given access to the owner's operating statement as shown below. All expenses appear to be reasonable, requiring only exclusion of improper expenses. In order to derive the overall capitalization rate (R_o) from the comparable sale, you must reconstruct the owner's operating statement and develop the appropriate information from which to derive the overall capitalization rate (R_o).

Owner's Operating Statement	
Gross Income	\$3,450,245
Vacancy & Collection Loss	<u>-\$172,512</u>
Effective Gross Income	\$3,277,733
Expenses:	
Maintenance	\$393,328
Water & Sewer	\$65,700
Pool Cleaning	\$15,500
Insurance	\$115,000
Contributions to political party	\$25,000
Interest expense	\$370,000
Electricity	\$491,500
Natural Gas	\$398,000
Management Expense	\$196,000
Depreciation	\$520,000
Federal Income Taxes	<u>\$378,000</u>
Total Expenses	\$2,968,028
Net Operating Income	\$309,705

PAGE LEFT BLANK INTENTIONALLY

D. Band-of-investment technique (weighted average) mortgage and equity components:

This technique uses the band-of-investment calculation to determine the weighted average of the mortgage (debt) and equity rates to obtain the overall capitalization rate (R_o). The debt and equity rates are weighted by the respective contributions of debt and equity to the overall financing of the total property value.

Financing	Portion		Rate		Product
Debt	%	X	Annual Mortgage Constant	=	Weighted #
Equity	%	X	Equity Dividend Rate	=	Weighted #
Total	1.00			=	Overall Cap Rate

1. Annual mortgage constant (or mortgage capitalization rate, R_M)
 - a. The mortgage capitalization rate (R_M) is also known as the annual mortgage constant. It is different from the mortgage yield rate (Y_M). The mortgage yield rate (Y_M) is the lender's return on the money borrowed, while the mortgage capitalization rate (R_M) is the ratio of the annual principal and interest payments to the amount borrowed.
 - b. Stated another way, the mortgage capitalization rate (R_M) is the percentage of the original loan that is required to be paid (principal and interest) annually. It is the ratio of the total mortgage payments for the year divided by the amount of money borrowed.
 - c. The mortgage capitalization rate can be found by using the compound interest tables, partial payment factor, column 6. If using the monthly compound interest table, the column 6 factor must be multiplied by 12 to make it an annual mortgage constant.
2. Equity dividend rate (or equity capitalization rate, R_E)
 - a. The equity dividend rate or equity capitalization rate (R_E) is the ratio of the annual equity income after debt service to total equity investment.
 - b. Stated another way, the equity dividend rate or equity capitalization rate (R_E) is the percentage of the original equity investment that is represented by the income available to the original equity investment.

3. When developing the capitalization rate using the band-of-investment technique with mortgage and equity components, it does not include an ETR component. Thus, for the assessor/appraiser to correctly use the capitalization rate, the ETR must be added to the rate before it is divided into the net operating income. In developing the Income Approach for taxation, a capitalization rate (R_o) without an ETR is called an "unloaded" cap rate, while a cap rate with an ETR applied is considered "loaded".

Illustration 3-4: Band of Investment Technique

For example, assume the typical mortgage financing requires an 8% loan with monthly payments for 20 years. Looking up the mortgage annual constant in the compound interest tables for this loan indicates a rate of 0.100368.* The equity capitalization rate (R_E) is 12%. Also assume that the debt represents 60% of value and the equity represents 40% of value. The effective tax rate is 2.5%. To determine the overall capitalization rate (R_O) using the band-of-investment methods with debt and equity components proceed as follows:

Financial Components	Percent of Investment		Rate		Product
Debt	0.60	X	0.100368 ¹	=	0.060221
Equity	0.40	X	0.120000	=	0.048000
Totals	1.00		Overall Rate	=	0.108221

¹The annual constant (RM) for debt is the ratio of the total mortgage payments for the year divided by the amount of money borrowed.

Unloaded Overall Capitalization Rate (Without ETR):

$$R_O = 0.1082 \text{ or } 10.82\%$$

Loaded Overall Capitalization Rate (With ETR):

$$R_O = 0.108221 + 0.025 = 0.133221 \text{ or } 13.32\%$$

* NOTE: The mortgage annual constant or mortgage capitalization rate (RM) can be computed with a financial calculator or found in the compound interest tables. For example, in the compound interest tables the partial payment factor (column 6) for a 20-year loan at an 8% interest rate with monthly payments is 0.008364. Because this partial payment factor is for monthly compounding it must be multiplied by 12 to get the total annual constant of **0.100368**. This means that 12 times the monthly partial payment factor is the annual amount that must be paid each year on the loan to fully amortize it over a 20-year (240-month) period.

Exercise 3-4: Development of an R_o Band of Investment Technique Using Mortgage & Equity Components

You have determined that typical properties, like the subject you are appraising, are financed with 75% debt and 25% equity. From discussions with lenders and investors you have found that typical financing terms for this type of property require 9.3% of the amount borrowed be paid annually to amortize the loan. The typical equity dividend rate (or equity capitalization rate – R_E) would be 10% for the equity investor. The subject property has an effective tax rate of 2.5%.

From this information develop the overall unloaded capitalization rate (R_o) applicable to the subject property.

Exercise 3-5: Development of an R_o Band of Investment Technique Using Mortgage & Equity Components

You have been asked to develop an overall capitalization rate (R_o) for use in a commercial appraisal. A recent market study shows that similar commercial properties are financed with 80% debt and 20% equity. From discussions with investors and lenders, you have determined investors require an equity capitalization rate (R_E) of 6% for properties such as the subject and lenders require an 8% yield on loans of this type of mortgage, financed for 25 years with monthly payments. From this information you know that the mortgage capitalization rate (R_M) is 0.092616. The effective tax rate for the jurisdiction is 2%. [*You determined this mortgage capitalization rate (R_M) by looking in the compound interest tables or using a financial calculator – discussed in the next chapter*]

From this information develop the overall unloaded capitalization rate (R_o) applicable to the subject commercial property.

PAGE LEFT BLANK INTENTIONALLY

E. Net income ratio method

1. Based on dividing the net operating income ratio (NIR) by the effective gross income multiplier (EGIM).

$$\text{NIR} / \text{EGIM} = \text{Capitalization Rate}$$

2. The net operating income ratio is 100 % minus the operating expense ratio. Following are two formulas to find the NIR:

$$\text{NIR} = 100\% - \text{Operating Expense Ratio}$$

$$\text{NIR} = \text{NOI} / \text{EGI}$$

3. To correctly use this capitalization rate, the assessor/appraiser must be sure how the property taxes were handled in the net operating income calculation. If the operating expenses and thus the OER did not include the property tax as a deduction, the resulting capitalization rate already has an ETR component and no adjustment is necessary. However, if the property tax was included in the expenses, the appropriate ETR must be added to the R_o before it is divided into the subject property's NOI to determine value.
4. To correctly use this capitalization rate, the assessor/appraiser must be sure how the property taxes were handled in the net operating income calculation.
 - a. If the operating expenses and thus the OER **did not include** the property tax as a deduction, the resulting capitalization rate already has an ETR (loaded) component and is appropriate for use in taxation.
 - b. However, if the property tax **was included** in the expenses, the indicated R_o is "unloaded" and appropriate for non-assessment valuation. The appropriate ETR would need to be added to the R_o if used in assessment to arrive at a "loaded" cap rate.

Demonstration 3-2: Development of an Overall Capitalization Rate (R_o) by the Net Income Ratio Method

For example, assume the effective gross income for a commercial property is \$234,000 and the operating expenses for similar properties typically amount to 40 % of effective gross income. Also assume the property sold recently for \$1,123,200. What is the overall capitalization rate?

The net income ratio is 0.60 which is obtained by deducting the operating expense ratio of 40% from 100%. The effective gross income multiplier of 4.80 is developed by dividing the sales price of \$1,123,200 by the effective gross income of \$234,000.

$$4.80 = 1,123,200 \div 234,000$$

With this information the overall capitalization rate (R_o) can be computed as follows:

$$R_o = 0.60 \div 4.80 = 0.125 \text{ or } 12.5\%$$

Exercise 3-6: Development of an R_o Using the Net Income Ratio Method

Using the given market data, answer the following questions.

A 10-unit apartment complex is receiving market rents of \$1,500 per unit per month. Vacancy and collection losses are projected to be 6%. Annual total operating expenses are forecast to be \$53,721. The property recently sold for \$1,100,000

1. What is the potential gross income?
2. What is the effective gross income?
3. What is the net operating income?
4. What is the operating expense ratio?
5. What is the net income ratio?
6. What is the effective gross income multiplier?
7. What is the overall capitalization rate?

PAGE LEFT BLANK INTENTIONALLY

Exercise 3-7: Development of an R_o Using Net Income Ratio Method

Develop the overall capitalization rate (R_o) from the following information which you have developed from sales comparables in the area of your subject property:

Potential gross income (PGI)	\$1,545,000
Vacancy and collection loss rate (V&C)	5%
Operating Expenses	\$543,068
Effective gross income multiplier (EGIM)	7.0

Note: Typically, the operating expense ratio is computed as percentage of effective gross income (EGI). However, the expense ratio can be computed as a percentage of potential gross income (PGI). It is important for the appraiser to be consistent in application. If the operating expense ratio is computed as a percentage of potential gross income, the potential gross income multiplier must be used. Alternatively, if the operating expense ratio is computed as a percentage of the effective gross income, the effective gross income multiplier must be used.

PAGE LEFT BLANK INTENTIONALLY

F. Debt coverage ratio method

1. Is often used by lending institutions that are sensitive to the safety of their loan investments. The lenders are concerned with profits expected by the property and providing a safety margin so that the borrower will be able to meet debt service obligations on the amount loaned on the property.
2. Formula is simple to use: $R_O = DCR \times M \times R_M$ where DCR equals the debt coverage ratio, M equals the loan-to-value ratio, and R_M equals the mortgage capitalization rate.
3. Formula for developing the debt coverage ratio (DCR) is:

$$DCR = \frac{NOI}{I_M}$$

- a. NOI = Net Operating Income
 - b. I_M = Income necessary to satisfy mortgage payments (annual debt service). This can also be computed by multiplying R_M times the amount borrowed.
 - c. The DCR expresses the relationship between NOI and total annual debt service (principal and interest).
4. Use of the DCR methodology does not normally include the ETR component. This is simply because most bankers or lending institutions develop their DCR calculations after property taxes have been deducted.

Illustration 3-5: Debt coverage ratio method

To illustrate, assume that a property's net operating income is \$700,000 and the annual debt service is \$511,740. The debt coverage ratio is calculated as follows:

$$\text{DCR} = \$700,000 \div \$511,740 = \mathbf{1.3679}$$

If a mortgage can be obtained to finance 75% of the value of the property with an annual mortgage requirement of 11.19% of the amount financed, the overall rate is calculated as follows:

$$R_o = 1.3679 \times 0.1119 \times 0.75$$

$$R_o = 0.1148$$

$$R_o = \mathbf{0.11 \text{ (rounded)}}$$

Exercise 3-8: Development of an R_o Using Debt Coverage Ratio Method

You have determined that a commercial property can expect a net operating income of \$930,000 from an analysis of market information. Further, it is expected that the annual debt service for the property will be \$620,000. You have determined that 75% of the property's value can be financed by debt with an annual mortgage requirement of 8% of the amount financed.

What is the overall capitalization rate applicable to this property?

PAGE LEFT BLANK INTENTIONALLY

Exercise 3-9: Development of an R_o Using Debt Coverage Ratio Method

Based on research and market analysis, you have learned that mortgage lenders in your area require a debt coverage ratio of 1.4 for the particular type of commercial property you are asked to appraise. The lenders have noted they will typically finance 60% of the investment for a term of 20 years in these commercial ventures, and the lenders require a 10% yield on their loans, which require level monthly payments of principal and interest. By using a financial calculator, you have determined that a 10% loan for 20 years with level monthly payments indicates a mortgage annual constant, or mortgage capitalization rate – R_M , of 11.58%.

Based upon the above data, what would be the indicated overall capitalization rate (R_o) indicated for the subject commercial property?

PAGE LEFT BLANK INTENTIONALLY

G. Band of investment technique using land and building components

1. Must know the land capitalization rate (R_L) and building capitalization rate (R_B). The land capitalization rate is comprised of the overall yield rate and effective tax rate. The building capitalization rate consists of the overall yield rate, recapture rate and effective tax rate.
2. Computed weighted average of land and building rates – The respective land and building rates are weighted by the percentage of value represented by land and building, respectively.
3. If both the land and improvement rates include the appropriate ETR, then the resulting overall capitalization rate also would include the ETR component.

Component	Portion		Rate		Product
Land	%	X	Land Capitalization Rate	=	Weighted #
Building	%	X	Building Capitalization Rate	=	Weighted #
Total	1.00			=	Overall Cap Rate

Illustration 3-6: Band of Investment, Land & Building

For example, assume the land capitalization rate (R_L) is 10% and the improvement capitalization rate (R_B) is 14%. Assume further that the land represents 25% of value and the improvements represent 75% of value. To determine the overall capitalization rate (R_O) using the band-of-investment method, proceed as follows:

Property Component	Percent of Investment		Cap Rate		Product
Land	0.25	X	Land Capitalization Rate	=	0.025
Improvement	0.75	X	Building Capitalization Rate	=	0.105
Total	1.00		Overall Rate (R_O)	=	0.130

Exercise 3-10: Development of an R_o Band of Investment Technique Using Land & Building Components

You are developing overall capitalization rates for retail stores as a part of your mass appraisal process. You have determined that typical retail developments similar to the stores you are appraising are made up of land and building components of 30% and 70%, respectively.

From market extraction techniques, you have learned that the land and building capitalization rates for similar properties are 8% and 12%, respectively.

Determine the overall capitalization rate using the band-of-investment method (weighted average of land and building capitalization rates).

PAGE LEFT BLANK INTENTIONALLY

Exercise 3-11: Development of an Overall Capitalization Rate (R_o) by Various Methods

Given the following information:

Sales Price	\$6,000,000
Land Value	\$1,250,000
First Mortgage (60% of total value)	60%
Equity Cap Rate	12.5%
Net Operating Income	\$660,000
Annual Mortgage Constant (R_M)	10%
Effective Gross Income	\$1,200,000
Operating Expenses Ratio	45%
Effective Tax Rate (ETR)	2%

Compute the overall unloaded capitalization rate (R_o) by using:

- Derivation from comparable sales (IRV)
- Debt coverage ratio method
- Net income ratio method
- Band of investment technique (mortgage & equity components)

PAGE LEFT BLANK INTENTIONALLY

H. Using Market Surveys as a Source of Capitalization Rates

Market surveys provide sources of information indicating rates of return and other variables for income-producing properties. Personal surveys done by the appraiser or others from the appraiser's office are usually the best source of investment data. These provide primary information coming directly from the buyer or others familiar with the market. Third party surveys from commercial services are helpful with a broad range of property types. Survey data can be used to supplement and test for reasonableness the previously discussed five primary methods of developing an overall capitalization rate (R_o).

1. Personal Surveys

Surveys done by the assessor's office serve as an excellent source of local activity. Individual investors offer their market expectations and rationale when purchasing income-producing properties. The motivations of buyers typically center on profit through cash flows. Most investors anticipate what they will realize in terms of annual NOI. Interviewing buyers and sellers involved with recent sales of income-producing properties can provide a wealth of information and market insight.

2. Third Party Surveys

Reports prepared by outside agencies can provide a good overview of certain types of property from a regional or national perspective. Care must be taken to thoroughly understand exactly what is being reported by these services. Published sources include real estate appraisal or brokerage firms, investment firms, and subscription services. The survey data can be obtained at no cost, paid subscription, and/or contracted out by an assessor's office to study a specific jurisdiction.

IV. Capitalization Rate Individual Components

The capitalization rate used in direct capitalization consists of three parts: overall yield rate (discount rate), recapture rate and effective tax rate. Development of all three parts will be reviewed in this section.

A. Overall Yield Rate (Discount Rate):

1. Development of the property's overall yield rate (Y_o) can be:
 - a. Developed by survey and opinion data from owners of similar property.
 - b. Extracted from comparable sales.
 - c. Derived from alternative investments of comparable risk.
 - d. Developed by band-of-investment technique using mortgage interest rate and equity yield rate.
2. The components of the overall yield rate are:
 - a. Safe Rate
 - b. Risk Rate
 - c. Non-Liquidity Rate
 - d. Management Rate
3. Band of Investment Method

The overall yield rate (Y_o) is the weighted average of the debt and equity rates, weighted by the respective proportions of total property value that each represents.

Exercise 3-12: Extracting the Property's Overall Yield Rate (Y_o) by the Band of Investment Method

You have been asked to derive the overall yield rate (Y_o) for a public utility property that is expected to be financed with interest only debt requiring a balloon payment at maturity to pay off the mortgage. Based on interviews with owners of this property type, financing is done with long-term bonds. Lenders have informed you that their yield requirement for this type of debt is 8% and you know that equity investors require a 12% yield for this level of risk.

Market information indicates that investors in this type of property typically finance 40% of this type of purchase with debt and 60% with equity. Further, there is no indication that there will be a change in value or income during the holding period because of regulatory restrictions.

What is the indicated overall yield rate (Y_o) for this investment?

PAGE LEFT BLANK INTENTIONALLY

B. Recapture Rate

The annual rate at which invested capital (improvement) is returned to the investor over a specified period. It is the ratio between the annual amount being recaptured and the original amount invested. The recapture rate can be developed using the remaining economic life technique (reciprocal of the remaining economic life) and the market comparison method (which will be discussed in Chapter 5).

1. Remaining Economic Life Technique

The remaining economic life technique requires that an estimate be made about the number of years the building will continue to produce income. The formula is:

$$\text{Recapture rate} = 1 \div \text{Remaining Economic Life}$$

OR

$$\text{Recapture rate} = \frac{1}{\text{Remaining Economic Life}}$$

Therefore, the remaining economic life for a property can be calculated using the recapture rate. The formula is:

$$\text{Remaining Economic Life} = 1 \div \text{Recapture rate}$$

OR

$$\text{Remaining Economic Life} = \frac{1}{\text{Recapture rate}}$$

Remember that if you know how to find one variable in an equation, you can find the others. Recall that if $a = b \div c$, then $c = b \div a$, and $b = a \times c$

For example, $0.25 = 1 \div 4$, then $4 = 1 \div 0.25$, and $1 = 0.25 \times 4$

It works the same way with these formulas, too.

Illustration 3-7: Recapture Rate

Assume that a study of comparable properties and information obtained from investors and lenders indicate that the remaining economic life of the subject building is 25 years and the improvements are expected to depreciate on a straight-line basis.

The indicated annual recapture rate for the building would then be:

$$1 / 25 = 0.04 \text{ or } 4\%$$

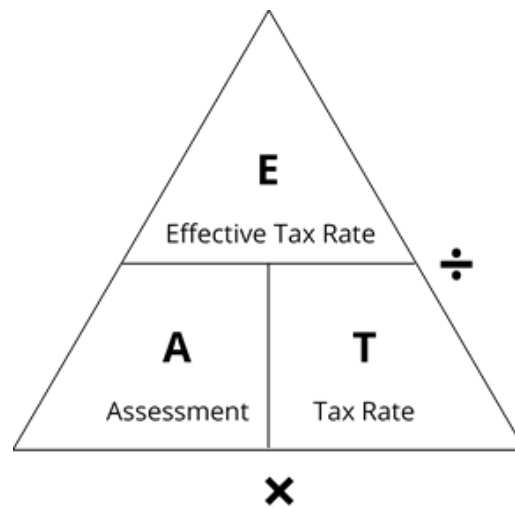
2. Market Comparison Technique

This method is based on use of the IRV formula. If the recapture income can be isolated from the property's NOI, then the recapture rate can be developed using the IRV formula relationships. (This technique will be discussed in more detail in Chapter 5.)

C. Effective Tax Rate

1. This is the percentage that annual real estate taxes represent in relation to total property value. When the level of assessment is 100% of appraised value or market value, the effective tax rate (ETR) and the tax rate are same. Where assessments are less than 100% (fractional assessments) then the effective tax rate is the ratio of tax rate to assessment level.
2. The effective tax rate can be computed by two methods – the EAT formula and market comparison.

a. EAT formula

i. **E** = the effective tax rate (ETR)

The income approach considers real estate taxes as a proper expense. However, the appraiser for ad valorem tax purposes is establishing value for this purpose. Therefore, it would be incorrect to deduct a predetermined amount for real estate tax based on a prior appraisal. Instead, the appraiser for ad valorem tax purposes should not deduct the current real estate tax liability from the operating income but instead develop an effective tax rate and include it as a component in the overall capitalization rate.

ii. **A** = the assessment level (assessment ratio)

The level of assessment is the ratio of assessed value to full market value. In many jurisdictions, the assessed value (the taxable value of the property) is equal to the appraised value (full value or market value) of the property. In other jurisdictions, the assessed value is a fractional amount of the appraised value as set by state statutes. When the appraised value is the same as the assessed value, the level of assessment is 100% of the appraised value. When the assessment level is a fractional amount of the appraised value, the level of assessment is less than 100%.

iii. T = the tax rate

Tax rate structures vary from jurisdiction to jurisdiction. Some jurisdictions refer to the tax rate in terms of mill rate or dollars per thousand of assessed valuation, while other jurisdictions refer to the tax rate in terms of dollars per hundred of assessed valuation.

Tax rates, as dollars per hundred or as mill rates, can be converted to percentages.

- Dollars per hundred are converted to a percentage by dividing by 100. This accomplishes the same thing as moving the decimal point two places to the left.
- Mill rate is divided by 1,000. This accomplishes the same thing as moving the decimal point three places to the left.
- When applying the tax rate, the appraiser must always express the rate as a decimal.
- The same tax rate can be expressed in a variety of ways. The following example uses an 8% tax rate.

\$8 per \$100	$(8 \div 100 = 0.08)$
\$80 per \$1,000	$(80 \div 1,000 = 0.08)$
\$0.08 per \$1	$(0.08 \div 1 = 0.08)$
80 mills per \$1	$(80 \div 1,000 = 0.08)$

b. Market comparison - like a recapture rate, the effective tax rate can be derived from market sales transactions.

- Uses IRV equation
- Known components of the IRV equation are:
 - Real estate taxes
 - Property value
- Real taxes $\div$ the market value = effective tax rate.

Illustration 3-8: Market Comparison

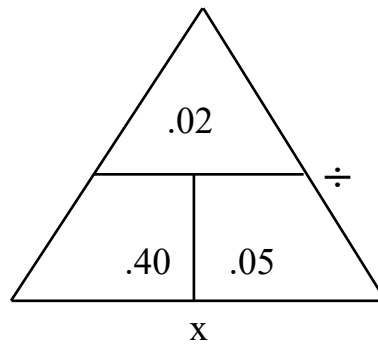
The effective tax rate can also be obtained when the dollar amount of the real estate taxes and the appraised or market value of the property are known. This is accomplished by dividing the dollar amount of real estate taxes by the appraised or market value of the property.

Current taxes	\$4,000
Property value	\$200,000
Effective tax rate	$(\\$4,000 \div \\$200,000) = 0.02$ or 2.0%

Demonstration 3-3: EAT Equation

The assessment level is 40% of appraised value and the current tax rate is \$5.00 per hundred. This provides an effective tax rate as follows:

Assessment level	0.40
Tax rate	$\times \quad \underline{0.05}$
Effective tax rate	0.02 or 2.0%



Exercise 3-13: How Tax Rates are Expressed

No.	Tax Rate	Decimal Equivalent
1	\$65.00/\$1,000	
2	72 Mills	
3	\$8.00/\$100	
4	\$80.00/\$1,000	
5	115 Mills	
6	\$4.50/\$100	
7	62.5 Mills	
8	\$75.25/\$1,000	
9	\$12.75/\$100	
10	\$6.25/\$100	

Exercise 3-14: Development of the Effective Tax Rate

1. Complete the following table of equivalencies:

Mills		\$ per \$100		Nominal Tax Rate*
	=	2.60	=	
35	=		=	
	=		=	0.0100
	=	7.25	=	0.0725
6	=		=	
	=		=	0.0150

*Express as decimal

2. Complete the following table. Utilize the EAT triangle formula.

Assessment Level	Nominal Rate \$ per \$100	Mills	Nominal Tax Rate	Effective Tax Rate
0.25	\$5.50			
		60.0		0.03600
0.35				0.02975
0.50		62.5		
0.40		70.0		0.02800
	\$7.75			0.02325

3. Complete the following table:

Market Value	Tax Dollars	Effective Tax Rate*
\$560,000	\$12,880	
\$900,000		0.033
\$680,000	\$19,040	
	\$32,500	0.025
\$940,000		0.026
	\$23,100	0.035

*express as decimal

Exercise 3-15: Development of the Effective Tax Rate by Market Comparison**A. Market Comparison**

The Smith's property sold recently for \$360,000. The assessment level in this area is 50%. The prior tax bill for the Smith's was \$5,400. What is the effective tax rate?

B. EAT Formula

The Jackson's property is located in an area of town that was appraised last year. The current assessment level is 50%. The current tax rate is \$3.00 per hundred. What is the effective tax rate?

PAGE LEFT BLANK INTENTIONALLY

Exercise 3-16: Development of the Effective Tax Rate by Market Comparison

You are developing effective tax rates for several tax jurisdictions. You have found the following sales with the taxes paid in the latest year for each sale.

Tax Jurisdiction: North Side

Sale A sold recently for \$420,000 and had an annual tax bill of \$7,560.

Sale B sold one month ago for \$375,000. The annual taxes were \$6,750.

Tax Jurisdiction: South Side

Sale C sold recently for \$165,000 and had an annual tax bill of \$2,063.

Sale D sold 18 days ago for \$184,000. The annual property taxes were \$2,300.

What were the effective tax rates for each sale?

PAGE LEFT BLANK INTENTIONALLY

REVIEW QUESTIONS – Chapter 3

1. Name five methods of developing an overall capitalization rate (R_o).

- _____
- _____
- _____
- _____
- _____

2. The summation method of estimating an overall yield rate (Y_o) [property discount rate] is a theoretical method which is based upon the sum of four components that affect the return on or requirements of an investment. The four components are:

- _____
- _____
- _____
- _____

3. When developing the overall capitalization rate from derivation from comparable sales, three critical items which must be highly comparable between comparable sales and the subject property are:

- _____
- _____
- _____

4. Another name for the mortgage capitalization rate (R_M) is the _____.
5. The _____ technique computes a weighted average of the mortgage capitalization rate (R_M) and equity capitalization rate (R_E). The resulting weighted average is called the overall capitalization rate (R_O).
6. The capitalization rate method most often use by lending institutions is the _____ method.
7. The _____ is the percentage that annual real estate taxes are in relation to total property value.
8. Name the two methods of developing an effective tax rate.
- _____
 - _____
9. The net income ratio method can be used to develop an overall capitalization rate when the net income ratio is divided by the _____.
10. What are the three components of an overall capitalization rate?
- _____
 - _____
 - _____

11. The _____
_____ for developing an overall capitalization rate is appropriate when the effective gross income multiplier is known.
12. The _____
_____ reflects the relationship between the land income and land value.
13. The weighted average of the land capitalization rate and the building capitalization rate is known as the: _____

_____.
14. Dividing the building (improvement) income by the _____
_____ will result in the building (improvement) value.
15. The assessment level is 30% of appraised value and the current tax rate is \$8.50 per hundred. What is the effective tax rate expressed as a decimal and a percentage? _____ or _____ (rounded).

PAGE LEFT BLANK INTENTIONALLY

Chapter 4 | Contemporary Capitalization Methods

APPENDIUM KNOWLEDGE AREAS



SUGGESTED READING

Source	Chapter	Pages
Property Assessment Valuation	Chapter 13	361-370
	Chapter 14	375-377

CHAPTER TOPICS AND TIMETABLE

Topic	Time	Percentage
Capitalization Methods	30 minutes	11
Application of Direct Capitalization	3 hours and 20 minutes	71
Introduction to Compound Interest Tables	30 minutes	11
Review Questions	20 minutes	7

OBJECTIVES

Upon completion of Chapter 4, the student should be able to:

- Define direct capitalization.
- Identify the mathematical relationships between income and value used in direct capitalization and explain how they are indicated in sales transactions and other market evidence.
- Define, calculate, and apply gross income multipliers.
- Derive the overall capitalization rate.
- Convert income estimates to value using direct capitalization method.
- Define yield capitalization.
- Differentiate between yield capitalization and direct capitalization.
- Define partial payment factor and apply it correctly in the development of a capitalization rate using band-of-investment (mortgage-equity) method.
- Identify uses of the compound interest tables for solving common appraisal problems.
- Differentiate between an effective interest rate and a nominal interest rate.

Chapter 4 | Contemporary Capitalization Methods

I. Capitalization Methods

- A. **Direct capitalization** - a method of converting an estimate of a single year's income into value in one direct step. This conversion is made in one direct step, either by dividing the income estimate by an appropriate rate or by multiplying the income by an appropriate income factor.
- B. **Yield capitalization** (discounting) - a method of converting future net benefits into present value where each future net benefit is discounted at a proper yield rate (discount rate). Yield capitalization can also be accomplished by developing an overall rate that specifically reflects the investment's pattern of income, change in value, and yield rate. Yield capitalization is discussed in IAAO Course 112.
- C. Direct capitalization is different from yield capitalization because direct capitalization does not directly consider the individual cash flows beyond the first year. However, yield capitalization explicitly calculates year-by-year effects of each year's cash flow, changes in income patterns, changes in the original investment's value, and other considerations. Yield capitalization is also more intensive to apply, requires additional labor and several significant assumptions such as the length of ownership and other data that is often difficult to gather for mass appraisal.
- D. Either direct capitalization or yield capitalization will produce a proper indication of value when correctly applied. For direct capitalization to be applied correctly, it must be based upon relevant market information derived from comparable properties, which should have similar income-expense ratios, land value-to-building value ratios, risk characteristics, and future expectations of income and value changes over a typical holding period.
- E. Characteristics of direct capitalization
 - 1. Converts a single year's income estimate into value indication.
 - 2. Rarely distinguishes between return of and return on investment.
 - 3. Does not identify investor assumptions or forecasts of the holding period, pattern of income, or changes in the value of the original investment.

4. Direct capitalization may be used on various income levels some of which are: potential gross income, effective gross income, net operating income, land income, and improvement income.

F. The generic formulas used in direct capitalization are:

- a. **Value = Income ÷ Rate (IRV)**
- b. **Value = Income x Factor (VIF)**

II. Application of Direct Capitalization

A. Overall capitalization rate (R_o)

1. Without property tax component

- a. When property is valued for ad valorem tax purposes, property taxes should not be shown as an operating expense because the actual taxes are not known as of the assessment date. In mass appraisal, the intended use of the appraisal is typically to establish value for ad valorem property tax purposes.
- b. The problem can be resolved by developing an effective tax rate (as shown in Chapter 3 of this manual and Chapter 13 of *Property Assessment Valuation*) and adding it to the overall capitalization rate (R_o) for the subject property.

2. With a property tax component

- a. The ad valorem tax appraiser must be aware of and correctly match the appropriate net operating income (which includes an amount to pay property taxes) with an overall capitalization rate (which includes an effective property tax rate component).
- b. Whatever income is in the numerator must also be included in the rate in the denominator. (See Chapter 13 of *Property Assessment Valuation* regarding matching of numerator and denominator.)

3. Direct capitalization of land

a. Required comparability criteria

- i. Amenities – similar locations, restrictions, investor (economic) desirability, utility, etc.
- ii. Expense ratios – Even if the amenities are similar, if the expense ratios are different for the comparables as compared to the subject, a proper overall rate cannot be derived.

b. Effective tax rate

- i. Included (loaded)
- ii. Excluded (unloaded)

Demonstration 4-1: Direct Capitalization of Land

You are appraising two unimproved storage lot properties in a core metropolitan area for an assessment office. The expense analysis excludes real estate taxes as an expense, as the values will be used for taxation. The exclusion of taxes as an expense will be offset by loading the capitalization rate with the effective tax rate for the area.

Subject Property 1 contains 21,500 square feet, has a rectangular shape, level topography, and an operating expense ratio of 10% (without taxes). Subject Property 1 has a net operating income of \$70,950 (excluding taxes as an expense) and is located in a jurisdiction that has an effective tax rate of 2%.

Subject Property 2 contains 27,000 square feet, has an irregular shape, level topography, and an operating expense ratio of 15% (without taxes). Subject Property 2 has a net operating income of \$98,280 (excluding taxes as an expense) and is located in a jurisdiction that has an effective tax rate of 2%.

You need to develop an overall rate to apply to your subjects' net operating incomes. A search of sales data produced four recent sales of parking lots that will not require any adjustment for market conditions. The four sales were confirmed as arms-length transactions, sold with typical financing, and were leased at current market rent.

The net operating incomes for the comparable sales were reported to include real estate taxes as an expense. To better compare the data with the subject's estimates, the real estate tax expense has been excluded from the expense ratio calculation. Data on the four sales is shown in the following grid.

	Sale A	Sale B	Sale C	Sale D
Sale Price	\$1,300,000	\$1,200,000	\$1,600,000	\$1,100,000
Square Foot Area	21,258	22,027	31,934	21,875
Shape	Rectangular	Irregular	Irregular	Rectangular
Topography	Level	Level	Level	Level
Operating Expenses Ratio (OER) <i>Excludes Taxes</i>	10.25%	14.5%	15.25%	9.75%
Net Operating Income (NOI) <i>Includes Taxes</i>	\$51,000	\$59,000	\$81,000	\$45,000

Step 1) Calculate the overall rate (R_o) for each sale.

Since the net operating income (NOI) of the sales *includes* real estate taxes as an expense, the indicated overall rate (R_o) is considered an "unloaded" capitalization rate. "Unloaded" indicates it is not loaded with an effective tax rate, which is most commonly reported by market participants and data sources.

	Sale A	Sale B	Sale C	Sale D
Net Operating Income (NOI)	\$51,000	\$59,000	\$81,000	\$45,000
÷ Sale Price	\$1,300,000	\$1,200,000	\$1,600,000	\$1,100,000
= Overall Rate (R_o Unloaded)	0.0392	0.0492	0.0506	0.0409

Step 2) Identify which comparable sales are most similar to the subject properties being appraised that can be weighted in the capitalization rate conclusion.

Property Data	Sale A	Sale D	Subject 1
Square Foot Area	21,258	21,875	21,500
Shape	Rectangular	Rectangular	Rectangular
Topography	Level	Level	Level
Operating Expenses Ratio (OER) <i>Excludes Taxes</i>	10.25%	9.75%	10.00%
Overall Rate (R_o , Unloaded)	0.0392	0.0409	0.0400

Subject Property 1 has a rectangular shape, level topography, and an operating expense ratio of 10.00% (without taxes), then an unloaded overall rate consistent with Sales A and D would be appropriate, or 0.04 (4%), rounded.

Property Data	Sale B	Sale C	Subject 2
Square Foot Area	22,027	31,934	27,000
Shape	Irregular	Irregular	Irregular
Topography	Level	Level	Level
Operating Expenses Ratio (OER) <i>Excludes Taxes</i>	14.50%	15.25%	15.00%
Overall Rate (R_o , Unloaded)	0.0492	0.0506	0.0500

Subject Property 2 contains 27,000 square feet, has an irregular shape, level topography, and an operating expense ratio of 15% (without taxes), then an unloaded overall rate consistent with Sales B and C would be appropriate, or 0.05 (5%), rounded.

Step 3) Add the subjects' effective tax rates (ETR) to the unloaded overall capitalization rates (R_o) to arrive at the "loaded" overall rate (R_o).

A capitalization rate must be loaded with the effective tax rate when the expenses exclude real estate taxes. If the effective tax rate (ETR) for the two subject properties is 2%, then the loaded overall capitalization rates (R_o) would be 6% (4% + 2%) and 7% (5% + 2%), respectively for the two subject properties.

Step 4) Apply the IRV formula ($\text{Income} \div \text{Rate} = \text{Value}$)

NOI Without Taxes			Loaded Cap Rate		Value
Subject Property 1	\$70,950	÷	0.06	=	\$1,182,500
Subject Property 2	\$98,280	÷	0.07	=	\$1,404,000

Exercise 4-1: Direct Capitalization of Land

Your assignment is to appraise two unimproved storage lot properties that are in the same general location and in the same tax area. You have found three recent sales not requiring any adjustment for market conditions. The three sales were confirmed as arms-length transactions, sold with typical financing, and were leased at current market rent.

However, the comparable sales are not in the same tax area as the subject. The net operating income confirmed with the sellers *includes* the real estate tax expense. To better compare the data with the subject's estimates, the real estate tax expense has been *excluded* from the expense ratio calculation.

The subjects' NOIs have been estimated to *exclude* real estate taxes because the valuations are for ad valorem purposes.

Develop an overall capitalization rate that can be applied to the subject properties and calculate an estimate of value using direct capitalization.

Data on the three sales is as follows:

	Sale 1	Sale 2	Sale 3
Sale Price	\$1,950,000	\$1,800,000	\$2,400,000
Square Foot Area	31,850	33,000	46,500
Shape	Irregular	Irregular	Rectangular
Topography	Level	Level	Level
Operating Expenses Ratio (OER) <i>Excludes Taxes</i>	21%	19%	15%
Net Operating Income (NOI) <i>Includes Taxes</i>	\$126,000	\$118,000	\$143,000

Data on the subject properties is as follows:

	Subject A	Subject B
Square Foot Area	32,400	45,000
Shape	Irregular	Rectangular
Topography	Level	Level
Operating Expenses Ratio (OER) <i>Excludes Taxes</i>	20%	15%
Net Operating Income (NOI) <i>Excludes Taxes</i>	\$161,500	\$183,600
Effective Tax Rate (ETR)	0.02	0.02

4. Direct capitalization of improved properties
 - a. Required comparability criteria
 - i. Amenities – similar locations, restrictions, investor (economic) desirability, utility, etc.
 - ii. Expense ratios
 - iii. Remaining economic life
 - iv. Land-to-improvement (building) ratios
 - b. Effective tax rate
 - i. Included (Loaded)
 - ii. Excluded (Unloaded)

Demonstration 4-2: Direct Capitalization of Improved Property

You are appraising an apartment building for tax purposes. You have conducted a market analysis and established market rent for the subject. You have also reconstructed the income/expense statement, which provided a net operating income of \$405,000 and an operating expense ratio of 36%, exclusive of real estate taxes as an expense. Land in this area represents 25% of a property's total value. Your estimate of the remaining economic life for this property is 14 years. The effective tax rate for the subject is 0.020.

You have found three recent sales of similar apartment complexes that do not require any adjustment for market conditions. The three sales were confirmed as arms-length transactions and sold with typical financing. A market analysis found the properties rented at market rent and you have reconstructed the income/expense statements for each comparable.

The net operating incomes for the comparable sales were reported to include real estate taxes as an expense. To better compare the data with the subject's estimates, the real estate tax expense has been excluded from the expense ratio calculation. Data on the three comparable sales is as follows:

	Sale 1	Sale 2	Sale 3
Sale Price	\$5,760,000	\$5,610,000	\$5,900,000
Land-Improvement Value Ratio	1:3	1:3	1:4
Remaining Economic Life	22	15	20
Operating Expenses Ratio (OER) <i>Excludes Taxes</i>	40%	35%	40%
Net Operating Income (NOI) <i>Includes Taxes</i>	\$430,000	\$391,000	\$445,000

Analyze the three comparables and select an appropriate capitalization rate for the subject property. In selecting the overall rate, it is necessary to consider the similarities and differences between the subject and the comparables.

Step 1) Calculate the overall rate (R_o) for each sale. Since the net operating income (NOI) includes real estate taxes as an expense, the indicated overall rate (R_o) is considered an "unloaded" capitalization rate, which is most commonly reported by market participants and data sources.

	Sale 1	Sale 2	Sale 3
Net Operating Income (NOI)	\$430,000	\$391,000	\$445,000
÷ Sale Price	\$5,760,000	\$5,610,000	\$5,900,000
= Overall Rate (R_o , Unloaded)	0.0747	0.0697	0.0754

Step 2) Identify which comparable sale(s) is most similar to the subject property being appraised that can be weighted in the capitalization rate conclusion. Similar items of comparability are noted in bold.

	Sale 1	Sale 2	Sale 3	Subject
Land-Improvement Value Ratio	1:3	1:3	1:4	1:3*
Remaining Economic life	22	15	20	14
Operating Expense Ratio (OER)	40%	35%	40%	36%
= Overall Rate (R_o , Unloaded)	0.0747	0.0697	0.0754	0.0700

* Land Value = 25% of total value; $1 \div 0.25 = 4$ total parts, or 1:3 ratio

- The subject's land-to-improvement ratio is the same as Sale 1 and Sale 2.
- The subject's remaining economic life is similar to Sale 2.
- The subject's expense ratio is similar to Sale 2.
- Considering all the items of comparability, Sale 2 is the most similar to the subject.

With weight on Sale 2, an unloaded capitalization rate of 0.07 (7%) is appropriate for the subject.

Step 3) Add the subject's effective tax rate (ETR) to the unloaded overall rate (R_o) to arrive at the loaded overall rate (R_o). If the effective tax rate (ETR) for the subject property is 2%, then the loaded overall rate (R_o) would be 9% (7% + 2%).

Step 4) Apply the IRV formula (Income ÷ Rate = Value)

NOI Without Taxes as an Expense			Loaded Cap Rate		Value
Subject Property	\$405,000	÷	0.09	=	\$4,500,000

Exercise 4-2: Direct Capitalization of Improved Property

The local assessor's office requests a value for an apartment complex in the jurisdiction. You have conducted a market analysis and established market rent for the subject. You have also reconstructed the income/expense statement, which provided a potential gross income of \$940,000, vacancy and collection loss of 5% and miscellaneous income of \$7,000. Operating expenses represent 30% of effective gross income with a reserve for replacements and *excluding* real estate taxes.

Land in this area represents 20% of a property's value. Your estimate of the remaining economic life for this property is 23 years. The assessment level is 50% of actual value and the tax rate is 30 mills.

You have found three recent sales of similar apartment complexes that do not require any adjustment for market conditions. The three sales were confirmed as arms-length transactions and sold with typical financing. A market analysis found the properties rented at market rent and you have reconstructed the income/expense statements for each comparable.

The real estate tax expense has been *excluded* from the expense ratio calculation for the comparables, but the effective tax rate for each sale was confirmed.

Develop an income and expense statement for the subject and the comparable properties. Analyze the sale data and select the appropriate unloaded overall capitalization rate for the subject property using the items of comparability. Calculate the loaded capitalization rate for subject and estimate the subject's value using direct capitalization.

Data on the three comparable sales is given on the following page.

	Sale 1	Sale 2	Sale 3
Sale Price	\$7,200,000	\$7,600,000	\$8,000,000
Potential Gross Income	\$900,000	\$980,400	\$1,121,000
Vacancy & Collection Loss	5%	5%	5%
Miscellaneous Income	\$7,800	\$8,220	\$10,400
Operating Expense Ratio (OER excludes taxes)	30%	30%	34%
Land-to-Improvements Ratio	1:4	1:3	1:4
Remaining Economic Life	23 Years	25 Years	27 Years
Effective Tax Rate	1.90%	1.60%	1.75%

Demonstration 4-3: Direct Capitalization with an Overall Rate by the Band of Investment Technique

You are appraising a small office complex with 20,000 square feet of gross area and 1,600 square feet of common area, in an area with very limited sales. However, you are able to obtain income and expense data for this type of property. You have conducted a market study and found market rent to be \$15.00 per square foot of net leasable area. Typical vacancy and collection losses are 5%. Expenses are 35% of effective gross income. The effective tax rate for the subject is 0.010. Using the data obtained from your study, you have reconstructed the income/expense statement. The reconstructed statement is as follows:

Potential Gross Income	\$276,000
Vacancy & Collection Loss (5%)	<u>-\$13,800</u>
Effective Gross Income	\$262,200
Expenses (35%)	<u>-\$91,770</u>
Net Operating Income	\$170,430

Since there are limited sales, you decide to use the Band of Investment method to develop an overall capitalization rate (R_o) because you obtained good financial information in your market study.

Lending institutions will finance 60% of the value of the property at a rate of 8% for 25 years. Equity has been receiving a cash flow (equity dividend rate) of return of 12%. The monthly partial payment factor for an 8% loan for 25 years is 0.007718. In order to obtain the mortgage annual constant or the mortgage capitalization rate the monthly partial payment factor of 0.007718 is multiplied by 12 ($0.007718 \times 12 = 0.092616$).

The overall capitalization rate (without a tax component) is computed as follows:

Financial Component	Percent of Investment		Rate		Product
Debt	0.60	X	0.092616	=	0.055570
Equity	0.40	X	0.12	=	0.048000
Totals	1.00	Overall Rate (R _o)		=	0.103570
Unloaded Overall Capitalization Rate (Without ETR)					0.104
Effective Tax Rate					0.010
Loaded Overall Capitalization Rate (With ETR)					0.114

Indicated value of the subject property: $\$170,430 \div 0.114 = \$1,495,000$

Exercise 4-3: Direct Capitalization with an Overall Rate by the Band of Investment Technique

You are appraising an office complex and have obtained the following market data for the property:

Gross Square Feet	80,000
Common Area	12%
Market Rent	\$18.00 per square foot of net leasable area
Vacancy Rate	10%
Expense Ratio	40%
Effective Tax Rate	0.014

Financing for this type of property can be obtained from local lenders for 70% of the value at a rate of 10%. Market analysis indicates an equity capitalization rate of 15%. The partial payment factor from the compound interest tables (column 6) for a 25-year loan at 10% with monthly payments is .009087.

Determine the indicated value of the subject property using a loaded overall capitalization rate developed by the Band of Investment method.

PAGE LEFT BLANK INTENTIONALLY

Demonstration 4-4: Direct Capitalization with an Overall Rate by Net Income Ratio

An office complex has a potential gross income of \$270,000, with vacancy and collection loss being 10% and expenses 40% of effective gross income. The net operating income includes income to property taxes.

An analysis of similar complexes provides an effective gross income multiplier of 7.5.

The net income ratio can be obtained by subtracting the expense ratio from 1.00. The subject property has expenses of 40% or an expense ratio of 0.40. Subtracting the expense ratio from 1.00 provides a net income ratio of 0.60.

The net income ratio can also be obtained by dividing the net operating income by the effective gross income. The subject has a potential gross income of \$270,000, less \$27,000 for vacancy and collection loss, providing an effective gross income (EGI) of \$243,000. Subtracting expenses, which are 40% of effective gross income, provides a net operating income (NOI) of \$145,800. Dividing the NOI \$145,800 by the EGI \$243,000 provides a net income ratio of 0.60.

Potential Gross Income	\$270,000
Vacancy & Collection Loss	<u>-\$27,000</u>
Effective Gross Income	\$243,000
Expenses @ 40% of EGI	<u>-\$97,200</u>
Net Operating Income	\$145,800
Effective Gross Income	<u>÷ \$243,000</u>
Net Income Ratio	0.60
Effective Gross Income Multiplier	<u>÷ 7.5</u>
Overall Rate	0.08

The overall capitalization rate is calculated as follows:

$0.60 \text{ (net operating income ratio)} \div 7.5 \text{ (effective gross income multiplier)} = 0.080$
(overall rate)

Indicated value of the subject property: $\$145,800 \div 0.080 = \$1,822,500$

Exercise 4-4: Direct Capitalization with an Overall Rate by Net Income Ratio

A retail store you are appraising has Potential Gross Income of \$260,000. Vacancy and collection loss is 5%. The expense ratio is 35%. The net operating income includes income to property taxes. The effective gross income multiplier for this property is 5.

Develop an overall capitalization rate (R_O) by the Net Income Ratio method and use it to estimate the value of the subject property.

Demonstration 4-5: Direct Capitalization with an Overall Rate by the Debt Coverage Ratio

You are appraising a commercial property in an area that does not have any sales. However, you are able to obtain income and financing information. This allows you to use the Debt Coverage Ratio method to develop an overall capitalization rate (R_o).

Your subject property has a net operating income of \$434,000, which includes income to property taxes.

A property comparable to your subject property is rented at market rent and has a net operating income of \$450,000 with annual debt service of \$360,000. The property was financed with a loan for 70% of its value, at 8%, for 20 years. The annual mortgage constant for this loan is 10%.

In order to develop an overall capitalization rate, the Debt Coverage Ratio must be calculated. It is calculated as follows:

$$\text{\$450,000} \div \text{\$360,000} = 1.25$$

The overall rate can now be calculated by the formula:

$$R_o = \text{DCR} \times R_M \times M$$

The calculation for the overall capitalization rate would be:

$$1.25 \times 0.10 \times 0.70 = 0.0875 \text{ or } 8.75\%$$

The indicated value for the subject property would be:

$$\text{\$434,000} \div 0.0875 = \text{\$4,960,000}$$

Exercise 4-5: Direct Capitalization with an Overall Rate by the Debt Coverage Ratio

Your appraisal assignment is to develop an overall capitalization rate (R_O) that can be used to value a commercial property. The subject property has a net operating income of \$420,000, which includes income to property taxes.

A property comparable to your subject property has a net operating income of \$600,000 with annual debt service of \$400,000. The property was financed with a loan for 70% of its value and a mortgage constant of 11.58%.

Develop an overall capitalization rate (R_O) using the Debt Coverage ratio method. Use this R_O to calculate an indicated value for the subject property.

B. Income multipliers

1. Residential properties (gross rent multiplier)

Residential Formula:

$$\text{Sale Price} \div \text{Monthly Income} = \text{GRM}$$

2. Commercial properties (gross income multiplier)

- a. Income multipliers can be developed using either potential gross income or effective gross income. There are times when one is more appropriate than the other depending on the application. Either application is correct mathematically as long as the appraiser is consistent in deriving and using the income multiplier. Consistency in application here is important.

Commercial Formula:

$$\text{Sale Price} \div \text{Annual Income} = \text{GIM}$$

- b. Normally, the gross income multiplier is based on effective gross income (EGIM) – just like expense ratios are normally based on the relationship between effective gross income and total operating expenses.
- c. However, like expense ratios, it is mathematically correct to base expense ratios on potential gross income as long as the appraiser is consistent in the application of the ratios and multipliers.

Demonstration 4-6: Gross Income Multipliers

Example 1:

You are developing a gross income multiplier (GIM) to use with a subject you are appraising. You have found the following sales and their corresponding potential gross incomes.

Sale Number	Sale Price	Annual Gross Income	GIM
1	\$2,100,000	\$300,000	7.00
2	\$2,245,500	\$320,000	7.02
3	\$2,415,000	\$350,000	6.90
4	\$2,660,000	\$380,000	7.00

The appropriate GIM would be 7.00.

The subject you are appraising has a potential gross income of \$225,000.

Application of the GIM to the subject is as follows: $\$225,000 \times 7.0 = \$1,575,000$

Example 2:

You are developing an effective gross income multiplier (EGIM) to use with a subject you are appraising. You have found the following sales and their corresponding incomes.

Sale Number	Sale Price	Annual Gross Income	Vacancy & Collection Loss	Effective Gross Income	EGIM
1	\$2,400,000	\$420,000	\$20,000	\$400,000	6.00
2	\$2,250,000	\$410,000	\$30,000	\$380,000	5.92
3	\$2,675,000	\$465,000	\$25,000	\$440,000	6.08
4	\$2,880,000	\$510,000	\$30,000	\$480,000	6.00

The appropriate EGIM would be 6.00.

The subject you are appraising has an effective gross income of \$450,000.

Application of the EGIM to the subject is as follows: $\$450,000 \times 6.00 = \$2,700,000$

Exercise 4-6: Development Gross Income Multipliers

A retail store containing 16,000 square feet of gross leasable area is rented for \$16.00 per square foot per year. Vacancy and collection loss is 4%. Allowable expenses are 40% of effective gross income. The land value is estimated to be \$384,000. The store sold recently for \$1,536,000.

Develop a potential gross income multiplier and an effective gross income multiplier.

PAGE LEFT BLANK INTENTIONALLY

III. Introduction to Compound Interest Tables

A. Compound interest tables

1. Expresses the time value of money – a dollar received today is worth more than a dollar to be received in the future.
2. Effective interest rate vs. nominal interest rate
 - a. The effective interest rate is the nominal interest rate divided by the number of conversion periods in one year. For example, if the nominal interest rate is 0.09, the effective interest rate would be 0.0075 for monthly compounding. ($0.09 \div 12 = 0.0075$)
 - b. The effective interest rate is used in all calculations for the compound interest tables.
3. Involves six interest functions which measure present worth and future worth
 - a. Future worth of \$1 is...
 - i. basis of all interest functions
 - ii. also known as the compound amount of \$1
 - iii. reciprocal of the present worth of \$1
 - iv. shows the growth of a single deposit over a specific time period
 - b. Future worth of \$1 per period is...
 - i. also known as the compound amount of \$1 per period
 - ii. reciprocal of the sinking fund factor
 - iii. shows the growth at compound interest of a series of deposits
 - c. Sinking fund factor is...
 - i. reciprocal of the future worth of \$1 per period
 - ii. Shows the amount of periodic payment necessary to accumulate to a specific sum
 - d. Present worth of \$1...
 - i. reciprocal of the future worth of \$1
 - ii. Shows the present worth of a single future payment

- e. Present worth of \$1 per period is...
 - i. also known as the annuity factor or the Inwood coefficient
 - ii. reciprocal of the partial payment factor
 - iii. Shows the present worth of a series of future level income payments (an annuity)
- f. Partial payment factor is...
 - i. also known as the installment to amortize \$1
 - ii. reciprocal of the present worth of \$1 per period
 - iii. Shows the periodic payment necessary to amortize a loan at a specific interest rate over a specific period of time

Demonstration 4-7: Description of the Compound Interest Tables (see appendix)

Column #1 - (Future Worth of \$1) - This factor is the basis for the other five columns. It shows the growth of a single deposit or investment over a specific time period. It is also known as the compound amount of \$1. It is the reciprocal of Column #4 (the reversion factor).

Column #2 - (Future Worth of \$1 per Period) - This factor shows the growth at compound interest of a series of deposits or investments. It is also known as the compound amount of \$1 per period. It is the reciprocal of Column #3 (Sinking Fund Factor).

Column #3 - (Sinking Fund Factor) - This factor shows the periodic deposit necessary to accumulate \$1 over a specific time period. It is the reciprocal of Column #2 (Future Worth of \$1 per Period).

Column #4 - (Reversion Factor) - This factor shows the present worth of a single future payment or deposit of \$1. It is also known as the reversion factor. It is the reciprocal of Column #1 (Future Worth of \$1).

Column #5 - (Present Worth of \$1 per Period) - Shows the present worth of a series of future payments or deposits. It is also known as the Inwood coefficient. It is the reciprocal of Column #6 (Partial Payment Factor).

Column #6 - (Partial Payment Factor) - Shows the periodic payment necessary to amortize a loan at a specified interest rate over a specific number of periods. It is also known as the Installment to Amortize \$1. It is the reciprocal of Column #5 (Present Worth of \$1 per Period).

Exercise 4-7: Development of Annualized Mortgage Constant

You are performing a reappraisal on all the offices in the central business district. You have decided to use the mortgage & equity band of investment technique. You have already calculated an equity capitalization rate of 12% and now need to calculate the mortgage constant. Lending institutions require a loan to value ratio of 70% for office properties and the typical terms of the loan for these offices are twenty years with monthly payments with an interest rate of 8%. What is the annual mortgage constant (R_M)?

Demonstration 4-8: Using Compound Interest Tables (See Appendix)

Column #1 - (Future Worth of \$1)

Example: A single deposit of \$5,000 is deposited in a savings account which pays 6% annually. If the deposit is left in the account for five years, to what amount would this investment grow?

Solution Using Compound Interest Table

6% annual compound table, column 1, at 5 years' factor = 1.338226

$\$5,000 \times 1.338226 = \$6,691.13$

Solution Using the HP-12C

Step	Explanation	Key Stroke	Display
1	Solve for Future Value (FV)		
2	Clear memory	GOLD f CLX	0.0000
3	Enter number of periods	5 n	5.0000
4	Enter interest rate	6 i	6.0000
5	Enter Present Value (PV)	5,000 CHS PV	-5,000.0000
6	Enter Payment (PMT)	0 PMT	0.0000
7	Calculate Future Value (FV)	FV	6,691.13

Column #2 - (Future Worth of \$1 per Period)

Example: An annual deposit of \$1,200 is deposited in a savings account which pays 6% annually. If no principal or interest is withdrawn, how much will be in the savings account at the end of five years?

Solution Using Compound Interest Table

6% annual compound table, column 2 (Future Worth of \$1 per Period), at 5 years' factor = 5.637093

$\$1,200 \times 5.637093 = \$6,764.51$

Solution Using the HP-12C

Step	Explanation	Key Stroke	Display
1	Solve for Future Value (FV)		
2	Clear memory	GOLD f CLX	0.0000
3	Enter number of periods	5 n	5.0000
4	Enter interest rate	6 i	6.0000
5	Enter Payment	1,200 CHS PMT	-1,200.0000
6	Calculate Future Value (FV)	FV	6,764.51

Column #3 - (Sinking Fund Factor)

Example: What annual deposit at 6% interest would be required to accumulate an amount of \$50,000 in 20 years?

Solution Using Compound Interest Table

6% annual compound table, column 3 (Sinking Fund Factor), at 20 years' factor = 0.027185

$\$50,000 \times 0.027185 = \$1,359.25$

Solution Using the HP-12C

Step	Explanation	Key Stroke	Display
1	Solve for Payment (PMT)		
2	Clear memory	GOLD f CLX	0.0000
3	Enter number of periods	20 n	20.0000
4	Enter interest rate	6 i	6.0000
5	Enter Future Value (FV)	50,000 CHS FV	-50,000.0000
6	Calculate Payment	PMT	1,359.23

Column #4 - (Present Worth of \$1)

Example: On your 30th birthday you will collect the proceeds from a small insurance policy with a cash-in value of \$10,000. If your 27th birthday is today, what is the present worth of your right to redeem the insurance policy assuming an annual yield of 10%?

Solution Using Compound Interest Table

10% annual compound table, Column 4 (Present Worth of \$1), at 3 years' factor = 0.751315

$\$10,000 \times 0.751315 = \$7,513.15$.

Solution Using the HP-12C

Step	Explanation	Key Stroke	Display
1	Solve for Present Value (PV)		
2	Clear memory	GOLD f CLX	0.0000
3	Enter number of periods	3 n	3.0000
4	Enter interest rate	10 i	10.0000
5	Enter Future Value (FV)	10,000 CHS FV	-10,000.0000
6	Enter Payment (PMT)	0 PMT	0.0000
7	Calculate Present Value (PV)	PV	7,513.15

Column #5 - (Present Worth of \$1 per Period)

Example: A small property you own produces an annual net income of \$10,000. Assuming a rate of return of 6% and a remaining economic life of 12 years, what is the present worth of this level, terminal income stream?

Solution Using Compound Interest Table

6% annual compound table, Column #5 (Present Worth of \$1 per Period), at 12 years' factor = 8.383844

$$\$10,000 \times 8.383844 = \$83,838.44$$

Solution Using the HP12C

Step	Explanation	Key Stroke	Display
1	Solve for Present Value (PV)		
2	Clear memory	GOLD f CLX	0.0000
3	Enter number of periods	12 n	12.0000
4	Enter interest rate	6 i	6.0000
5	Enter Future Value (FV)	0 FV	0.0000
6	Enter Payment (PMT)	10,000 CHS PMT	-10,000.0000
7	Calculate Present Value (PV)	PV	83,838.44

Column #6 - (Partial Payment Factor)

Example: What is the annual payment on a mortgage of \$550,000 at 10% interest over a 15-year term?

Solution Using Compound Interest Table

10% annual compound table, Column #6 (Partial Payment Factor), at 15 years' factor = 0.131474

$$\$550,000 \times 0.131474 = \$72,310.70$$

Solution Using the HP12C

Step	Explanation	Key Stroke	Display
1	Solve for Payment (PMT)		
2	Clear memory	GOLD f CLX	0.0000
3	Enter number of periods	15 n	15.0000
4	Enter interest rate	10 i	10.0000
5	Enter Present Value (PV)	550,000 CHS PV	-550,000.0000
6	Calculate Payment (PMT)	PMT	72,310.58

REVIEW QUESTIONS – Chapter 4

1. _____ is a method of converting an estimate of a single year's income into value in one direct step.
2. The two generic formulas used in direct capitalization are:
 - _____
 - _____
3. _____ is a method of converting future net benefits into present value where each future net benefit is discounted at a proper yield rate (discount rate).
4. The _____ shows the periodic payment necessary to amortize a loan at a specified interest rate over a specific number of periods. It is also known as the installment to amortize \$1.
5. The _____ shows the present worth of a series of future payments or deposits. It is also known as the Inwood coefficient.
6. The _____ shows the present worth of a single future payment or deposit of \$1.
7. Write the formula for developing a gross income multiplier.

8. List three critical comparability criteria necessary when utilizing direct capitalization.

- _____
- _____
- _____

9. A comparable sale property to the subject has a net operating income of \$150,000 inclusive of taking real estate taxes as an expense. The property sold for \$1,500,000 and is located in the subject's tax jurisdiction, which has a 1% effective tax rate. The subject property has an expected net operating income of \$132,000 exclusive of taking real estate taxes as an expense.

What is the subject property's value? _____

10. A property recently sold for \$2,000,000. At the time of sale, the property had a potential gross income of \$270,000, and an expected vacancy loss of \$20,000.

What is the effective gross income multiplier? _____

11. When property is being appraised for ad valorem tax purposes the _____ must be included as a part of the overall capitalization rate.

12. List four types of amenities that must be considered when selecting vacant land sales.

- _____
- _____
- _____
- _____

13. List the four major categories of required comparability that must be considered when selecting improved property sales to use in the development of an overall capitalization rate.

- _____
- _____
- _____
- _____

14. _____ is different from yield capitalization, in that it does not directly consider the individual cash flows beyond the first year.

15. The formula value equals _____ $\div$ _____ is one of the generic formulas used in direct capitalization.

PAGE LEFT BLANK INTENTIONALLY

Chapter 5 | Historical Capitalization Methods (Straight-Line Capitalization)

APPENDIUM KNOWLEDGE AREAS



SUGGESTED READING

Source	Chapter	Pages
Property Assessment Valuation	Chapter 14	390-396

CHAPTER TOPICS AND TIMETABLE

Topic	Time	Percentage
Classic Straight-Line Capitalization Method	30 minutes	7
Straight-Line Capitalization Assumptions	15 minutes	3
Components of the Land and Building Capitalization Rates	55 minutes	12
Methods of Developing the Land Capitalization Rate and the Building Capitalization Rate	1 hour	13
Land Residual Technique	1 hour, 45 minutes	24
Building Residual Technique	1 hour, 45 minutes	24
Review Questions	20 minutes	4
Quiz #2	1 hour	13

OBJECTIVES

Upon completion of Chapter 5, the student should be able to:

- Demonstrate competency by the selection of the correct capitalization rate for both land and building.
- Find value of income-producing properties using the building residual technique.
- Find value of income-producing properties using the land residual technique.
- Analyze the shape and behavior of the income stream.
- Understand the structure of straight-line capitalization.

Chapter 5 | Historical Capitalization Methods (Straight-Line Capitalization)

I. Classic Straight-Line Capitalization Method

- A. Shape and behavior of the income stream – Straight-line (downward)
- B. Manner or method of handling recapture - Recapture is received in equal amounts during the economic life of the improvement (straight-line).
- C. Proportion of the investment upon which the Overall Yield (discount) is earned - non-depreciated balance of investment
- D. Known value of either land or improvement
 - 1. Land residual
 - 2. Building residual
- E. These techniques are useful in specialized situations, such as financial feasibility tests in highest and best use, value estimates in condemnation, valuation of right of way easements, and land lease values where it is required to estimate separate market values for land and improvements. Residual techniques can be replaced by or supplemented with a discounted cash flow analysis, which is addressed in greater detail in IAAO Course 112.

II. Straight-Line Capitalization Assumptions

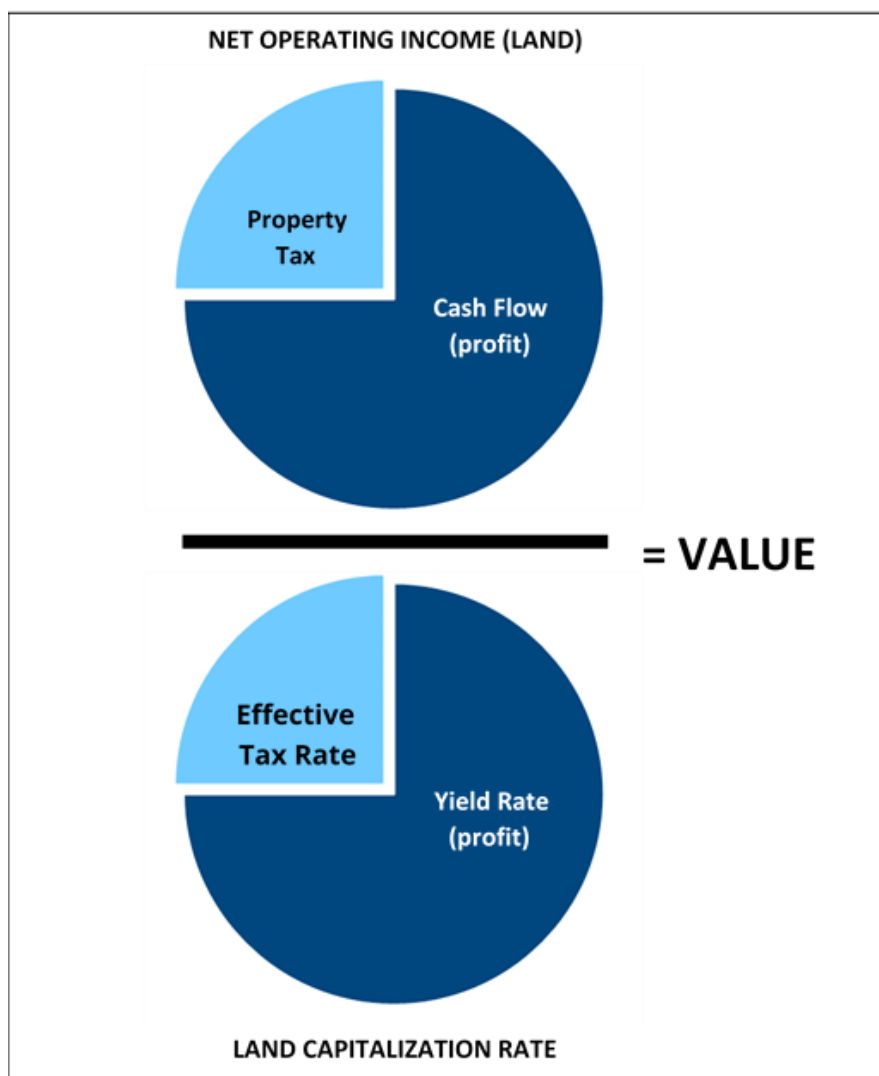
- A. According to Property Assessment Valuation, 3rd edition, p 387, Straight-line capitalization is appropriate under the following set of circumstances:
 - 1. The income is likely to decline over the economic life of the improvement.
 - 2. The improvement is recaptured in equal amounts over its economic life.
 - 3. The discount is received each year on the remaining balance of the investment.
 - 4. The tenant ranks financially average to poor.
 - 5. The lease is month to month or short term.
 - 6. Conditions 4 & 5 contribute to condition 1.
- B. Recapture received in equal amounts (straight-line basis) during the remaining economic life of the improvements

- C. The property's overall yield (discount) is received on the balance of the investment, after a periodic recapture of a portion of the improvement value.

III. Land Value Calculation: components of the Land Capitalization Rate and Building Capitalization Rate

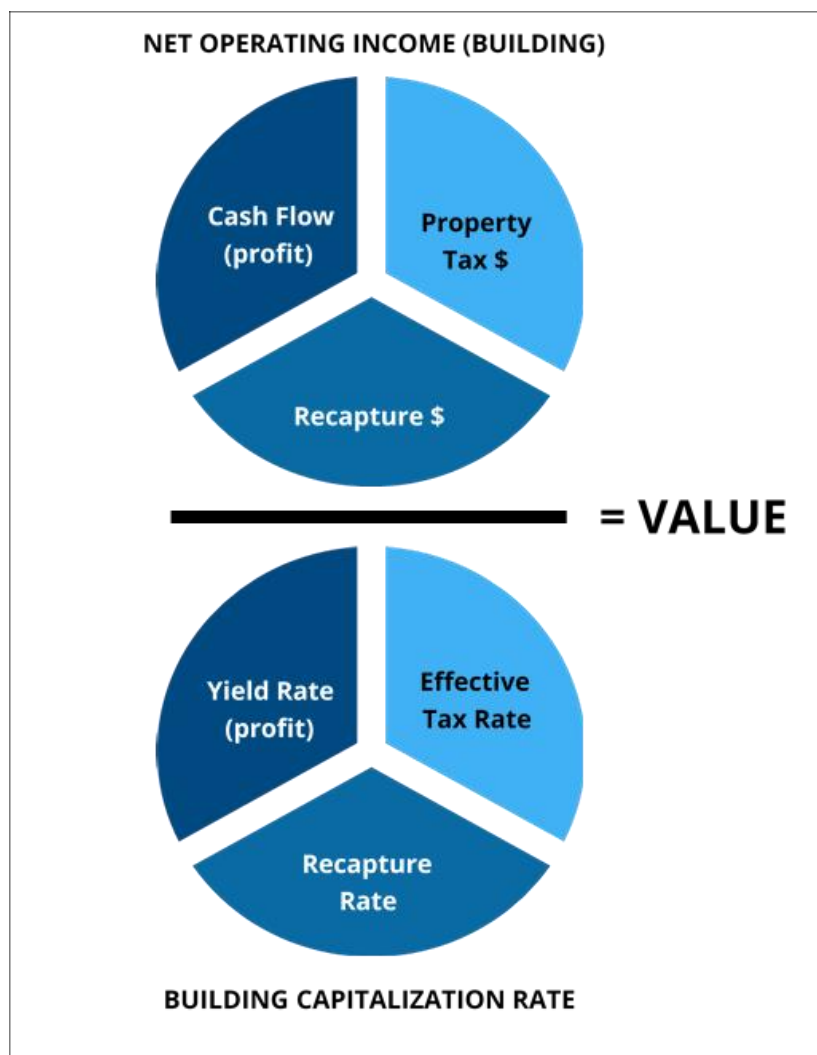
A. Land capitalization rate (R_L)

1. Overall yield rate (Y_o)
2. Effective tax rate (ETR)



B. Building capitalization rate (R_B)

1. Overall yield rate (Y_O)
2. Effective tax rate (ETR)
3. Recapture rate
 - a. Reciprocal ($1 \div x$) of remaining economic life (REL) technique of developing the recapture rate
 - b. Market comparison technique of developing the recapture rate
 - c. The land and building capitalization rates share two of the same capitalization rate components: yield (discount) rate and Effective Tax Rate (ETR). Remembering that land is not a wasting asset, and consequently does not physically deteriorate like a building, the only difference is the recapture rate.



Demonstration 5-1: Remaining Economic Life Technique

Change years of remaining economic life to recapture rates and recapture rates to years of remaining economic life by using the reciprocal ($1 \div x$) of the information provided.

Problem No.	Remaining Economic Life	Recapture Rate
1	25 Years	
2		0.1000
3	50 Years	
4	30 Years	
5		0.0625
6	20 Years	
7		0.0220
8	60 Years	
9		0.0250

Solution:

Problem No.	Remaining Economic Life	Recapture Rate
1	25 Years	$1 \div 25 = 0.0400$
2	$1 \div 0.10 = 10$ years	0.1000
3	50 Years	$1 \div 50 = 0.0200$
4	30 Years	$1 \div 30 = 0.0333$
5	$1 \div 0.0625 = 16$ years	0.0625
6	20 Years	$1 \div 20 = 0.050$
7	$1 \div 0.0220 = 45$ years	0.0220
8	60 Years	$1 \div 60 = 0.0167$
9	$1 \div 0.0250 = 40$ years	0.0250

Demonstration 5-2: Market Comparison Technique

The net operating income produced by any property can be allocated to satisfy the yield rate, recapture rate, and effective tax rate. When any two of these are known, the third can readily be obtained.

A recapture rate can be developed from comparable sale properties when the following information is known:

- Sale Price
- Land or improvement value
- Net operating income
- Overall yield rate (discount rate)
- Effective tax rate

Given the following data:

Sale Price	\$1,600,000
Land Value	\$400,000
Net Operating Income	\$198,000
Overall Yield Rate (Y_o)	0.085
Effective Tax Rate	0.020

Derive the recapture rate using the market comparison method.

Calculation of Net Operating Income After Deduction of Discount and Taxes

Net Operating Income		\$198,000
Discount Dollars	$\$1,600,000 \times 0.085$	-\$136,000
Tax Dollars	$\$1,600,000 \times 0.020$	<u>-\$32,000</u>
Net Operating Income After Deduction for Discount and Taxes		\$30,000

Calculation of Recapture Rate

Recapture Income		\$30,000
Sale Price (Improvements)		<u>$\div \\$1,200,000$</u>
Recapture Rate		0.025

Exercise 5-1: Developing a Recapture Rate by the Market Comparison Technique

Given the following data:

Sale Price	\$1,800,000
Improvement Value	\$1,200,000
Net Operating Income	\$240,000
Overall Yield (discount) Rate	8.5%
Assessment Level	25%
Tax Rate	80 mills

Calculate the recapture rate using the market comparison method.

PAGE LEFT BLANK INTENTIONALLY

IV. Methods of Developing the Land Capitalization Rate and the Building Capitalization Rate

A. Summation method

1. Land capitalization rate (R_L) components
 - a. Overall yield rate (Y_O)
 - b. Effective tax rate (ETR)
2. Building capitalization rate (R_B) components
 - a. Overall yield rate (Y_O)
 - b. Effective tax rate (ETR)
 - c. Recapture rate
3. The land and building capitalization rates share two of the same capitalization rate components: yield (discount) rate and Effective Tax Rate (ETR).
 - a. Land is not a wasting asset, and consequently does not physically deteriorate like a building, the only difference is the recapture rate.
 - b. If you have the land capitalization rate, simply add the recapture rate to calculate the building capitalization rate.
 - c. If you have the building capitalization rate, subtract the recapture rate to arrive at the land capitalization rate.

B. Market comparison (using IRV formula)

1. Land income $\div$ land value = land capitalization rate (R_L)
2. Building income $\div$ building value = building capitalization rate (R_B)

Demonstration 5-3: Land Capitalization Rates

Developing a land capitalization rate from market sales transactions requires the appraiser to identify the land income and the land value, which can be used with the IRV equation to determine the land capitalization rate. Developing a land capitalization rate by the summation method requires the appraiser to find the sum of the overall property yield rate and the effective tax rate. Both methods are demonstrated below.

Given the following:

Sale Price	\$2,000,000
Improvement Value	\$1,600,000
Land Income	\$40,000
Assessment Level	50%
Tax Rate	\$4.00 per \$100
Overall Property Yield Rate (Y_o)	8%

Market Comparison Method:

Land income ÷ land value = land rate = \$40,000 ÷ \$400,000 = 10%

(Land Value = Sale Price \$2,000,000 Less Improvement Value \$1,600,000)

OR

Summation Method

Overall Property Yield (Discount) Rate = 0.08

Effective Tax Rate: (Assessment Level) 0.50 x 0.04 (Tax Rate) = 0.02

Land Capitalization Rate = 0.10

Exercise 5-2: Developing a Land Capitalization Rate by the Market Comparison & Summation Methods

Given the following:

Land Value	\$500,000
Assessment Level	50%
Tax Rate	40 mills
Overall Yield Rate (Y_o)	8.5%
Remaining Economic Life	25 years
Land Income	\$52,500

- A. Determine the land capitalization rate (R_L) by summing the overall property yield rate and effective tax rate.
- B. Determine the land capitalization rate (R_L) by using the IRV equation.

PAGE LEFT BLANK INTENTIONALLY

Demonstration 5-4: Improvement Capitalization Rate

Developing an improvement (building) capitalization rate from market sales transactions requires the appraiser to identify the improvement income and the improvement value, which can be used with the IRV equation to determine the improvement capitalization rate. Developing an improvement capitalization rate by the summation method requires the appraiser to find the sum of the Y_o (discount rate), recapture rate, and the effective tax rate. Both methods are demonstrated below.

Sale Price	\$2,000,000
Land Value	\$400,000
Improvement Income	\$240,000
Assessed Value	\$1,000,000
Tax Rate	\$4.00 per \$100
Remaining Economic Life	20 years
Overall Yield Rate (Y_o)	8%

Market Comparison Method:

Improvement income ÷ improvement value = improvement rate

$\$240,000 \div \$1,600,000 = 0.15$ or **15%**

Summation Method:

Y_o (Discount Rate): **0.08**

Effective Tax Rate:

Assessment Level = $\$1,000,000 \text{ Assessed Value} \div \$2,000,000 \text{ Sale Price} = 0.50$

Tax Rate = $4 \div 100 = 0.04$

Effective Tax Rate (ETR) = $0.50 \text{ (Assessment Level)} \times 0.04 \text{ (Tax Rate)} = \mathbf{0.02}$

Recapture Rate:

$1 \div 20 \text{ (Remaining Economic Life)} = \mathbf{0.05}$

Improvement Capitalization Rate = $0.08 + 0.02 + 0.05 = 0.15$ or 15%

Exercise 5-3: Developing a Building Capitalization Rate by the Market Comparison & Summation Methods

Given the following:

Sale Price	\$1,125,000
Land Value	\$225,000
Assessed Value	\$281,250
Tax Rate	80 mills
Overall Yield Rate (Y_o)	10%
Remaining Economic Life	25 years
Net Operating Income	\$171,000

- A. Determine the building/improvement capitalization by summing the overall yield rate, recapture rate, and effective tax rate.
- B. Determine the building/improvement capitalization rate by using the IRV equation.

V. Residual Valuation Techniques: Land Residual Technique

- A. Modern capitalization theory defines the residual techniques as techniques where the investment is divided into components and allows for the capitalization of the income allocated to the component of unknown value after all investment components of known value have been satisfied.
- B. The residual technique is appropriately applied through land residual, building residual, property residual, equity residual and mortgage residual. (Property residual, equity residual and mortgage residual are not a part of this course.)
- C. A supportable improvement value must be developed for the application. The annual net return to the improvement is deducted from the total annual net operating income. The remaining income, which is the residual amount, is attributable to the land. This income is capitalized into a value indicator for the land.
- D. Requirements for application:
 - 1. Known improvement value is supported by market data.
 - 2. Improvement must be new (or hypothetically new).
 - 3. Improvement must represent the highest and best use of the site.
 - 4. Develop an annual net operating income.
 - 5. Develop an overall yield (discount) rate, recapture rate, and effective tax rate.
- E. Application is appropriate when:
 - 1. Land consists of a large parcel with no comparable property available
 - 2. Land sales are not available.
 - 3. Determining highest and best use as vacant. The highest residual land value when comparing different legally permissible uses on a site can be a good indicator for a site's most likely use.
 - 4. Improvements are adequate in condition and suitable to the site.
 - 5. Building is new (or hypothetical), and the proper use and cost are known.

6. Results of the land residual technique can vary dramatically depending upon the assumptions made in the analysis. The appraiser should make sure at least one of the above criteria is present before considering its application.

F. Steps in the land residual technique:

1. Develop a reconstructed operating statement.
2. Determine annual net operating income.
3. Develop capitalization rates for land and improvement (building).
 - a. Land capitalization rate = sum of:
 - i. Overall Yield (Discount) rate
 - ii. Effective tax rate
 - b. Improvement (building) capitalization rate = sum of:
 - i. Overall Yield (discount) rate
 - ii. Recapture rate (Remaining Economic Life)
 - iii. Effective tax rate
4. Calculate income attributable to improvements. ($I_B = R_B \times V_B$)
5. Calculate income attributable to land.
6. Calculate indicated land value.
7. Add land and improvement values to obtain total property value.

Demonstration 5-5: Land Residual Technique

Your assignment is to appraise a parcel of vacant land, in the middle of a business block in a desirable community. Zoning is for commercial purposes, and building codes allow full coverage of the site. The land has 100 front feet on the street and a uniform depth of 150 feet for rectangular shape.

A study of the highest and best use of the site indicated that a one-story brick structure housing two retail stores would be most profitable. The proposed building would cover the entire site, using all the space. A contractor reports the cost of the proposed structure is \$125.00 per square foot. Net operating income is estimated to be \$368,750 for the proposed improved property.

Analysis indicates the following rates are appropriate: overall yield rate, 12%; recapture, 2%; and effective tax rate, 2%. Further analysis indicates that straight-line capitalization should be employed with this proposed improved property.

Because of the absence of comparable land sales, the land residual technique is used to arrive at an estimate of value for this parcel.

Application of Land Residual Technique		
Net Operating Income		\$368,750
Less Income Attributable to Building	15,000 SQFT @ \$125.00 = \$1,875,000 x 16%	<u>-\$300,000</u>
Residual Income to Land		\$68,750
Land Capitalization Rate (R_L):		
Property Discount Rate (Y_o) or overall yield on investment		0.12
Effective Tax Rate		<u>+ 0.02</u>
Land Capitalization Rate		0.14
Residual Income to Land	\$68,750 ÷ 0.14	\$491,071
Rounded to \$491,000		

Demonstration 5-6: Residual Template

Net Operating Income	\$	
Less: Income to land (or improvement)		
(Land value x land rate)		
(Or Improvement value x improvement rate)	-	
Residual income to improvement (or land)		
Improvement income capitalized at improvement cap rate	\$	
(Land capitalized at land cap rate)		
 Property yield rate (discount rate)		
Effective tax rate		
Total land capitalization rate	=	
Recapture rate (return of improvement investment)		
Total improvement capitalization rate	=	
Improvement income		
	=	
Improvement cap rate		
Improvement value		
	\$	
Land income		
	=	
Land cap rate		
Land value		
	\$	
Total property value	\$	

Demonstration 5-7: Residual Technique Matrix

An alternative solution to land and building residual problems is to use the following matrix to solve the problem.

	Income	÷ Rate	= Value
Building	I_B	R_B	V_B
+ Land	I_L	R_L	V_L
= Total	I_O	R_O	V_O

Where....

	Income		Value
Building	I_B		V_B
+ Land	$+I_L$		$+V_L$
= Total	$=I_O$		$=V_O$

And conversely ...

	Income		Value
Total	$=I_O$		$=V_O$
- Building	$-I_B$		$-V_B$
= Land	$=I_L$		$=V_L$

Or...

	Income		Value
Total	$= I_O$		$= V_O$
- Land	$- I_L$		$- V_L$
= Building	$= I_B$		$= V_B$

Using the IRV formula:

Income $\div$ Rate = Value and Rate $\times$ Value = Income,
the value each of the land, building, and total value components can be isolated using the components available. Then the residual value of either the land or the building can be solved after processing the IRV formula on those property components known.

Using the residual matrix, the math operation would begin with applying the IRV formula by multiplying the building value by building cap rate ($V_B \times R_B$) to arrive at the building's income (I_B), moving right to left in "building" row of the matrix:

	Income		Value
Building	I_B	 $= R_B$  $\times$  V_B	
+ Land	I_L		V_L
= Total	I_O		V_O

If the property's total income (I_O) is known, then the difference between the total income (I_O) and the building's income (I_B) is the land's income (I_L), or total income less building income equals land income ($I_O - I_B = I_L$), which is performed in the "income" column of the matrix.

	Income		Value
Building	I_B	R_B	V_B
+ Land	$I_O - I_B = I_L$		V_L
= Total	I_O		V_O

After the income to the land is calculated, the IRV formula is applied by taking the land's income and dividing by the land cap rate ($I_L \div R_L$) to arrive at the land value (V_L), moving left to right in "land" row of the matrix:

	Income		Value
Building	I_B	R_B	V_B
+ Land	 I_L  $\div R_L$ 		$= V_L$
= Total	I_O		V_O

Example: The following information is known about an office building:

Total Property NOI	\$300,000
Building Value	\$2,000,000
Building Cap Rate	0.12
Land Cap Rate	0.09

What is the residual land value of the property?

Step 1) Insert dollar figures that are known into the Matrix

	Income	$\div$ Rate	= Value
Building			\$2,000,000
+ Land			
= Total	\$300,000		

Step 2) Apply the IRV formula by multiplying the building value by building cap rate ($V_B \times R_B$) to arrive at the building's income (I_B), moving right to left in "building" row of the matrix:

	Income	$\div$ Rate	= Value
Building	\$240,000  $= 0.12$  $\times \$2,000,000$		
+ Land			
= Total	\$300,000		

Step 3) The difference between the total income ($I_o = \$300,000$) and the building's income ($I_B = \$240,000$) is the land's income (I_L), or total income less building income equals land income ($\$300,000 - \$240,000 = \$60,000$), which is performed in the "income" column of the matrix.

	Income	÷ Rate	= Value
Building	\$240,000	= 0.12	X \$2,000,000
+ Land	\$60,000		
= Total	\$300,000		

Step 4) Apply the IRV formula is applied by taking the land's income and dividing by the land cap rate ($I_L \div R_L$) to arrive at the land value (V_L), moving left to right in "land" row of the matrix:

	Income	÷ Rate	Value
Building	\$240,000	= 0.12	X \$2,000,000
+ Land	\$60,000	→ ÷ 0.09	→ = \$667,000
= Total	\$300,000		

The residual land value is \$667,000.

Exercise 5-4: Straight-Line Capitalization-Land Residual Technique

You are appraising a downtown parking lot that has recently been paved with asphalt over a crushed stone base at a cost \$3.00 per square foot. The paving has an estimated life expectancy of 10 years. The lot has a frontage of 250 feet and a depth of 400 feet.

Annual anticipated operating expenses are listed below.

Insurance	\$2,400
Management	4% of effective gross income
Snow removal and lot maintenance	\$18,000
Legal and accounting	1% of effective gross income
Miscellaneous expenses	\$3,600
Real estate taxes	2.5% of the value
Typical investor expected rate of return	9.5%
Effective gross income	\$225,000

Using straight-line capitalization, estimate the value of the property.

PAGE LEFT BLANK INTENTIONALLY

Exercise 5-5: Straight-Line Capitalization-Land Residual Technique

The appraisal assignment is a parcel of vacant land located in an area of improved strip commercial property. Zoning is for commercial use, and building codes allow for full improvement development on the site. The lot is rectangular in shape, with a frontage of 100 feet and a depth of 125 feet.

A highest and best use analysis indicates the most profitable use of the site is retail, with a one-story building utilizing the entire site.

Contractor estimates for cost of construction of the proposed building is \$140.00 per square foot, with an economic life of 40 years.

Area market rent and expense analysis produced the following:

Market rent	\$35 per square foot annually
Vacancy and collection loss	2%
Management fees	5%
Insurance	\$7,700
Maintenance and repair	\$15,750
Utilities	\$12,600
Trash service	\$2,100
Legal and accounting	\$4,200
Miscellaneous expense	\$5,600
Replacement reserves	\$7,000

Further analysis produced the following:

Overall yield rate for property (Y_o)	12%
Assessment level	25%
Tax rate	60 mills

Using the above information, develop a value estimate for the land.

PAGE LEFT BLANK INTENTIONALLY

VI. Residual Valuation Techniques: Building Residual Technique

- A. The application requires the land to be valued independently of the improvement (building). The annual net return to the land is deducted from the estimated total annual net operating income. The remaining income, the residual amount, is attributable to the improvement and is capitalized into a value indicator for the building.
- B. Requirements for application
 - 1. Known land value supported by market data
 - 2. Realistic net operating income
 - 3. Property's Overall Yield Rate YO (discount rate), recapture rate, and effective tax rate are known
- C. Application - The residual building value indication is most meaningful with a well-supported land value and improvements that do not significantly contribute to the overall value of the property. This technique is the first step in market-extracted depreciation using improved sales, where the depreciated value of the improvements (building residual) is compared with the replacement cost new to obtain an indication of depreciation from all sources. Apply the Building Residual Technique when at least 2 of the following criteria are present:
 - 1. Land value estimate may be supported from computations involving a hypothetical proper improvement to the land.
 - 2. Vacant land sales are available.
 - 3. Improvement is not new.
 - 4. Improvement is not adequate for the site.
 - 5. Determining highest and best use as improved. The value contribution of the building improvements can be measured and contrasted with other options for the improvements (or change in use).

D. Steps in the building residual technique

1. Develop a reconstructed operating statement.
2. Determine annual net operating income.
3. Develop land and improvement (building) capitalization rates.
 - a. Land capitalization rate = sum of:
 - i. Overall Yield (Discount) rate
 - ii. Effective tax rate
 - b. Improvement (building) capitalization rate = sum of:
 - i. Overall Yield (Discount) rate
 - ii. Recapture rate ($1 \div \text{Remaining Economic Life}$)
 - iii. Effective tax rate
4. Calculate income attributable to land.
5. Calculate income attributable to improvement (building).
6. Calculate indicated building/improvement value.
7. Add land and improvement values to obtain total property value.

Demonstration 5-8: Building Residual Technique

The subject property of this appraisal is a supermarket containing 12,000 square feet, with a land value of \$800,000 derived from recent sales. The property is zoned commercial and is currently leased at market rent to a national food store chain on a 10-year lease at \$30 per square foot per year. A vacancy and collection loss allowance of 5% is indicated to reflect the credit rating of the tenant and the term of the lease.

Allowable expenses consist of:

Insurance	\$4,280
Maintenance and repairs	\$7,200
Management of property	5% of effective gross income
Legal and accounting fees	1% of effective gross income
Parking lot maintenance	\$4,800 per year

Real estate taxes amount to 2.6% of the value of the property. The land should return 6% yield plus the 2.6% for taxes.

The capitalization rate for the building should include 6% overall yield, 2% for recapture (50 year estimated remaining life of the building), plus the 2.6% for taxes.

Given this data, you have reconstructed the income and expense statement as shown below:

Potential Gross Income	12,000 SF @ \$30.00	\$360,000
Less Vacancy and Collection Loss (5%)		<u>-\$18,000</u>
Effective Gross Income		\$342,000
Less Expenses:		
Management	\$342,000 x 5%	\$17,100
Legal & Accounting	\$342,000 x 1%	\$3,420
Parking Lot Maintenance		\$4,800
Maintenance and Repair		\$7,200
Insurance		\$4,280
Total Expenses		<u>-\$36,800</u>
Net Operating Income		\$305,200

The next step is to value the property by the straight-line capitalization method, building residual technique as shown on the following page.

Net operating Income		\$305,200
Less income attributable to land	\$800,000 x 0.086	<u>-\$68,800</u>
Residual income attributable to building		\$236,400
Capitalized at 0.106		\$2,230,189
Overall yield rate	0.060	
Recapture of building portion of investment	+ 0.020	
Effective tax rate	<u>+ 0.026</u>	
Building capitalization rate	0.106	
Plus land value		<u>+ \$800,000</u>
Total indicated property value		\$3,030,189 Rounded to \$3,030,000

ALTERNATE WORKING using matrix (begin after calculating capitalization rates)

Step 1) Insert known figures into the matrix.

	Income	÷ Rate	= Value
Building		0.106	
+ Land		0.086	\$800,000
= Total	\$305,200		

Step 2) Apply the IRV formula by multiplying the land value by land cap rate ($V_L \times R_L$) to arrive at the land's income (I_L), moving right to left in "land" row of the matrix:

	Income	÷ Rate	= Value
Building		0.106	
+ Land	\$68,800	= 0.086	X \$800,000
= Total	\$305,200		

Step 3) The difference between the total income ($I_o = \$305,200$) and the land's income ($I_L = \$68,800$) is the building's income (I_B), or total income less land income equals building income ($\$305,200 - \$68,800 = \$236,400$), performed in the "income" column of the matrix:

	Income	÷ Rate	= Value
Building	\$236,400	0.106	
+ Land	\$68,800	0.086	\$800,000
= Total	\$305,200		

Step 4) Apply the IRV formula by taking the building's income and dividing by the building cap rate ($I_B \div R_B$) to arrive at the building value (V_B), moving left to right in "building" row of the matrix:

	Income	$\div$ Rate	= Value
Building	\$236,400	$\div 0.106$	= \$2,230,189
+ Land	\$68,800	0.086	\$800,000
= Total	\$305,200		

Step 5) Add the building and land values to find the total property value:

	Income	$\div$ Rate	= Value
Building	\$236,400	0.106	\$2,230,189
+ Land	\$68,800	0.086	\$800,000
= Total	\$305,200		\$3,030,189

Exercise 5-6: Straight-Line Capitalization-Building Residual Technique

The Orange Grove office building, containing 20 equal size offices of 1,200 square feet each, was constructed 10 years ago. After reviewing today's market, you have estimated the remaining economic life of the building to be 25 years. The current market rent for this type of office space is \$25 per square foot. The typical operating expenses for this property amount to \$171,000 annually. Vacancy and collection loss is expected to be 5%, similar to other office buildings in the area.

After reviewing the real estate market, you find the typical overall property yield rate for similar investments is 9% and real estate taxes are 2.5%. The land on which the Orange Grove office building sits has recently been valued at \$600,000 using comparable sales.

Using straight-line capitalization and the building residual technique, compute the total indicated value of the improved property.

PAGE LEFT BLANK INTENTIONALLY

Exercise 5-7: Straight-Line Capitalization-Building Residual Technique

You are appraising an older industrial building in an urban industrial district, which has a remaining economic life of 20 years. The land where the building is situated is valued by comparable sales at \$2,500,000. The potential gross income for the property is expected to be \$2,354,000 and the vacancy and collection loss for this type of building is typically 4%. Operating expenses are expected to be \$660,000.

Typical investors require an overall yield rate of 8% to invest in this type of property and property taxes are based on a \$4.00/per \$100 tax rate with an assessment level of 50%.

This property fits the requirements for straight-line capitalization. Your assignment is to appraise the property using straight-line capitalization with the building residual technique.

PAGE LEFT BLANK INTENTIONALLY

REVIEW QUESTIONS – Chapter 5

1. If the land capitalization rate in straight-line capitalization (including the effective tax rate) is 9%, and the remaining economic life of the improvement is 25 years, the building capitalization rate is _____.
2. In straight-line capitalization, _____ is received in equal amounts during the economic life of the improvement.
3. Straight-line capitalization assumes a _____ income stream during the remaining economic life of the improvements.
4. Straight-line capitalization is best utilized when the property has:
 - _____
 - _____
 - _____
5. The building residual technique should be used when:
 - _____
 - _____
 - _____
 - _____
6. The land residual technique should be used when:
 - _____
 - _____
 - _____
 - _____

7. The land capitalization rate is composed of what components?

- _____
- _____

8. The building capitalization rate is composed of what components?

- _____
- _____
- _____

9. Overall yield rate (Y_o) is another name for a property's _____ rate.

10. In the building residual technique, the land capitalization rate is _____ by the _____ value to obtain the land income.

11. To obtain the _____ capitalization rate, one can divide the building income by the building _____.

12. The _____ residual technique requires the development of the discount rate, recapture rate, effective tax rate, and a supportable market value for land.

- a. Land
- b. Building
- c. Both A and B
- d. None of the above

Addendum | Fee Simple and Related Information

ESTATES IN REAL ESTATE

An “estate” is defined as, The amount, degree, nature and quality of a personal interest in land or other property.” Black’s Law Dictionary 10th Ed. (Garner, 2014)

The primary types of estates in real property are as follows: fee simple absolute, fee simple upon condition subsequent, fee simple determinable, life estate, and remainders.

Fee Simple Absolute

The estate that gives the owner the most control over the disposition of the property and the length of ownership is the *fee simple absolute* estate (often shortened to *fee simple*). The holder of the fee simple estate has complete control over the disposition of the property after death and there is no event set forth in the deed that could result in a loss of the property prior to the death of the owner.

Black’s Law Dictionary defines fee simple as,

An interest in land that, being the broadest property interest allowed by law, endures until the current holder dies without heirs; esp., a fee simple absolute. (Garner, 2014, p. 733)

In *Introduction to The Law of Real Property*, fee simple is described as,

... the largest estate known to law. It denotes the maximum of legal ownership, the greatest possible aggregate of rights, powers, privileges, and immunities which a person may have in land. It is an estate of potentially infinite duration in the holder’s successors who acquire the holder’s interest in the property either by conveyance, devise, or inheritance. The three hallmarks of the estate are that it is alienable, devisable, and descendible. (Moynihan and Kurtz, 2002, p. 34)

Life Estate/Remainders

In a life estate, the owner of the life estate owns the property for the “measuring life” which is typically the life span of the holder of the life estate. In other words, Anna Alpha receives a life estate in property from her parents. She owns the property for as long as she lives. However, after she dies, the property then transfers to the *remainder* (which we will discuss in a minute). Anna Alpha could theoretically sell her life estate, but the buyer of Anna Alpha’s property (Barry Beta) would only own the property for as long as *Anna* lives. After the death of the life estate holder, the property then automatically transfers to the *remainder*, who then owns the property in fee simple absolute. Anna does not get to choose who gets the property after her death.

Fee simple determinable and fee simple subject to condition subsequent

A fee simple determinable and a fee simple subject to condition subsequent simply provide that the holder of the estate only owns the property until a certain event takes place (i.e. The Cleveland Indians win the World Series). Once that event takes place, the property transfers to the already identified remainder. The remainder can sell his remainder interest in the property as well. Again, Anna Alpha could sell her fee simple determinable (or fee simple subject to condition subsequent) to Barry Beta, but Barry Beta would only own the property until the Cleveland Indians win the World Series.

COMMON MISUNDERSTANDINGS REGARDING FEE SIMPLE

The student will likely encounter definitions of fee simple that differ from the legal definition of fee simple, particularly in the field of appraisal. *The Appraisal Institute* defines fee simple absolute as “absolute ownership unencumbered by any other interest or estate subject only to the limitations imposed by the governmental powers of taxation, eminent domain, police power, and escheat.” However, this definition has caused some problems in the appraisal world.

In particular, *unencumbered by any other interest or estate* is a problematic phrase that has moved the definition away from the emphasis on infinite duration and inheritability to an implication that unspecified interests and encumbrances will result in something other than the fee simple estate.

A property held in fee simple may convey in a sale and transfer title encumbered with a lease or other interest without changing the fee simple estate. Although an interest encumbering the estate, such as an easement, a restrictive covenant, or a lease may lessen or enhance the value of the estate, it does not change the fact that a property is held in fee simple.

Fee simple estate has nothing to do with leases, mortgages, liens, and deed restrictions or easements or any other encumbrance or distribution of property rights to others. The typical homeowner owns a home in fee simple absolute, and the deed reflects that estate. The existence of a mortgage does not mean the owner has less than a fee simple absolute estate. The home also has utility easements for water, power, and cable; however, the owner still holds the property in fee simple absolute. More specifically, the property is owned in fee simple absolute *subject to* the mortgage and the utility easements. And if the home is leased, then the property is owned in fee simple absolute *subject to* the lease. A fee simple estate or any other estate is not defeated by the existence of encumbrances, including a lease.

Fee Simple is Not a Value Concept

Fee simple is not synonymous with market value. Fee simple is an estate and a property rights concept, not a value concept, although some appraisers have mistakenly used fee simple to imply “at market.” Market rent or market value is precisely that, and fee simple should not be used interchangeably with those terms. The fee simple estate can be valued assuming market rents just as an appraiser can value the fee simple estate of a property assuming the rents in place and the vacant units at market rent. The valuation performed is based on the scope of work and what is required by a taxing jurisdiction. More appropriately, an appraiser would define the scope of work accordingly, i.e., market value of the fee simple estate or fee simple estate assuming the subject property is vacant and available, if that is what is required. For the majority of taxing jurisdictions, what is required is fee simple absolute considering rents at market.

LEASED FEE AND LEASEHOLD INTERESTS

The term leased fee is an appraisal term defined in The Dictionary of Real Estate (Appraisal Institute 2015). It is not a legal term and is rarely used by market participants in the sale transaction market. Leasehold is a legally defined term as well as an appraisal term. Black's Law Dictionary (Garner 2014) defines leasehold as, "a tenant's possessory estate in land or premises... ." The terms are used as follows:

- **Leased fee.** The ownership interest held by the lessor, which includes the right to receive the contract rent specified in a lease plus the reversionary right when the lease expires. (Appraisal Institute 2015, 128). The term is used by appraisers as a basis to estimate the lessor's value subject to a lease. It is based usually on the capitalization of net operating income (NOI) or the sum of the present value of the forecast NOI over a holding period and the present value of the reversion. In reality, leased fee is synonymous with fee simple, subject to a lease when possession but not the ownership is temporarily transferred to another.
- **Leasehold.** This is the possessory interest held by a tenant. The term is used by appraisers as a basis to estimate the value of the lessee's interest, usually calculated by capitalizing the difference between market rent and contract rent. If a lease exists that reflects market characteristics, including market rent, then the leasehold has no market value. However, if the tenant pays less than market, the difference between the present value of what is paid, and the present value of market rents would be a positive leasehold value in the real estate for the tenant.

Technically what is being referred to is a fee simple estate subject to a lease. However, the term leased fee is common appraisal terminology used to refer to contract rent or rents in-place. When arriving at a fee simple valuation value for ad valorem taxation, the appraiser must recognize that leases, easements, and estates other than fee simple exist in the real world of comparables the appraiser considers. A lease fulfills the basic wish of an owner to receive rent—it is a contract for the use of the property to provide rental income to the owner. The appraiser must be able to make any necessary market-based adjustments to those comparables in order for them to be useful in arriving at the appropriate valuation goal required by the law of the jurisdiction.

If the appraiser is considering using leased-fee sales, then it must be determined whether the contract terms and contract rents are equivalent to market terms and market rents as of the valuation date or whether supportable adjustments can be made to the leased-fee sales.

THE BUNDLE OF RIGHTS/STICKS

Real property consists of land and improvements to the land. However real property can also be described from the perspective of the legal rights the owner has in the land and improvements. These rights are often described as the *bundle of rights* or the *bundle of sticks*. (When describing property as legal rights, it is often referred to as “real estate”.) The bundle of rights metaphor depicts real estate as containing a mixture of individual rights, such as the right of possession, right to sell, right to mortgage, right to lease, right to use the property, and so forth. While there are many rights associated with real property, the primary rights associated with land ownership are the right to:

- a) Sell
- b) Lease
- c) Use
- d) Give away
- e) Enter or leave
- f) Refuse to do any of these

Note that sometimes a fee simple absolute estate is described as the “entire bundle of rights” however, there are certain rights that may not be present, even in a fee simple estate. For example, a fee simple estate subject to a lease is still a fee simple estate (i.e. the duration of ownership has not changed) even though the right to use the property has been transferred to another.

PAGE LEFT BLANK INTENTIONALLY

Appendix

Rates and Relationships

IRV

$$I = R \times V$$

$$R = I \div V$$

$$V = I \div R$$

VIF

$$V = I \times F$$

$$I = V \div F$$

$$F = V \div I$$

EAT

$$E = A \times T$$

$$A = E \div T$$

$$T = E \div A$$

$$R_O = Y_O - CR$$

$$Y_O = R_O + CR$$

$$CR = Y_O - R_O$$

$$DCR = NOI \div I_M$$

$$R_O = DCR \times M \times R_M$$

$$R_O = NIR \div EGIM$$

$$R_O = M \times R_M + (1 - M) \times R_E$$

$$R_L = I_L \div V_L$$

$$R_B = I_B \div V_B$$

$$R_E = I_E \div V_E$$

$$R_M = I_M \div V_M$$

$$NIR = 1 - OER$$

$$OER = 1 - NIR$$

$$GIM = \text{Property Value} \div \text{Gross Income}$$

$$EGIM = \text{Property Value} \div \text{Effective Gross Income}$$

Description of Various Rates

Overall Capitalization Rate (R_O) – The R_O reflects the relationship between a single year's net operating income and the total property value. It is typically derived from market comparisons with highly comparable sales, band of investment calculations, the DCR technique, $NIR \div GIM$, yield change techniques. It also reflects the weighted average of the R_L and R_B weighted by the respective contributions to value of the land and building. R_O can be either greater than, equal to, or less than Y_O depending on whether the property value is expected to decrease, remain stable, or increase, respectively.

Overall Yield Rate (Y_O) – Sometimes referred to as the property discount rate or “r” or the “return on” rate, internal rate of return, or property interest rate. The overall yield rate is expressed as a compound annual percentage rate. It considers all expected benefits of property ownership, including the proceeds from the sale at the termination of the investment. It can be obtained by extraction from sales of similar properties, comparison with alternative investments of comparable risk, surveys of property investors, or the yield change formula – $Y_O = R_O + CR$.

[CAUTION: Using the band of investment technique to blend Y_M and Y_E is mathematically incorrect unless the loan is “interest only” and there is no change in income or value over the holding period. Conceptually, the Y_O is a weighted average of the equity yield rate and the mortgage interest rate. However, in practice the Y_O is not usually calculated by the BOI technique because the ratio of debt and equity changes each year as the loan is amortized and the property value changes.]

Land Capitalization Rate (R_L) – The R_L reflects the relationship between a single year's net income attributable to the land and the value of the land. It is usually extracted from sales using the relationship $R_L = I_L \div V_L$. R_L can be either greater than, equal to, or less than Y_L depending on whether the land value is expected to decrease, remain stable, or increase, respectively. [Note: Within the context of classic straight-line capitalization where land income and value are not expected to change over the holding period, the land capitalization rate (including a property tax component) is the sum of the Y_O and the effective tax rate (because in that situation $R_L = Y_L$).]

Land Yield Rate (Y_L) – The Y_L reflects the rate of return on capital invested in the land portion of real property. It is expressed as a compound annual rate and considers all expected benefits of land ownership, including the proceeds from the sale of the land at the termination of the investment. Since $Y_L = Y_O = Y_B$ is virtually always true (because investors rarely assign a different risk rate to the land and building), the same methods used to derive Y_O are valid for deriving Y_L .

Building (Improvement) Capitalization Rate (R_B) – The R_B reflects the relationship between a single year's net income attributable to the building and the value of the building. It is usually extracted from sales using the relationship $R_B = I_B \div V_B$. R_B can be either greater than, equal to, or less than Y_B depending on whether the building value is expected to decrease, remain stable, or increase, respectively. [Note: Within the context of classic straight-line capitalization where building "return on" income is earned only on the undepreciated balance of the building investment during the holding period, the building capitalization rate (including a property tax component) is the sum of the Y_O , the recapture rate, and the effective tax rate (because in that situation, $R_B = Y_B$).]

Building (Improvement) Yield Rate (Y_B) – The Y_B reflects the rate of return on capital invested in the building portion of real property. It is expressed as a compound annual rate and considers all expected benefits of building ownership, including the proceeds from the sale of the building at the termination of the investment. Since $Y_B = Y_O = Y_L$ is virtually always true (because investors rarely assign a different risk rate to the land and building), the same methods used to derive Y_O are valid for deriving Y_B .

Mortgage Capitalization Rate (R_M) – Also known as the mortgage annual constant, the R_M is the ratio of the annual debt service (principal & interest). It reflects the relationship between mortgage income (I_M) and mortgage principal. It is one of the components used in the band of investment technique to get an R_O . It can be obtained from the compound interest tables by multiplying the partial payment factor (Column #6) by the number of conversion periods in one year.

Mortgage Yield Rate (Y_M) – The Y_M is also known as the mortgage interest rate. It is the lender's rate of return on a loan (if no points or other charges increase the lender's yield). It is usually obtained by interviewing lenders or can be extracted from the terms of the mortgage.

Equity Capitalization Rate (R_E) – The R_E is also known as the equity dividend rate or the cash on cash rate. It reflects the relationship between a single year's equity income and the value of the equity. It is typically extracted from sales using the relationship $R_E = I_E \div V_E$. It is extremely sensitive to loan-to-value ratios. [CAUTION: Unlike the Y_E , the R_E should not be derived from sales of stocks or bonds because of dissimilarity of investments, i.e. liquidity, holding term, etc.]

Equity Yield Rate (Y_E) – The Y_E is the “return on” rate for equity and is sometimes referred to as the equity discount rate or the equity interest rate. It considers all the expected cash flows attributable to the equity investment, including proceeds from sale at the termination of the investment. The Y_E can be extracted from sales of similar properties, from alternative investments of comparable risk such as stocks and bonds, or from surveys of market participants.

Loan to Value Ratio (M) – This is the ratio between a mortgage loan and the value of the property, usually expressed as a percentage. It is usually obtained from interviews with active lenders. It can be extracted from sales by dividing the loan amount by the selling price.

Debt Coverage Ratio (DCR) – This is the ratio of net operating income to annual debt service (principal & interest). It measures the ability of a property to meet its annual debt requirements out of net operating income. The DCR can be extracted from sales by dividing the net operating income by the annual debt service. The DCR is an ideal test of reasonableness and can be multiplied by the R_M and the M to obtain R_O . ($R_O = DCR \times R_M \times M$)

Effective Tax Rate (ETR) – This is the ratio of property taxes to total property value. It can be derived by the EAT formula by multiplying the assessment level times the tax rate as long as all components are expressed as proper decimals. ($ETR = A \times T$) It can also be extracted from sales by dividing the income to property taxes by total property value. It can be directly added to or subtracted from the R_O to get an R_O with or without a property tax component. Conceptually, the ETR cannot be directly added to or subtracted from the Y_O to get a Y_O with or without a property tax component because Y_O is a yield rate used in discounting the cash flows in yield capitalization and the ETR is a capitalization rate. An exception to this is when using classic straight-line capitalization where the $Y_O = R_L$.

Operating Expense Ratio (OER) – The OER reflects the relationship between operating expenses and the Effective Gross Income (EGI) or between the relationship between operating expenses and Potential Gross Income. Consistency is paramount to using this ratio properly. If the OER is represented to be a percentage of EGI it must be applied consistently with the manner in which it was derived. Likewise, if the OER is represented to be a percentage of Potential Gross Income (PGI) it must be applied consistently with the manner in which it was derived.

Net Income Ratio (NIR) – The NIR reflects the relationship between net operating income and the Effective Gross Income (EGI) or between the relationship between net operating income and Potential Gross Income (PGI). Consistency is paramount to using this ratio properly. If the NIR is represented to be a percentage of EGI it must be applied consistently with the manner in which it was derived. Likewise, if the NIR is represented to be a percentage of (PGI) it must be applied consistently with the manner in which it was derived.

Interest Tables

ANNUAL COMPOUND INTEREST TABLE							
Effective Rate = 6.00%		Base =		1.06		Ann Convers. Periods =	
6.00%	1	2	3	4	5	6	6.00%
Annual	Amount of	Accumulation	Sinking	Pres Val	Pres Val	Installment	Annual
	1 @ Comp	of 1 Per	Fund	Reversion	Annuity	to	
Years	Interest	Period	Factor	of 1	1/Period	Amortize	Years
1	1.060000	1.000000	1.000000	0.943396	0.943396	1.060000	1
2	1.123600	2.060000	0.485437	0.889996	1.833393	0.545437	2
3	1.191016	3.183600	0.314110	0.839619	2.673012	0.374110	3
4	1.262477	4.374616	0.228591	0.792094	3.465106	0.288591	4
5	1.338226	5.637093	0.177396	0.747258	4.212364	0.237396	5
6	1.418519	6.975319	0.143363	0.704961	4.917324	0.203363	6
7	1.503630	8.393838	0.119135	0.665057	5.582381	0.179135	7
8	1.593848	9.897468	0.101036	0.627412	6.209794	0.161036	8
9	1.689479	11.491316	0.087022	0.591898	6.801692	0.147022	9
10	1.790848	13.180795	0.075868	0.558395	7.360087	0.135868	10
11	1.898299	14.971643	0.066793	0.526788	7.886875	0.126793	11
12	2.012196	16.869941	0.059277	0.496969	8.383844	0.119277	12
13	2.132928	18.882138	0.052960	0.468839	8.852683	0.112960	13
14	2.260904	21.015066	0.047585	0.442301	9.294984	0.107585	14
15	2.396558	23.275970	0.042963	0.417265	9.712249	0.102963	15
16	2.540352	25.672528	0.038952	0.393646	10.105895	0.098952	16
17	2.692773	28.212880	0.035445	0.371364	10.477260	0.095445	17
18	2.854339	30.905653	0.032357	0.350344	10.827603	0.092357	18
19	3.025600	33.759992	0.029621	0.330513	11.158116	0.089621	19
20	3.207135	36.785591	0.027185	0.311805	11.469921	0.087185	20
21	3.399564	39.992727	0.025005	0.294155	11.764077	0.085005	21
22	3.603537	43.392290	0.023046	0.277505	12.041582	0.083046	22
23	3.819750	46.995828	0.021278	0.261797	12.303379	0.081278	23
24	4.048935	50.815577	0.019679	0.246979	12.550358	0.079679	24
25	4.291871	54.864512	0.018227	0.232999	12.783356	0.078227	25
26	4.549383	59.156383	0.016904	0.219810	13.003166	0.076904	26
27	4.822346	63.705766	0.015697	0.207368	13.210534	0.075697	27
28	5.111687	68.528112	0.014593	0.195630	13.406164	0.074593	28
29	5.418388	73.639798	0.013580	0.184557	13.590721	0.073580	29
30	5.743491	79.058186	0.012649	0.174110	13.764831	0.072649	30

MONTHLY COMPOUND INTEREST TABLE							
Effective Rate = 0.50%		Base = 1.005		Ann Convers. Periods		12	
6.00%	1	2	3	4	5	6	6.00%
Mths	Amount of 1 @ Comp Interest	Accumulation of 1 Per Period	Sinking Fund Factor	Pres Val Reversion of 1	Pres Val Annuity 1/Period	Installment to Amortize	Mths
1	1.005000	1.000000	1.000000	0.995025	0.995025	1.005000	1
2	1.010025	2.005000	0.498753	0.990075	1.985099	0.503753	2
3	1.015075	3.015025	0.331672	0.985149	2.970248	0.336672	3
4	1.020151	4.030100	0.248133	0.980248	3.950496	0.253133	4
5	1.025251	5.050251	0.198010	0.975371	4.925866	0.203010	5
6	1.030378	6.075502	0.164595	0.970518	5.896384	0.169595	6
7	1.035529	7.105879	0.140729	0.965690	6.862074	0.145729	7
8	1.040707	8.141409	0.122829	0.960885	7.822959	0.127829	8
9	1.045911	9.182116	0.108907	0.956105	8.779064	0.113907	9
10	1.051140	10.228026	0.097771	0.951348	9.730412	0.102771	10
11	1.056396	11.279167	0.088659	0.946615	10.677027	0.093659	11
Years							Mths
1	1.061678	12.335562	0.081066	0.941905	11.618932	0.086066	12
2	1.127160	25.431955	0.039321	0.887186	22.562866	0.044321	24
3	1.196681	39.336105	0.025422	0.835645	32.871016	0.030422	36
4	1.270489	54.097832	0.018485	0.787098	42.580318	0.023485	48
5	1.348850	69.770031	0.014333	0.741372	51.725561	0.019333	60
6	1.432044	86.408856	0.011573	0.698302	60.339514	0.016573	72
7	1.520370	104.073927	0.009609	0.657735	68.453042	0.014609	84
8	1.614143	122.828542	0.008141	0.619524	76.095218	0.013141	96
9	1.713699	142.739900	0.007006	0.583533	83.293424	0.012006	108
10	1.819397	163.879347	0.006102	0.549633	90.073453	0.011102	120
11	1.931613	186.322629	0.005367	0.517702	96.459599	0.010367	132
12	2.050751	210.150163	0.004759	0.487626	102.474743	0.009759	144
13	2.177237	235.447328	0.004247	0.459298	108.140440	0.009247	156
14	2.311524	262.304766	0.003812	0.432615	113.476990	0.008812	168
15	2.454094	290.818712	0.003439	0.407482	118.503515	0.008439	180
16	2.605457	321.091337	0.003114	0.383810	123.238025	0.008114	192
17	2.766156	353.231110	0.002831	0.361513	127.697486	0.007831	204
18	2.936766	387.353194	0.002582	0.340511	131.897876	0.007582	216
19	3.117899	423.579854	0.002361	0.320729	135.854246	0.007361	228
20	3.310204	462.040895	0.002164	0.302096	139.580772	0.007164	240
21	3.514371	502.874129	0.001989	0.284546	143.090806	0.006989	252
22	3.731129	546.225867	0.001831	0.268015	146.396927	0.006831	264
23	3.961257	592.251446	0.001688	0.252445	149.510979	0.006688	276
24	4.205579	641.115782	0.001560	0.237779	152.444121	0.006560	288
25	4.464970	692.993962	0.001443	0.223966	155.206864	0.006443	300
26	4.740359	748.071876	0.001337	0.210954	157.809106	0.006337	312
27	5.032734	806.546875	0.001240	0.198699	160.260172	0.006240	324
28	5.343142	868.628484	0.001151	0.187156	162.568844	0.006151	336
29	5.672696	934.539150	0.001070	0.176283	164.743394	0.006070	348
30	6.022575	1004.515042	0.000996	0.166042	166.791614	0.005996	360

ANNUAL COMPOUND INTEREST TABLE							
Effective Rate = 8.00%		Base = 1.08		Ann Convers. Periods = 1			
8.00%	1	2	3	4	5	6	8.00%
Annual	Amount of 1 @ Comp Interest	Accumulation of 1 Per Period	Sinking Fund Factor	Pres Val Reversion of 1	Pres Val Annuity 1/Period	Installment to Amortize	Annual
Years							Years
1	1.080000	1.000000	1.000000	0.925926	0.925926	1.080000	1
2	1.166400	2.080000	0.480769	0.857339	1.783265	0.560769	2
3	1.259712	3.246400	0.308034	0.793832	2.577097	0.388034	3
4	1.360489	4.506112	0.221921	0.735030	3.312127	0.301921	4
5	1.469328	5.866601	0.170456	0.680583	3.992710	0.250456	5
6	1.586874	7.335929	0.136315	0.630170	4.622880	0.216315	6
7	1.713824	8.922803	0.112072	0.583490	5.206370	0.192072	7
8	1.850930	10.636628	0.094015	0.540269	5.746639	0.174015	8
9	1.999005	12.487558	0.080080	0.500249	6.246888	0.160080	9
10	2.158925	14.486562	0.069029	0.463193	6.710081	0.149029	10
11	2.331639	16.645487	0.060076	0.428883	7.138964	0.140076	11
12	2.518170	18.977126	0.052695	0.397114	7.536078	0.132695	12
13	2.719624	21.495297	0.046522	0.367698	7.903776	0.126522	13
14	2.937194	24.214920	0.041297	0.340461	8.244237	0.121297	14
15	3.172169	27.152114	0.036830	0.315242	8.559479	0.116830	15
16	3.425943	30.324283	0.032977	0.291890	8.851369	0.112977	16
17	3.700018	33.750226	0.029629	0.270269	9.121638	0.109629	17
18	3.996019	37.450244	0.026702	0.250249	9.371887	0.106702	18
19	4.315701	41.446263	0.024128	0.231712	9.603599	0.104128	19
20	4.660957	45.761964	0.021852	0.214548	9.818147	0.101852	20
21	5.033834	50.422921	0.019832	0.198656	10.016803	0.099832	21
22	5.436540	55.456755	0.018032	0.183941	10.200744	0.098032	22
23	5.871464	60.893296	0.016422	0.170315	10.371059	0.096422	23
24	6.341181	66.764759	0.014978	0.157699	10.528758	0.094978	24
25	6.848475	73.105940	0.013679	0.146018	10.674776	0.093679	25
26	7.396353	79.954415	0.012507	0.135202	10.809978	0.092507	26
27	7.988061	87.350768	0.011448	0.125187	10.935165	0.091448	27
28	8.627106	95.338830	0.010489	0.115914	11.051078	0.090489	28
29	9.317275	103.965936	0.009619	0.107328	11.158406	0.089619	29
30	10.062657	113.283211	0.008827	0.099377	11.257783	0.088827	30

MONTHLY COMPOUND INTEREST TABLE							
Effective Rate = 0.67%		Base = 1.006667		Ann Convers. Periods		12	
8.00%	1	2	3	4	5	6	8.00%
Mths	Amount of 1 @ Comp Interest	Accumulation of 1 Per Period	Sinking Fund Factor	Pres Val Reversion of 1	Pres Val Annuity 1/Period	Installment to Amortize	Mths
1	1.006667	1.000000	1.000000	0.993377	0.993377	1.006667	1
2	1.013378	2.006667	0.498339	0.986799	1.980176	0.505006	2
3	1.020134	3.020044	0.331121	0.980264	2.960440	0.337788	3
4	1.026935	4.040178	0.247514	0.973772	3.934212	0.254181	4
5	1.033781	5.067113	0.197351	0.967323	4.901535	0.204018	5
6	1.040673	6.100893	0.163910	0.960917	5.862452	0.170577	6
7	1.047610	7.141566	0.140025	0.954553	6.817005	0.146692	7
8	1.054595	8.189176	0.122112	0.948232	7.765237	0.128779	8
9	1.061625	9.243771	0.108181	0.941952	8.707189	0.114848	9
10	1.068703	10.305396	0.097037	0.935714	9.642903	0.103703	10
11	1.075827	11.374099	0.087919	0.929517	10.572420	0.094586	11
Years							Mths
1	1.083000	12.449926	0.080322	0.923361	11.495782	0.086988	12
2	1.172888	25.933190	0.038561	0.852596	22.110544	0.045227	24
3	1.270237	40.535558	0.024670	0.787255	31.911806	0.031336	36
4	1.375666	56.349915	0.017746	0.726921	40.961913	0.024413	48
5	1.489846	73.476856	0.013610	0.671210	49.318433	0.020276	60
6	1.613502	90.25325	0.010867	0.619770	57.034522	0.017533	72
7	1.747422	112.113308	0.008920	0.572272	64.159261	0.015586	84
8	1.892457	133.868583	0.007470	0.528414	70.737970	0.014137	96
9	2.049530	157.429535	0.006352	0.487917	76.812497	0.013019	108
10	2.219640	182.946035	0.005466	0.450523	82.421481	0.012133	120
11	2.403869	210.580392	0.004749	0.415996	87.600600	0.011415	132
12	2.603389	240.508387	0.004158	0.384115	92.382800	0.010825	144
13	2.819469	272.920390	0.003664	0.354677	96.798498	0.010331	156
14	3.053484	308.022574	0.003247	0.327495	100.875784	0.009913	168
15	3.306921	346.038222	0.002890	0.302396	104.640592	0.009557	180
16	3.581394	387.209149	0.002583	0.279221	108.116871	0.009249	192
17	3.878648	431.797244	0.002316	0.257822	111.326733	0.008983	204
18	4.200574	480.086128	0.002083	0.238063	114.290596	0.008750	216
19	4.549220	532.382966	0.001878	0.219818	117.027313	0.008545	228
20	4.926803	589.020416	0.001698	0.202971	119.554292	0.008364	240
21	5.335725	650.358746	0.001538	0.187416	121.887606	0.008204	252
22	5.778588	716.788127	0.001395	0.173053	124.042099	0.008062	264
23	6.258207	788.731114	0.001268	0.159790	126.031475	0.007935	276
24	6.777636	866.645333	0.001154	0.147544	127.868388	0.007821	288
25	7.340176	951.026395	0.001051	0.136237	129.564523	0.007718	300
26	7.949407	1042.411042	0.000959	0.125796	131.130668	0.007626	312
27	8.609204	1141.380571	0.000876	0.116155	132.576786	0.007543	324
28	9.323763	1248.564521	0.000801	0.107253	133.912076	0.007468	336
29	10.097631	1364.644687	0.000733	0.099033	135.145031	0.007399	348
30	10.935730	1490.359449	0.000671	0.091443	136.283494	0.007338	360

ANNUAL COMPOUND INTEREST TABLE							
Effective Rate = 10.00%		Base = 1.1		Ann Convers. Periods= 1			
10.00%	1	2	3	4	5	6	10.00%
Annual	Amount of 1 @ Comp Interest	Accumulation of 1 Per Period	Sinking Fund Factor	Pres Val Reversion of 1	Pres Val Annuity 1/Period	Installment to Amortize	Annual
Years							Years
1	1.100000	1.000000	1.000000	0.909091	0.909091	1.100000	1
2	1.210000	2.100000	0.476190	0.826446	1.735537	0.576190	2
3	1.331000	3.310000	0.302115	0.751315	2.486852	0.402115	3
4	1.464100	4.641000	0.215471	0.683013	3.169865	0.315471	4
5	1.610510	6.105100	0.163797	0.620921	3.790787	0.263797	5
6	1.771561	7.715610	0.129607	0.564474	4.355261	0.229607	6
7	1.948717	9.487171	0.105405	0.513158	4.868419	0.205405	7
8	2.143589	11.435888	0.087444	0.466507	5.334926	0.187444	8
9	2.357948	13.579477	0.073641	0.424098	5.759024	0.173641	9
10	2.593742	15.937425	0.062745	0.385543	6.144567	0.162745	10
11	2.853117	18.531167	0.053963	0.350494	6.495061	0.153963	11
12	3.138428	21.384284	0.046763	0.318631	6.813692	0.146763	12
13	3.452271	24.522712	0.040779	0.289664	7.103356	0.140779	13
14	3.797498	27.974983	0.035746	0.263331	7.366687	0.135746	14
15	4.177248	31.772482	0.031474	0.239392	7.606080	0.131474	15
16	4.594973	35.949730	0.027817	0.217629	7.823709	0.127817	16
17	5.054470	40.544703	0.024664	0.197845	8.021553	0.124664	17
18	5.559917	45.599173	0.021930	0.179859	8.201412	0.121930	18
19	6.115909	51.159090	0.019547	0.163508	8.364920	0.119547	19
20	6.727500	57.274999	0.017460	0.148644	8.513564	0.117460	20
21	7.400250	64.002499	0.015624	0.135131	8.648694	0.115624	21
22	8.140275	71.402749	0.014005	0.122846	8.771540	0.114005	22
23	8.954302	79.543024	0.012572	0.111678	8.883218	0.112572	23
24	9.849733	88.497327	0.011300	0.101526	8.984744	0.111300	24
25	10.834706	98.347059	0.010168	0.092296	9.077040	0.110168	25
26	11.918177	109.181765	0.009159	0.083905	9.160945	0.109159	26
27	13.109994	121.099942	0.008258	0.076278	9.237223	0.108258	27
28	14.420994	134.209936	0.007451	0.069343	9.306567	0.107451	28
29	15.863093	148.630930	0.006728	0.063039	9.369606	0.106728	29
30	17.449402	164.494023	0.006079	0.057309	9.426914	0.106079	30

MONTHLY COMPOUND INTEREST TABLE							
10.00%	Effective Rate = 0.83%		Base =	1.008333	Ann Convers. Periods		12
	1	2	3	4	5	6	10.00%
	Amount of 1 @ Comp Interest	Accumulation of 1 Per Period	Sinking Fund Factor	Pres Val Reversion of 1	Pres Val Annuity 1/Period	Installment to Amortize	
Mths							Mths
1	1.008333	1.000000	1.000000	0.991736	0.991736	1.008333	1
2	1.016736	2.008333	0.497925	0.983539	1.975275	0.506259	2
3	1.025209	3.025069	0.330571	0.975411	2.950686	0.338904	3
4	1.033752	4.050278	0.246897	0.967350	3.918036	0.255230	4
5	1.042367	5.084031	0.196694	0.959355	4.877391	0.205028	5
6	1.051053	6.126398	0.163228	0.951427	5.828817	0.171561	6
7	1.059812	7.177451	0.139325	0.943563	6.772381	0.147659	7
8	1.068644	8.237263	0.121400	0.935765	7.708146	0.129733	8
9	1.077549	9.305907	0.107459	0.928032	8.636178	0.115792	9
10	1.086529	10.383456	0.096307	0.920362	9.556540	0.104640	10
11	1.095583	11.469985	0.087184	0.912756	10.469296	0.095517	11
Years							Mths
1	1.104713	12.565568	0.079583	0.905212	11.374508	0.087916	12
2	1.220391	26.446915	0.037812	0.819410	21.670855	0.046145	24
3	1.348182	41.781821	0.023934	0.741740	30.991236	0.032267	36
4	1.489354	58.722492	0.017029	0.671432	39.428160	0.025363	48
5	1.645309	77.437072	0.012914	0.607789	47.065369	0.021247	60
6	1.817594	98.111314	0.010193	0.550178	53.978665	0.018526	72
7	2.007920	120.950418	0.008268	0.498028	60.236667	0.016601	84
8	2.218176	146.181076	0.006841	0.450821	65.901488	0.015174	96
9	2.450448	174.053713	0.005745	0.408089	71.029355	0.014079	108
10	2.707041	204.844979	0.004882	0.369407	75.671163	0.013215	120
11	2.990504	238.860493	0.004187	0.334392	79.872986	0.012520	132
12	3.303649	276.437876	0.003617	0.302696	83.676528	0.011951	144
13	3.649584	317.950102	0.003145	0.274004	87.119542	0.011478	156
14	4.031743	363.809201	0.002749	0.248032	90.236201	0.011082	168
15	4.453920	414.470346	0.002413	0.224521	93.057439	0.010746	180
16	4.920303	470.436376	0.002126	0.203240	95.611259	0.010459	192
17	5.435523	532.262780	0.001879	0.183975	97.923008	0.010212	204
18	6.004693	600.563216	0.001665	0.166536	100.015633	0.009998	216
19	6.633463	676.015601	0.001479	0.150751	101.909902	0.009813	228
20	7.328074	759.368836	0.001317	0.136462	103.624619	0.009650	240
21	8.095419	851.450244	0.001174	0.123527	105.176801	0.009508	252
22	8.943115	953.173779	0.001049	0.111818	106.581856	0.009382	264
23	9.879576	1065.549097	0.000938	0.101219	107.853730	0.009272	276
24	10.914097	1189.691580	0.000841	0.091625	109.005045	0.009174	288
25	12.056945	1326.833403	0.000754	0.082940	110.047230	0.009087	300
26	13.319465	1478.335767	0.000676	0.075078	110.990629	0.009010	312
27	14.714187	1645.702407	0.000608	0.067962	111.844605	0.008941	324
28	16.254954	1830.594523	0.000546	0.061520	112.617635	0.008880	336
29	17.957060	2034.847258	0.000491	0.055688	113.317392	0.008825	348
30	19.837399	2260.487925	0.000442	0.050410	113.950820	0.008776	360

ANNUAL COMPOUND INTEREST TABLE							
Effective Rate = 12.00%		Base =		1.12	Ann Convers. Periods =		1
12.00%	1	2	3	4	5	6	12.00%
Annual	Amount of	Accumulation	Sinking	Pres Val	Pres Val	Installment	Annual
	1 @ Comp	of 1 Per	Fund	Reversion	Annuity	to	
Years	Interest	Period	Factor	of 1	1/Period	Amortize	Years
1	1.120000	1.000000	1.000000	0.892857	0.892857	1.120000	1
2	1.254400	2.120000	0.471698	0.797194	1.690051	0.591698	2
3	1.404928	3.374400	0.296349	0.711780	2.401831	0.416349	3
4	1.573519	4.779328	0.209234	0.635518	3.037349	0.329234	4
5	1.762342	6.352847	0.157410	0.567427	3.604776	0.277410	5
6	1.973823	8.115189	0.123226	0.506631	4.111407	0.243226	6
7	2.210681	10.089012	0.099118	0.452349	4.563757	0.219118	7
8	2.475963	12.299693	0.081303	0.403883	4.967640	0.201303	8
9	2.773079	14.775656	0.067679	0.360610	5.328250	0.187679	9
10	3.105848	17.548735	0.056984	0.321973	5.650223	0.176984	10
11	3.478550	20.654583	0.048415	0.287476	5.937699	0.168415	11
12	3.895976	24.133133	0.041437	0.256675	6.194374	0.161437	12
13	4.363493	28.029109	0.035677	0.229174	6.423548	0.155677	13
14	4.887112	32.392602	0.030871	0.204620	6.628168	0.150871	14
15	5.473566	37.279715	0.026824	0.182696	6.810864	0.146824	15
16	6.130394	42.753280	0.023390	0.163122	6.973986	0.143390	16
17	6.866041	48.883674	0.020457	0.145644	7.119630	0.140457	17
18	7.689966	55.749715	0.017937	0.130040	7.249670	0.137937	18
19	8.612762	63.439681	0.015763	0.116107	7.365777	0.135763	19
20	9.646293	72.052442	0.013879	0.103667	7.469444	0.133879	20
21	10.803848	81.698736	0.012240	0.092560	7.562003	0.132240	21
22	12.100310	92.502584	0.010811	0.082643	7.644646	0.130811	22
23	13.552347	104.602894	0.009560	0.073788	7.718434	0.129560	23
24	15.178629	118.155241	0.008463	0.065882	7.784316	0.128463	24
25	17.000064	133.333870	0.007500	0.058823	7.843139	0.127500	25
26	19.040072	150.333934	0.006652	0.052521	7.895660	0.126652	26
27	21.324881	169.374007	0.005904	0.046894	7.942554	0.125904	27
28	23.883866	190.698887	0.005244	0.041869	7.984423	0.125244	28
29	26.749930	214.582754	0.004660	0.037383	8.021806	0.124660	29
30	29.959922	241.332684	0.004144	0.033378	8.055184	0.124144	30

MONTHLY COMPOUND INTEREST TABLE							
12.00%	Effective Rate = 1.00%		Base =	1.01	Ann Convers. Periods		12
	1	2	3	4	5	6	12.00%
	Amount of 1 @ Comp Interest	Accumulation of 1 Per Period	Sinking Fund Factor	Pres Val Reversion of 1	Pres Val Annuity 1/Period	Installment to Amortize	
Mths							Mths
1	1.010000	1.000000	1.000000	0.990099	0.990099	1.010000	1
2	1.020100	2.010000	0.497512	0.980296	1.970395	0.507512	2
3	1.030301	3.030100	0.330022	0.970590	2.940985	0.340022	3
4	1.040604	4.060401	0.246281	0.960980	3.901966	0.256281	4
5	1.051010	5.101005	0.196040	0.951466	4.853431	0.206040	5
6	1.061520	6.152015	0.162548	0.942045	5.795476	0.172548	6
7	1.072135	7.213535	0.138628	0.932718	6.728195	0.148628	7
8	1.082857	8.285671	0.120690	0.923483	7.651678	0.130690	8
9	1.093685	9.368527	0.106740	0.914340	8.566018	0.116740	9
10	1.104622	10.462213	0.095582	0.905287	9.471305	0.105582	10
11	1.115668	11.566835	0.086454	0.896324	10.367628	0.096454	11
Years							Mths
1	1.126825	12.682503	0.078849	0.887449	11.255077	0.088849	12
2	1.269735	26.973465	0.037073	0.787566	21.243387	0.047073	24
3	1.430769	43.076878	0.023214	0.698925	30.107505	0.033214	36
4	1.612226	61.222608	0.016334	0.620260	37.973959	0.026334	48
5	1.816697	81.669670	0.012244	0.550450	44.955038	0.022244	60
6	2.047099	104.709931	0.009550	0.488496	51.150391	0.019550	72
7	2.306723	130.672274	0.007653	0.433515	56.648453	0.017653	84
8	2.599273	159.927293	0.006253	0.384723	61.527703	0.016253	96
9	2.928926	192.892579	0.005184	0.341422	65.857790	0.015184	108
10	3.300387	230.038689	0.004347	0.302995	69.700522	0.014347	120
11	3.718959	271.895856	0.003678	0.268892	73.110752	0.013678	132
12	4.190616	319.061559	0.003134	0.238628	76.137157	0.013134	144
13	4.722091	372.209054	0.002687	0.211771	78.822939	0.012687	156
14	5.320970	432.096982	0.002314	0.187936	81.206434	0.012314	168
15	5.995802	499.580198	0.002002	0.166783	83.321664	0.012002	180
16	6.756220	575.621974	0.001737	0.148012	85.198824	0.011737	192
17	7.613078	661.307751	0.001512	0.131353	86.864707	0.011512	204
18	8.578606	757.860630	0.001320	0.116569	88.343095	0.011320	216
19	9.666588	866.658830	0.001154	0.103449	89.655089	0.011154	228
20	10.892554	989.255365	0.001011	0.091806	90.819416	0.011011	240
21	12.274002	1127.400210	0.000887	0.081473	91.852698	0.010887	252
22	13.830653	1283.065279	0.000779	0.072303	92.769683	0.010779	264
23	15.584726	1458.472574	0.000686	0.064165	93.583461	0.010686	276
24	17.561259	1656.125905	0.000604	0.056944	94.305647	0.010604	288
25	19.788466	1878.846626	0.000532	0.050534	94.946551	0.010532	300
26	22.298139	2129.813909	0.000470	0.044847	95.515321	0.010470	312
27	25.126101	2412.610125	0.000414	0.039799	96.020075	0.010414	324
28	28.312720	2731.271980	0.000366	0.035320	96.468019	0.010366	336
29	31.903481	3090.348134	0.000324	0.031345	96.865546	0.010324	348
30	35.949641	3494.964133	0.000286	0.027817	97.218331	0.010286	360

Abbreviations used in IAAO Course 102

BOI	Band of Investment
CR	Constant Rate of Change
DCR	Debt Coverage Ratio
EAT	Effective Tax Rate, Assessment Level, & Tax Rate formula
EGI	Effective Gross Income
EGIM	Effective Gross Income Multiplier
ETR	Effective Tax Rate
GIM	Gross Income Multiplier (sometimes known as Gross Rent Multiplier)
GLA	Gross Leasable Area
GRM	Gross Rent Multiplier (sometimes known as Gross Income Multiplier)
I _B	Income to the Building (Improvement)
I _E	Income to the Equity
I _L	Income to the Land
I _M	Income to the Mortgage (income necessary to pay principal & interest)
IRV	Income Rate & Value formula
I _v	Income to total property value (Net Operating Income)
NIR	Net Income Ratio (also known as Net Operating Income Ratio or 1 - OER)
NLA	Net Leasable Area
NOI	Net Operating Income
OER	Operating Expense Ratio (or 1 - NIR)
PAV	Property Assessment Valuation (published 1996)
PGI	Potential Gross Income
PW	Present Worth

R_B	Building Capitalization Rate (also known as the Improvement Capitalization Rate)
R_E	Equity Capitalization Rate (also known as the Equity Dividend Rate or Cash on Cash Rate)
R_L	Land Capitalization Rate
R_M	Mortgage Capitalization Rate
R_O	Overall Capitalization Rate (also known as OAR)
V_B	Value of the Building (Improvement)
V_E	Value of Equity
VIF	Value Income & Factor formula
V_L	Value of the Land
V_M	Value of the Mortgage (Mortgage Principal)
V&C	Vacancy & Collection Loss
Y_B	Building Yield Rate
Y_E	Equity Yield Rate
Y_L	Land Yield Rate
Y_M	Mortgage Yield Rate (Mortgage Interest Rate or lender's yield on the loan)
Y_O	Overall Yield Rate (also known as the Discount Rate or "r")

A Note on Rounding

Rounding is a way of simplifying a number while keeping its value relatively unchanged. The resulting number is not as accurate, but much easier to use.

The two most common questions raised by rounding are: 1) When should a number be rounded? and 2) To what extent? While these questions may seem simple, it's difficult to provide a conclusive response that covers every eventuality. The degree of precision required will vary depending on the purpose of a calculation.

For example, calculating the number of staff persons needed to accomplish a specific function within a given time period is only meaningful when the result is rounded to the nearest whole number. On the other hand, using state plan coordinates to locate a parcel may only be meaningful using numbers carried out to six decimal places.

Unless there are instructions to the contrary, students should avoid rounding within the intermediate steps of a multi-step calculation. Rounding should be left for the final result and should conform to the customary presentation of that particular calculation.

Glossary

Note:

Terms in the Glossary are taken from the textbook, Fundamentals of Tax Policy. Most of these definitions were developed by previous contributors to the 2003 version of the Tax Policy Course. The contributions of earlier authors are gratefully acknowledged.

A

Abatement—(1) An official reduction or elimination of one's assessed valuation after completion of the original assessment. (2) An official reduction or elimination of one's tax liability after completion of the assessment roll.

Acquisition Value System of Property Tax Assessment—A system of valuing property at its market value as of the last transfer of ownership or of the last major physical change. A property is placed on the tax roll at its acquisition value. Values usually are permitted only limited annual increases but may be updated when major physical changes occur or when the property is sold. The system established by California's Proposition 13 is an example.

Ad Valorem Tax—A tax levied in proportion to the value of the thing being taxed. The property tax is an ad valorem tax.

Appeal—A process in which a property owner contests an assessment either informally or formally.

Appraisal—The act of estimating the money value of property.

Assessment—The official act of discovering, listing, and appraising property.

Assessment Level—The common or overall ratio of assessed values to market values.

Assessment Progressivity (Regressivity)—An appraisal bias such that high-value properties are appraised higher (or lower) than other properties in relation to market values.

Assessment Ratio—(1) The fractional relationship an assessed value bears to the market value of the property in question. (2) By extension, the fractional relationship the total of the assessment roll bears to the total market value of all taxable property in a jurisdiction.

Audit—A systematic investigation or appraisal of procedures or operations for the purpose of determining conformity with specifically prescribed criteria.

Audit, Performance—An analysis of an organization to determine whether or not the quantity and quality of work performed meets standards. Ratio studies are an important part of performance audits of an assessing organization.

Audit, Procedural—An examination of an organization to determine whether established or recommended procedures are being followed.

Average Tax Rate—Total taxes paid divided by the total base.

Average Total Cost—The total cost to produce something divided by the number of units produced.

B

Benefit-Based Taxes—Taxes linked, often by earmarking, to the consumption of particular government services.

Block Grants—Grants that are designated for a broad set of uses, such as economic development. They are not as restrictive as categorical grants, with broader goals, more discretion, and fewer administrative requirements. However, block grants are not as flexible as revenue sharing.

Border Effect—Tax administration problems that arise because shoppers or commuters can evade or avoid certain taxes by crossing the border from one taxing jurisdiction to another.

Buoyancy—The ability of tax yields to rise (and fall) with the economy and with revenue needs. Buoyancy is a characteristic of value-based property tax systems but assessed valuations must be updated as the underlying market values change.

C

Capitalization—Used in terms of the value of land being lower in a jurisdiction with a high tax/low public service fiscal structure; alternatively, in terms of the value of land being higher in a jurisdiction with a low-tax/high-public-service fiscal structure.

Categorical Grants—Grants that are designated for a particular use, such as development of municipal parks.

Charges—Non-tax revenue from fees and reimbursements for current services.

Circuit Breaker—For qualifying property owners, a credit or rebate of specified amounts of property taxes incurred whenever such taxes exceed specified percentages or amounts of household income. In instances in which renters are included, rent, or rent equivalents substitute for property taxes.

Classification—The act of segregating property into two or more classes for the application of different effective tax rates.

Classified Property Tax System—A system intended by law to tax various kinds of property at different effective tax rates. Common approaches are establishing by law either that property is to be uniformly assessed but taxed differently by class, or that property is to be taxed at a uniform rate but assessed at different assessment ratios by class.

Coefficient of Dispersion (COD)—In ratio studies, a measure of uniformity calculated as the average percentage deviation from the median ratio.

Coefficient of Variation (COV)—A standard statistical measure of relative dispersion of sample data about the mean of the data. In ratio studies, a measure of uniformity calculated by expressing the standard deviation of the ratios as a percentage of the mean ratio.

Consensus Forecasting—Collective agreement by a group of knowledgeable individuals in the finance department about a future revenue/expenditure stream.

Consumer Surplus—The amount by which consumers' willingness to pay for a commodity exceeds the sum they actually have to pay.

Credit, Property Tax—An offset against the property tax payment for taxpayers who meet certain criteria or whose properties have certain characteristics or are used for specified purposes; a direct reduction in a tax payment rather than in a tax base.

D

Delphi Forecasting—A multiple-step method that involves predictions of expert individuals that are then reconsidered after other expert individuals comment on the forecasts.

Differential Taxation—A legislated difference in effective tax rates between groups of properties classified according to use, value, or some other criterion. Differentials may be affected by applying different legal assessment ratios, tax rates, or both. Systems that provide differentials are generally known as classified property tax systems in the United States.

Dillon's Rule—Legal interpretation that maintains that local governments are not power centers in their own right but derive all their powers from state governments.

Direct Taxation—Taxes that are imposed on the household that is meant to bear the burden.

E

Economic Incidence—A change in the distribution of real income induced by the tax.

Economies of Scale—Unit costs fall as the scale of production rises. For example, if it costs \$8,000 per student to educate a high school student in a school with 500 to 800 students, but only \$6,000 in a school with 1,000 to 2,500 students, then economies of scale exist.

Economies of Scale in Production—The average total cost to produce something continually falls as more of it is produced, and it can be beneficial to have only one producer (a natural monopoly). This is usually due to a large fixed cost of production.

Effective Tax Rate—For property tax, the tax rate expressed as a percentage of market value; will be different from the nominal tax rate when the assessment ratio is not equal to 1.

Efficiency—An allocation of resources such that no one person can be made better off without making another person worse off; includes both productive efficiency (no wasted resources) and allocative efficiency (resources are allocated to meet consumer demand).

Elasticity (Tax)—A measure of the responsiveness of tax yields to changes in economic conditions. The yield of an elastic tax increases rapidly in a growing economy. The yield of an inelastic tax increases slowly. Often measured by the formula:

Percentage change in tax ÷ Percentage change in personal income

Equalization—The process by which an appropriate governmental body attempts to ensure that all property under its jurisdiction is assessed at the same assessment ratio or at the statutorily required ratio(s). This process may be direct, involving adjustments to taxable values of individual properties, or indirect, usually involving adjustments to overall values for property classes or jurisdictions. Such adjusted overall values are then used to determine the proper distribution for state aid to local governments or to create uniform effective tax rates.

Equity—(1) In assessment, the degree to which assessments bear a consistent relationship to market value. Measures include the coefficient of dispersion, coefficient of variation, and price-related differential. (2) In popular usage, a synonym for tax fairness. (3) In ownership, the net value of property after liens and other charges have been subtracted.

Exemption—A specified category of income or spending that is not included in the tax base. In property taxation, an exclusion of part or all of the value of a category of property from taxation.

Expert Forecasting—Collective agreement by a group of knowledgeable individuals that extends outside of the finance department about a future revenue/expenditure stream.

Externality—Produced when the costs (benefits) faced by a producer (consumer) of a good/service do not represent all of the costs (benefits) generated from production (consumption).

F

Fairness—See Equity.

Federal Conformity—The extent to which a feature of state or local tax law is the same as federal tax law.

Federalism—Form of government in which power is constitutionally divided between a center or national government, and states or regional centers.

Federalism, Coercive—A degenerative form of cooperative federalism in which the Federal Government dominates state and local governments.

Federalism, Competitive—A system of government in which competition among governments is a major means of coordinating actions by the governments. The national government competes with states, states compete with each other, and local governments compete with each other.

Federalism, Cooperative—A system of government in which there is interdependence among the different types of government (for example, between national and state governments).

Federalism, Dual—A system of government under which the national government and the states have separate sets of responsibilities.

Federalist System of Government—A system of government in which the distribution of power is shared between a central authority (Federal Government) and the constituent units; each of the constituent units (state and local governments) is more or less self-governing.

Fiscal Capacity—A measure of the own-source revenues a state could raise if it had a “representative tax system” and applied the national average tax rate to its own amounts of the major sources of taxes and charges.

Fiscal Comfort—An index that combines fiscal capacity and fiscal need measures into a single measure. Determined from the ratio of fiscal capacity to fiscal need.

Fiscal Effort—"The ratio of each state's actual tax collections to the taxes it would have collected under the representative tax system" (Tannenwald 1999).

Fiscal Federalism—An approach to federalism that emphasizes the nature of the goods and services that government provides and derives the structure of government from that analysis. (Sometimes the term is used in a more general sense to refer to the financial analysis of federal systems of government.)

Fiscal Need—A measure of would-be revenue requirements if a state had a "representative expenditure system" and supplied the national average level of services to its own amounts of "workload" variables.

Flat Tax—A tax with a single rate applying to all income for a specified tax-exempt level. Note that the existence of a tax-exempt level of income means that the burden of the tax can still be progressive.

Fractional Assessments—Assessments that by law or by practice have assessment ratios different from 1. Usually the assessment ratio is less than 1, and if assessment biases are present, different classes of property may have different fractional ratios. Fractional assessments are often condemned as offering a way to obscure assessment biases.

G

General Equilibrium—The study of the interrelationships among various sectors.

General Revenue—All government revenue except from government-operated public utilities, government-operated liquor store revenue, and for unemployment compensation, employee retirement, workers' compensation, or other government trust funds.

General Sales Tax—A tax levied as a fixed percentage on the dollar value of a broad category of retail purchases.

H

Homestead Exemption—Exclusion of part or all of the value of a residence, usually an owner-occupied primary residence; a reduction in the property tax base.

Horizontal Equity—Individuals/households in equal economic situations should be treated the same. In property assessment, properties of the same value should be valued the same.

I

Incidence—Usually refers to the persons who ultimately bear a tax burden as opposed to the persons on whom the tax is initially imposed.

Income Incidence—The determination of the percentage of a person's income that is devoted to paying a tax. For a given tax, if this percentage rises (falls) as income rises, the tax is considered progressive (regressive).

Indirect Taxation—Taxes that are imposed on a commodity that are then shifted to the household.

Individual Income Tax—A tax levied on the net income of an individual. In most cases these are broad-based and include income from labor and other sources, but the Census definition includes distinctive taxes on income from interest or dividends.

Intergovernmental Revenue—"Revenue received from other governments as grants-in-aid, shared revenues, payments in lieu of taxes, or a reimbursement for the performance of services for the paying government" (Hoffman 2002).

L

Level of Assessment—The common or overall ratio of assessed values to market value. This may be a statutory ideal or may be determined by application of inferential statistics developed using ratio studies.

Leviathan—Often used as a metaphor for a government striving to grow as large as possible. Dictionary definitions include a sea monster mentioned in the Old Testament; a huge marine mammal, such as a whale, or anything of huge size.

Levy, Property Tax—(1) The total amount of money to be raised from the property tax as set forth in the budget of a taxing jurisdiction. (2) Loosely, by extension, the millage rate or the property tax bill sent to an individual property owner.

Lump-Sum Grant—A grant that does not depend upon how much the recipient government spends on a particular service.

M

Mandates—Federal regulations that impose costs on state and local governments. There are also state mandates on local governments.

Marginal Tax Rate—The proportion of the last dollar of income taxed by the government.

Market Value—Market value is the major focus of most real property appraisal assignments. Both economic and legal definitions of market value have been developed and refined. A current economic definition agreed upon by agencies that regulate federal financial institutions in the United States of America is:

The most probable price (in terms of money) that a property should bring in a competitive and open market under all conditions requisite to a fair sale, the buyer and seller each acting prudently and knowledgeably, and assuming the price is not affected by undue stimulus. Implicit in this definition is the consummation of a sale as of a specified date and the passing of title from seller to buyer under conditions whereby:

- The buyer and seller are typically motivated;
- Both parties are well informed or well advised, and acting in what they consider their best interests;
- A reasonable time is allowed for exposure in the open market;
- Payment is made in terms of cash in United States dollars or in terms of financial arrangements comparable thereto;
- The price represents the normal consideration for the property sold unaffected by special or creative financing or sales concessions granted by anyone associated with the sale.

Matching Grants—A matching grant provides a certain amount for each dollar spent by the recipient government. For example, a 50 percent matching grant would provide \$1 to the recipient government for each \$1 it spent on a particular function.

Millage, Mill Rate—A tax rate expressed as mills per dollar. For example, a 2 percent tax rate is \$2 per \$100, \$20 per \$1,000, or 20 mills per \$1. One mill is one-thousandth of \$1, or one-tenth of one cent.

Miscellaneous Revenue—Nontaxed, noncharged revenue derived from interest earnings, special assessments, sale of property, or other sources.

N

Naïve Forecasting—Assuming that historical relationships hold constant, this year's revenues are a constant function of last year's revenues.

Natural Monopoly—A monopoly, or a single producer of a good, that arises due to economies of scale throughout the relevant range of production of the good. This results in a continually declining average total cost to produce and the least expensive way to produce the good being a single producer.

Neutral/Neutrality—Refers to a tax that does not distort economic decisions. Closely related to the concept of efficiency, the situation in which patterns of consumption maximize the general welfare of society.

Nexus—The legal question of what constitutes sufficient presence in a state or jurisdiction to be held responsible for paying or collecting a tax.

Normative Statement—A statement that involves the use of value judgments and whose merits cannot be evaluated on facts alone.

O

Own-Source Revenue—Government funding that only comes from within the jurisdiction under consideration. For local governments, this means that it excludes revenue received from federal and state intergovernmental grants; it may include taxes, current charges, and miscellaneous revenue collected by the jurisdiction. Alternatively, general revenue minus intergovernmental revenue.

P

Partial Equilibrium—Models that study only one market (or sector) and ignore possible spillovers into other sectors.

Per Capita—Divided by the number of people in the population.

Positive Statement—A statement that involves the use of no value judgments and whose merits can be evaluated on facts alone.

Price-Related Differential (PRD)—A statistical measure of vertical property tax equity. The PRD is calculated by dividing the mean ratio by the weighted mean ratio in a ratio study. If the result exceeds 1.03, assessments are considered regressive. If the result is less than 0.98, assessments are considered progressive.

Privilege Tax—A special tax on the right to start, expand, or continue a business (syn. franchise tax).

Progressive Taxation—A tax system under which a taxpayer's average tax rate increases with income.

Progressivity—*Income Progressivity* refers to a situation in which people with higher incomes pay a higher percentage of income in taxes than persons with lower incomes. *Assessment Progressivity* refers to a situation in which effective property tax rates on higher value properties are greater than effective property tax rates on lower value properties. The opposite of Regressivity.

Proportional Taxation—A tax system under which a taxpayer's average tax rate is the same at each level of income.

Proposition 13—The 1978 California property tax limit proposition that replaced current market value with acquisition value as the basis for property taxation.

Public Good—A good/service that once produced can be jointly consumed, and it is difficult to keep a nonpayer from consuming. This is the opposite of a private good (such as an apple) that as a whole cannot be jointly consumed, and another can easily be stopped from consuming.

R

Rate-Driven Levy—The property tax rate to be applied is specified in the budget or tax levy ordinance of a taxing jurisdiction, in contrast to the usual situation in which the total revenue to be raised is specified, and the rate is calculated.

Ratio Study—A statistical study of the relationship between appraised or assessed values and market values; based on an analysis of the ratio derived by dividing the appraised or assessed values of property by the market values of such property. Sale prices or independent appraisals are used as proxies for market values.

Real Dollars—Adjusted for inflation relative to a reference or "base" year. Divided by the price index for the current year relative to the price index of the base year.

Regressive Taxation—A tax system under which a taxpayer's average tax rate decreases with income.

Regressivity—*Income Regressivity* refers to a situation in which people with lower incomes pay a higher percentage of income in taxes than persons with higher incomes. *Assessment Regressivity* refers to situations in which effective property tax rates on lower value properties are greater than effective property tax rates on higher value properties. The opposite of Progressivity.

Representative Expenditure System—A method of constructing national average rates of expenditure for many categories of outlays as the ratio of each type of spending to a corresponding "workload" measure such as population, number of pupils, vehicle-miles traveled, or poverty counts.

Representative Tax System—A method of constructing national average rates of tax for many types of taxes and charges as the ratio of each type of revenue to a corresponding measure of the tax base.

Revenue Elasticity—How sensitive the revenue from a particular tax is to (a) cyclical changes or (b) trends in overall economic activity.

Revenue-Sharing Grants—Grants used for almost any governmental purpose. For example, a municipality receiving a revenue-sharing grant might use the grant for fire prevention, welfare, or garbage pickup.

S

School Aid Equalization—A system of state revenue sharing to local schools based upon guaranteeing that each school district has the same property tax base per student upon which to levy local property taxes to fund per-student expenditure on public education.

Selective Sales Tax—A tax levied on the purchase of a specific type of good or service.

Specific Taxation—A tax imposed per unit of the product.

Spillover Effects—Benefits or costs of a good or service that accrue to individuals who do not pay for that good or service (same as externality). Spillovers can be positive (beneficial) or negative.

Statutory Incidence—The legal burden of the tax.

Sumptuary Taxes—Excise taxes designed to discourage the consumption of specific items, such as cigarettes or liquor. Colloquially referred to as sin taxes.

Sunset Provision—A provision within a statute creating a law or agency and providing for the automatic termination of that law or agency at a fixed date in the future.

T

TANF—Temporary Aid to Needy Families (TANF) is commonly referred to as welfare and replaced Aid to Families with Dependent Children (ADFC) in 1996. TANF is administered through a federal block grant to states that requires state welfare programs where most recipients must work within two years of receiving assistance, limits most assistance to five years total, and lets states establish “family caps” to deny additional benefits to mothers for children born while the mothers are already on public assistance.

Tax Assignment—A theory that determines which tax or revenue instruments should best be employed by the national government, which should best be employed by state governments, and which should best be employed by local governments.

Tax Avoidance—Altering behavior in such a way as to reduce legal tax liability.

Tax Burden—Economic costs or losses resulting from the imposition of a tax. Burden can be determined only by detailed economic analysis of all economic changes resulting from the tax. In popular usage, the term often refers to the initial incidence rather than ultimate economic costs.

Tax Capacity—The amount of tax that would be collected using an overall average tax rate. For example, for the tax capacity of a particular state, the overall amount raised nationally by the tax would be divided by total personal income (or population). The resulting national average tax rate would then be multiplied by the total personal income (or population) of the state to determine the tax capacity of that state.

Tax Effort—The percent of tax capacity utilized by a state. The actual tax collections divided by the state's tax capacity.

Tax Elasticity—Measure of the responsiveness of tax yield to changes in economic conditions; often measured by the formula:

percentage change in tax ÷ percentage change in economic base
with the economic base often being personal income.

Tax Evasion—Not paying taxes legally due.

Tax Incidence—Refers to the eventual distribution of the burden of a tax.

Tax Incidence Analysis—Economic analysis that compares the way different taxes affect the distribution of income; requires analysis of the impact of taxes on the market for the taxed item and the market for all factors (land, labor, and capital) used in producing the taxed item.

Tax Increment Financing (TIF)—The idea that property taxes, or other revenue, resulting from the increase in a tax base (for example, property values or retail sales) in a specific area can be used to repay the costs of investment in that area. Funds may be invested in various programs, such as public infrastructure improvements or land write-down subsidies to private investors.

Tax Policy Analysis—The process of gathering and interpreting economic data to provide information that can be used by policy makers to formulate tax policy.

Truth-in-Taxation (Full Disclosure) Requirements—Legal obligations for local government officials to make taxpayers aware of assessment increases, levy increase proposals, and the like and to give taxpayers an opportunity to participate in public hearings on the changes.

U

Uniformity—In property taxation, the principle that the same tax rate is applied to every property.

Unit Tax—A tax levied on the quantity purchased as opposed to the dollar amount purchased. Examples are cigarette taxes per pack and gasoline taxes per gallon.

Unitary Government—A hierarchical system of government in which state and local governments are de facto departments of the national government.

Use Tax—A companion to the general sales tax designed to cover out-of-state purchases.

Use-Value Assessment Laws—Laws that require or permit appraisal and assessment of property based on a particular restriction or use. Typically, use value laws are applied to agricultural land, timber land, or historical sites, permitting assessments to be lower than market value and thereby lessening the proportional share of property tax paid by these classes of property.

V

Vertical Equity—Distributing tax burdens fairly across people with different abilities to pay.

W

Wealth—Valuable material objects that are owned, either individually or collectively; that is, all tangible property. Note: In popular usage the term “wealth” is synonymous with “property” and, as such, embraces intangibles as well as tangible property. This usage is considered incorrect by economists and is not recommended. Intangible property, with the possible exception of goodwill, patents, and the like, is not a real source of income, but only a means of distributing income derived from the two primary sources, tangible property and persons. The adding together of tangible property and intangibles to secure total wealth results in multiple counting of the same values. Some authorities consider nonrepresentative intangible property as wealth, but this usage has received only limited acceptance.

Welcome Stranger Assessment—The practice of systematically assessing recently sold properties on the basis of their sales prices, while failing to reassess similar properties that have not recently sold.

References

- Hoffman, D. 2002. *Facts and figures on government finance*, 36th ed. Washington, DC: Tax Foundation.
- Tannenwald, R. 1999. Fiscal disparity among the states revisited. *New England Economic Review*. (Boston: Federal Reserve Bank of Boston) (July/August):3–25,
<http://www.bos.frb.org/economic/neer/neer1999/neer499a.htm> (accessed January 24, 2008).

PAGE LEFT BLANK INTENTIONALLY

We are IAAO ...

INTERNATIONAL ASSOCIATION OF ASSESSING OFFICERS

We're a professional membership organization of government assessment officials and others interested in the administration of the property tax. We were founded in 1934, and have more than 8,000 members worldwide from governmental, business, and academic communities.

WE are the internationally recognized leader and preeminent source for innovation, education, and research in property appraisal, assessment administration and property tax policy.

WE are IAAO, and **WE** value the world!



IAAO Designations

Who should earn IAAO designation?

Any professional who aspires to advance their career in appraising, mapping, property assessment, and property tax policy at any level - in government or in the private sector - should have an IAAO designation.

Why pursue an IAAO Designation?

A designation can elevate your career. Designated members can earn an increase in salary and responsibility. A designation instantly establishes your qualifications and credibility and is a universal, portable measure of your capabilities recognized around the world. Designees report gaining confidence in their abilities and the respect of their peers and those they serve. Earning a designation is a rewarding experience, demonstrating that you have attained the highest level of professionalism.

What is required?

To earn an IAAO Designation, one must be an IAAO member in good standing and complete the requirements for the particular designation pursued. Each designation has experiential, educational, demonstration project, and examination requirements.

IAAO Designations



CAE

The Certified Assessment Evaluator

designation is IAAO's flagship designation, denoting the highest level of education and experience in residential and commercial real-property appraisal, assessment administration, and tax policy. The CAE is for anyone who holds, or aims to hold, a senior position in their office.



RES

The Residential Evaluation Specialist

designation recognizes professionalism and competency in valuing residential real property for tax purposes. The RES is designed for those who are well-practiced in appraising a wide range of residential property for their office.



MAS

The Mass Appraisal Specialist

designation demonstrates both the valuation and technical skills necessary for today's more advanced mass appraisal practices. The MAS represents professionalism and competency in a wide range of matters covering mass appraisal theories, techniques, and application.



AAS

The Assessment Administration Specialist

designation denotes expertise in administering various functions for property tax purposes. The AAS suits those who have experience working in an assessment office, especially with administrative or tax policy matters.



CMS

The Cadastral Mapping Specialist

designation symbolizes professionalism and competency in the cadastral mapping techniques that support proper valuation of property for tax purposes, property tax administration, and property tax policy. Those who produce maps and other graphic images in support of their offices' property appraisal and assessment administration functions should possess the CMS.



PPS

The Personal Property Specialist

designation signifies advanced expertise in the valuation of personal property for tax purposes. The PPS is held by those who wish to stand-out for their unique expertise in assessing personal property.

If you must defend your work, an IAAO Designation instantly establishes your qualifications and credibility.

Send your questions to: designations@iaao.org

Courses offered by IAAO

□ Course 101 - Fundamentals of Real Property Appraisal

The Fundamentals of Real Property Appraisal course is designed to provide students with an understanding and working knowledge of the procedures and techniques required to estimate the market value of vacant and improved properties. This course concentrates on the skills necessary for estimating the market value of properties using two approaches to value: the cost approach and the sales comparison approach. The Fundamentals of Real Property Appraisal utilizes lectures, classroom discussion, and homework problems to emphasize the main concepts and procedures taught in the course.

Recommended: Property Assessment Valuation (PAV) textbook (3rd edition)

Approved: AQB 33.50 hours CE with exam, 30 hours CE no exam / IAAO 30 hours CE

□ Course 102 - Income Approach to Valuation

The Income Approach to Valuation course is designed to provide students with an understanding and working knowledge of the procedures and techniques required to estimate the market value of vacant or improved properties by the income approach. The material covers real estate finance and investment, capitalization methods and techniques, analysis of income and expenses to estimate operating income, selection of capitalization rates, and application of the approach. The Income Approach to Valuation utilizes lectures, classroom discussion, and homework problems to emphasize the main concepts and procedures taught in the course.

Recommended: Course 101, Property Assessment Valuation (PAV) textbook (3rd edition)

Approved: AQB 33.50 QE, 33.50 hours CE with exam, 30 hours CE no exam / IAAO 30 hours CE

□ Course 112 - Income Approach to Valuation II

The Income Approach to Valuation II course is an intermediate level program designed for the practicing appraiser. It builds on the training of IAAO Course 102. An understanding of the income approach to value and practical experience with income capitalization are essential prerequisites. Problem solving requires a calculator with a power key. This course makes extensive use of financial compounding and discounting techniques. Emphasis is on developing financial factors by way of electronic devices with only slight references to preprinted tables. Forecasting income patterns and property value changes are prominent variables used in estimating present values. Formula driven models are the primary appraisal tools versus look up tables and vertical calculations. Market value estimates through yield capitalization are the principal focus of the course.

The five chapters are a combination of lecture aided in outline form, example Practical Applications, Student Exercises and Review Questions. PowerPoint slides assist in conveying the material. Case studies are included to help the student relate the problems to real life situations. Two quizzes help the students measure their progress. All exercises include suggested solutions.

Recommended: Course 101, Course 102, Property Assessment Valuation (PAV) Textbook (3rd edition)

Approved: AQB 33.50 QE, 33.50 hours CE with exam / 30 hours CE no exam / IAAO 30 hours CE

□ Course 201 - Appraisal of Land

The Appraisal of Land course is designed to provide students with an understanding and working knowledge of the procedures and techniques required to estimate the market value of land. This course concentrates on the skills necessary for estimating land value primarily using the sales comparison approach.

Recommended: Course 101, Course 102, Property Assessment Valuation (PAV) Textbook (3rd edition)

Approved: AQB 33.50 hours CE with exam, 30 hours CE no exam / IAAO 30 hours CE

Pursue your designation at: www.iaao.org

Courses Required for Designations

	Course 101	Course 102	Course 112	Course 201*	Course 300	Course 311	Course 331	Course 332	Course 333	Course 400	Course 402	Course 500	Course 600	Course 601	Workshop 151**	Workshop 171
CAE	X	X	X			X				X					X	X
RES	X	X		X	X	X									X	X
AAS	X	X			A	A				X	X				X	X
CMS	X												X	X	X	X
PPS	X	X										X			X	X
MAS	X	X			X		X	X	X						X	X

Notes:

For AAS, only one of the "A" courses is required. Not all required education may be listed for any designation. Project and exam requirements apply. Inquire for details.

**Only required if a case study exam is chosen over the demonstration report option.*

***Can substitute TAF 15 Hr National USPAP from any approved provider.*

□ Course 300 - Fundamentals of Mass Appraisal

The Fundamentals of Mass Appraisal course provides an introduction to mass appraisal and is a prerequisite for the 300 series of courses offered by the IAAO. Topics covered include single-property appraisal versus mass appraisal, components of a mass appraisal system, data requirements and analysis, introduction to statistics, use of assessment ratio studies in mass appraisal, modeling of the three approaches to value, and selection of a mass appraisal system.

Recommended: Course 101, Course 102, Fundamentals of Mass Appraisal (FMA) textbook

Approved: AQB 33.50 hours CE with exam, 30 hours CE no exam / IAAO 30 hours CE

□ Course 311 – Real Property Modeling Concepts

The Real Property Modeling Concepts course presents a detailed study of the mass appraisal process as applied to residential and income-producing properties. Topics covered include a comparison of single-property appraisal and mass appraisal, the major steps in the mass appraisal process, data requirements, market analysis, use of sales ratio studies, cost approach, sales comparison approach, gross and net income analysis, capitalization rate development, model specification and calibration, valuation review techniques and maintenance.

Recommended: Course 300

Approved: IAAO 30 hours CE

□ Course 331 - Mass Appraisal Practices and Procedures

The Mass Appraisal Practices and Procedures course is designed to build on the subject matter covered in Course 300 – Fundamentals of Mass Appraisal and prepare the student to take the more advanced mass appraisal courses. It teaches the student how they can use Excel and SPSS to analyze data and apply it. Much of the emphasis will be on data accumulation and analysis primarily directed towards the cost approach. Along the way the student will learn how to use the graphing and analysis tools within Excel for ratio studies in addition to supporting existing cost schedules or building new ones.

Recommended: Course 300, Fundamentals of Mass Appraisal textbook (FMA)

Approved: AQB 33.50 hours CE with exam, 30 hours CE no exam /
IAAO 30 hours CE

□ Course 332 - Modeling Concepts

The Modeling Concepts course introduces and explains fundamentals of mass appraisal model building. Chapter 1 explains the role of models in mass appraisal and describes basic model structures and steps in model development. Chapter 2 covers exploratory data analysis, including descriptive statistics, charts and graphs, and spatial analyses. Chapter 3 describes data transformations, which largely determine the accuracy achieved within a given database. Chapter 4 introduces multiple regression analysis, including regression statistics and options, the interpretation of results, and separation of land and building values. Chapter 5 covers the review and support of mass appraisal values and chapter 6 describes mass appraisal reporting.

(continues next page)

Pursue your designation at: www.iaao.org

(continued from previous page)

The course includes a large number of demonstrations and labs using Excel, which is capable of producing basic models for a limited number of variables. Follow-up courses 333 and 334 cover residential and commercial modeling, respectively, in more depth and use the powerful but user-friendly statistical package, SPSS, in labs and demonstrations. Thus, this course serves as a foundation and bridge to IAAO's two follow-up model building courses.

Recommended: Course 300, Fundamentals of Mass Appraisal textbook (FMA), A solid working knowledge of Excel

Approved: IAAO 30 hours CE

□ Course 333- Residential Modeling Applications

The Residential Modeling Applications course focuses on mass appraisal model building for residential properties. Chapter 1 explains the different types of mass appraisal models, types of data used in models, steps in model development, software tools available for modeling, and key regression concepts and statistics. Chapter 2 introduces the student to features contained in common statistical software packages. Chapter 3 covers basic statistical analyses, data transformations, charts and graphs, and data filters and sub-files. Chapter 4 focuses on the development and application of price trends and vacant land models. Chapter 5 focuses on the development of improved residential models. Chapter 6 introduces automated comparable sales models. Along with problems and illustrations, more than half of the course is based on hands-on labs in which students work through exercises with real world data to answer questions and construct data transformations, price trends, land and improved residential models, and ratio analyses.

The course uses IBM's user-friendly statistical software package, SPSS (Statistical Package for the Social Sciences), the most common statistical software package used by assessors. Follow-up course 334 covers the development of non-residential mass appraisal models.

Recommended: Course 300, Fundamentals of Mass Appraisal textbook (FMA), A solid working knowledge of SPSS

Approved: IAAO 30 hours CE

□ Course 400 - Assessment Administration

The Assessment Administration course provides fundamental management concepts for management and supervisory personnel in the assessor's office. The course begins by emphasizing the need for management, and the various roles placed on the assessor and all supervisory personnel. The course then introduces the four major management functions (planning, organizing, directing, and controlling). Although the four functions are interrelated, a separate chapter is devoted to each one. This provides for a greater understanding of the major functions.

Recommended: Course 101, The Appraisal Foundation's Uniform Standards of Professional Appraisal Practice (USPAP) and Assessment Administration textbook

Approved: IAAO 30 hours CE

□ Course 402 – Tax Policy

The Tax Policy course offers strategies for assisting in the effective formulation and implementation of tax policies and provides the context in which property tax policy is established. During this course, students will use analytical tools, including sound theory to guide decision-making and creating workable solutions, to explain the effects of proposed property tax changes. This course is designed for assessment administrators and students of taxation, as well as professional policy advisors who guide and make decision in the area of tax policy on a regular basis.

Recommended: Course 101, Fundamentals of Tax Policy textbook

Approved: IAAO 30 hours CE

□ Course 500 – Assessment of Personal Property

The Assessment of Personal Property course is designed to provide students with an understanding and working knowledge of the procedures and techniques required to assess personal property. This course concentrates on the skills necessary for listing, appraising, and assessing the market value of properties using the three approaches to value: the cost approach, income approach and sales comparison approach. This course offers a broad mixture of theory and practical application.

Recommended: Course 101, Property Assessment Valuation (PAV) textbook (3rd edition)

Approved: IAAO 30 hours CE

Pursue your designation at: www.iaao.org

□ **Course 501 – Personal Property Auditing – Basic to Advanced**

The Personal Property Auditing course combines Workshop 552 and 553 material and provides a foundation for understanding basic financial records, specifically related to fixed assets. The balance sheet, depreciation schedule, and income statement are described and discussed in detail. Emphasis is placed on the pertinent data relevant to the personal property appraiser. In addition, the course expands to advanced discussions of accounting and auditing theory as it relates to fixed assets. Capitalization techniques are also discussed. A case study that includes sample financial records is used to "discover" fixed assets that may otherwise be hidden from the appraiser.

Recommended: Course 101, Course 500

Approved: 30 hours CE

□ **Course 600 – Cadastral Mapping Principles and Techniques**

The Cadastral Mapping Principles and Techniques course is designed as a comprehensive and interactive program that introduces entry-level map maintenance personnel and assessment technicians to the cadastral mapping field. Students will learn basic mapping principles and techniques and are expected to demonstrate basic skills that will allow you to plot deeded descriptions in both the metes and bounds land description and Public Land Survey systems.

Recommended: Course 101

Approved: 30 hours CE

□ **Course 601 – Cadastral Mapping Methods and Applications**

The Cadastral Mapping Methods and Applications course is a continuation of the mapping science curriculum and exposes students to aspects of protocol and legal principles that are not featured in Course 600.

Recommended: Course 101, Course 600

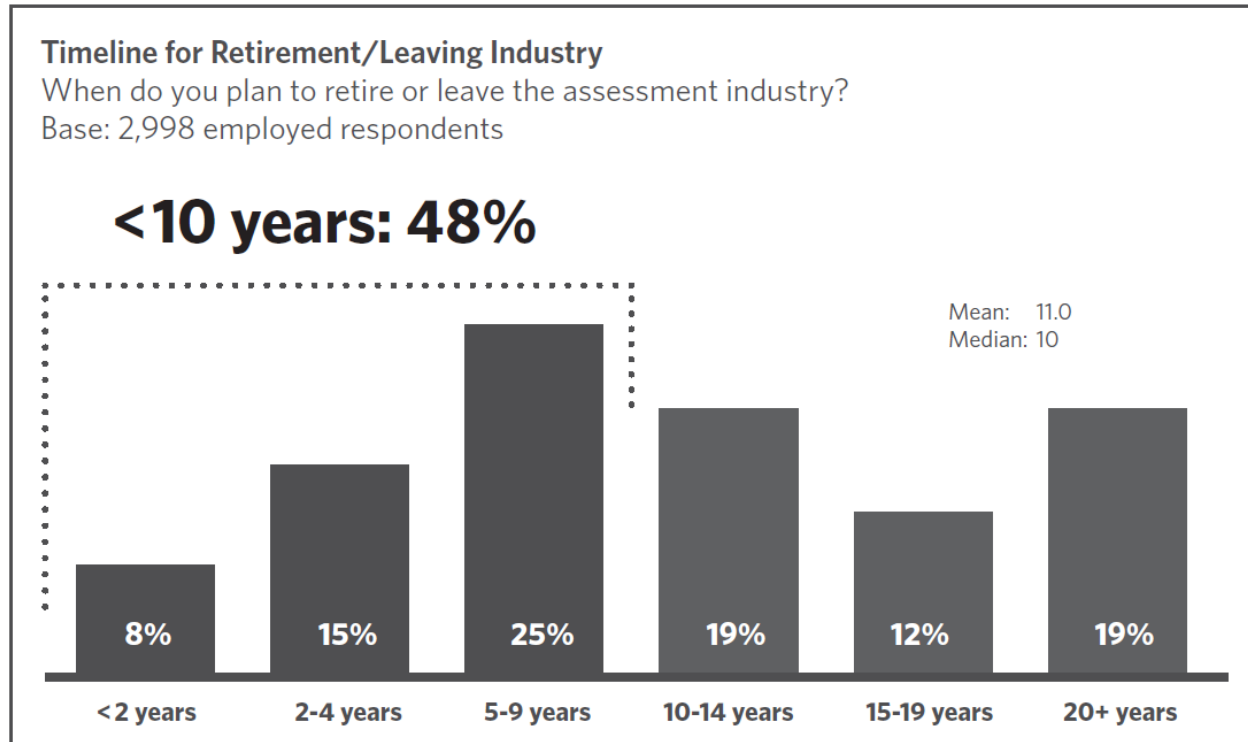
Approved: 30 hours CE

Pursue your designation at: www.iaao.org



Are you positioned to advance your career?

Nearly half of employed assessment professionals plan to retire or leave the assessment industry within 10 years.



Source: IAAO 2016 Assessment Industry Compensation Survey

Prepare now by earning an IAAO designation.



Discover how and apply at www.iaao.org/designation

